DISEASES OF MANGO

1. Powdery mildew: *Oidium mangiferae* (Perfect stage: *Erysiphe polygoni)*

Economic importance: The disease is worldwide in distribution. Reported from India, Pakistan, Ceylon and South Africa. In India the disease is particularly destructive in U.P. Maharashtra and Karnataka severe particularly during the months of December-March, i.e. cooler months.

Symptoms

- The disease is easily recognized by whitish or grayish powdery growth on the inflorescence and tender leaves
- Generally the infection starts from the inflorescence and spreads downwards covering the floral axis, tender leaves and stem. Infected leaves become twisted, curled and fall
- Infected floral parts are severely damaged and drop off. If the fruits are set, they do not grow in size and may drop before attaining pea size. Fruits are sometimes malformed, discolored due to severe mildew attack
- Because of poor fruits set and heavy flower and fruits drop, the loss due to the disease may be as high as 70-80%

Survival and spread

Primary: **Dormant mycelium and conidia** in infected plant debris
Secondary: Wind borne **oidium type conidia**

Favourable conditions

Disease spread is favored by warm humid weather with cool nights

Disease Cycle

Spores blown wind from infected areas readily adhere to hairy, unopened flowers near tip of the inflorescence and germinate in five to seven hours. Fungus grows rapidly during cloudy weather accompanied with heavy morning mist. Warm, humid weather and low night temperatures favour dissemination of the pathogen. Overall disease development is favoured by high humidity.

Management

- Two preventive sprays with wettable sulphur 0.3% before flowers open and after fruit set
- Dusting twice or thrice with fine sulphur
- Spraying with Karathane 0.1% before flowering and after fruit set (peanut stage).
- Resistant varieties: Neelum, Zardalu, Banglora, Torapari-khurd and Janardhan pasand

2) Anthracnose: *Colletotrichum gloeosporioides* (PS: *Glomerella cingulata*)

Economic importance: Reported from several districts of Punjab in 1939. Now prevalent in all mango – growing tracts of the country. The disease is a problem both in the field and storage. Neelum and Banglora are more susceptible.

Symptoms

- The fungus produces leaf spots, blossom blight, wither tip, twigs blight and fruit rot
- On leaves circular, light yellow brown spots are produced initially which soon enlarge with dark brown to black margins. If infection is severe individual spots coalesce leading to twisting of foliage and premature defoliation
- The disease spreads rapidly during rainy season and covers the tender twigs causing them to wither
- On inflorescence stalks and flowers minute spots are produced which increase in size and cause flower drop – the symptom hence is called **blossom blight**
- **Sunken spots** appear on fruits near the stem end as small brown areas that enlarge rapidly and become black. In some cases the areas involved are in the form of streaks running down from the stem end. Fruit pulp beneath the spots become hard followed by cracking and decay at ripening
- The decay is confined to the skin of the fruit except in late stages where it penetrates the flesh in shallow areas. Infected fruits may also drop off prematurely
- **Latent infection** is carried from the field and develops further which causes **rotting in the storage**

Survival and spread

Primary: **Saprophytic mycelium and conidia** in infected plant debris

Secondary: Wind borne **and rain splashed conidia**

Favourable conditions

- Temperature 25° C and R.H 95-97%
- Injuries to fruits at harvest and in storage

Disease cycle

The survival of pathogen in detached diseased twigs and leaves lying on surface of soil and in diseased twigs attached to the tree. They successfully reproduced the disease by inoculating leaves, petioles, stems and fruits. The optimum temperature for infection was found o be 25˚C. The disease spreads rapidly in the rainy season. Cloudy and misty weather during
flowering favors damage to the infected floral parts.

The pathogen causes severe leaf spotting. The appearance of spots in more concentration at the stem-end and sometimes in stripes down the sides of the fruits suggested distribution of spores by rain water over surface of the fruit. The fungus can enter the pores of green fruits. The latent infection of mature fruits may take place through lenticels. The fungus apparently infects the fruit while it is green and develops in flesh during ripening.

Etiology: *Colletotrichum gloeosporioides*.

- Acervuli developed on diseased parts of the plants.
- They are irregular and appear as brown to black dots. Setae are common on twigs but not on fruits.
- The acervuli when mature exude pink masses of conidia under moist conditions. Marginal setae are rare.
- Conidia are borne on hyaline conidiophores. The conidia are straight, cylindrical or oval, hyaline with two oil drops and are non-septate with round ends

Management

- Avoid over-crowding in orchards
- Tree sanitation is important. Diseased twigs are to be pruned and burnt along with fallen leaves and fruits
- Spray carbendazim or thiophanate methyl (0.1%) or Chlorothalonil (0.2%) at 14 days intervals from fruit set until harvest
- Before storage, treat the fruits with hot water (50-55°C) for 15 minutes or dip in Benomyl solution (500 ppm) or Thiobendazole (1000 ppm) for 5 minutes or expose them to ammonia and sulphur dioxide gases

3) Mango Malformation: ***Fusarium moliliforme* var. *subglutinans***

Economic importance

- This is one of the severe diseases of mango and is important in North India. It appears in and around Hyderabed & Medak Districts
- In A.P. this was first noticed in Aragonda village in 1971
- In India it is known to occur in U.P., Maharashtra, Haryana, Bihar, Punjab and A.P. particularly severe in U.P. causing much damage
- Coastal areas are free from the disease

Symptoms

There are two types of symptoms namely **vegetative** malformation and **floral** malformation because of hypertrophic effect of Gibberellic acid produced by the pathogen

Vegetative malformation

- There is a proliferation of infected tissue. The affected plants develop excessive vegetative branches which are of limited growth, swollen and have short internodes
- These dwarf branches are of various sizes which are often produced on the top of young seedlings giving a **bunchy top** appearance. The axillary buds of dwarf branches are unusually enlarged
- Vegetative malformation is more pronounced in young seedlings and seedling trees than in grafted plants

Floral malformation

- The flowering panicles instead of coming out as a normal one turns into compact bunch of hard flowers. **Individual flower is greatly enlarged** and has a large disc
- Based on compactness of the inflorescence three types of malformations have been described
  Light malformation – light compaction of inflorescence but bears fruits
  Medium malformation – moderately compact inflorescence with discontinuous growth and no fruit set
  Heavy malformation – entire inflorescence is converted in to a compact mass of sterile flowers. Inflorescence continues to grow with flowers having **hypertrophied bracts**. The percentage of bisexual flowers in malformed panicles is very low
- The malformed heads dry up in black masses and persist on the trees for a long time
- Affected inflorescences generally are aborted and bear no fruit

Survival and spread

Primary: **Saprophytic mycelium and chlamydospores** in infected plant debris
Secondary: Wind borne and mite disseminated **micro** and **macro conidia**

Favourable conditions: and cool and

- Prevalence of eryophyid mite
- Misty wet and cool weather

Etiology:

- *Fusarium moniliformae* var. *subglutinans*. Micro conidia are unicellular, oval to fusiform and produced from polyphialides.
- Macro conidia are 2-3 celled and falcate.
- Asexual fruiting body of the fungus is sporodochium.

Management

- Use of disease free planting material and prophylactic spray of insecticides and fungicides
- Pruning of diseased parts along with basal 15-20 cm apparently healthy portions
- Spray Benomyl or Carbendazim (0.1%).
- Spray napthelene acetic acid (NAA) 200 ppm during first week of October

4) Sooty mould: *Capnodium ramosum*

Symptoms

- Black superficial encrustation of mass of spores of the fungus on upper surface of leaves affecting the photosynthetic activity. The fungus has no parasitic relationship with host
- Black encrustation forms on the fruits affects appearance and hence market price
- Pathogen grows and thrives on the sugary or honey dew secretions of **plant hoppers**

Survival and spread

Primary: **Saprophytic mycelium and spores** in infected leaves
Secondary: **Conidia** disseminated by wind and scale insects and aphids, *Lecanium hemisphaericum, L. viride, L. nigrum, Hemilecanium imbricans* and *Pulvinaria psidii*

Favourable conditions

Infestation by honey dew secreting sucking insects particularly plant hoppers

Management

- Controlling of plant hoppers by spraying Carbaryl (2g/lt) or Phosphomidon 0.03%
- Spray dilute solution of starch 5% (Boil 1 kg of starch in 5 litres of water, cool and dilute to 20 litres). The dried starch comes off in flakes, removing along with it the sooty mold growth

5) Red rust: ***Cephaleuros virescens* (Green algae)**

Economic importance: In India it has been observed in Bihar, Karnataka and U.P. The disease appeared in an epidemic form in orchards of Tarai in 1956

Symptoms

- The disease is characterized by red rusty spots on the leaves and young twigs
- The spots are initially circular, slightly elevated and later coalesce and become irregular
- The upper surface of the spot consists of numerous unbranched sterile or fertile filaments of the pathogen which project through the cuticle

- Initially the spots are greenish-grey in colour but later on the surface bears reddish-brown appearance because the algal cells are impregnated with pigment which masks the normal green chlorophyll

Survival and spread
Primary: **Algal filaments and sporangia** in infected leaves
Secondary: Rain splashed **sporangia** and **zoospores**

Favourable conditions
- High density planting
- Frequent rains with high humidity

Management
- Avoid close plantations
- Prune unproductive branches to improve air circulation within the canopy of a tree
- Spray Bordeaux mixture 1% or copper oxychloride 0.3%

6. Stem end rot: *Diplodia natalensis*
Symptoms
The dark epicarp around the base of the pedicel. In the initial stage the affected area enlarges to form a circular, black patch. Under humid atmosphere extends rapidly and turns the whole fruit completely black within two or three days. The pulp becomes brown and somewhat softer. Dead twigs and bark of the trees, spread by rains
Pathogen
The fungus produces brown to black, globose to sub globose, pyriform, erumpent pycnidia that are ostiolate. Two types of conidia are produced within a pycnidium. One is hyaline, thin walled and unicellular. The another one is thick walled and bicelled with four to six longitudinal striations.
Mode of spread and survival
The fungus persists in infected plant parts which serve as source of inoculum.

7. Gummosis/ stem bleeding/ Die-back (*Botryodiplodia theobromae*)
Die-back is prevalent in mango growing states of India.

B. theobromae causing a variety of diseases including die back, fruit rot, dry rot, root rot and wood stain on different hosts. It is quite capable of causing browning of soft wood tissue and cracking of hard wood followed by gummosis and death. Graft union infection may cause early stage death of grafts. If the infection in young twigs occurs at nodal points, infected portion becomes dead resulting into death of rest of the apical portion. Its infection is mostly facilitated by wounding. Trees weakened due to water stress, extreme summer and winter temperatures, sun scorch, high humidity in propagation nurseries, hardpan soils and
nutritional deficiencies are more susceptible to infection

Symptoms:
The disease is characterized by dying back of twigs from top downwards, particularly in the older trees, followed by complete defoliation giving an appearance of scorching by fire.
Dark lesions on bark appear which increase in size resulting in death of young twigs.
The upper leaves lose their healthy green colour and gradually turn brown accompanied by upward rolling of leaf margin.
In advanced stages, such leaves shrivel, fall off in a month or more, leaving the shriveled twigs almost bare.
Cracks appear on branches which exude gum before they die.
When the graft union of young plant is affected, it usually dies
The infected twigs show brown streaks of vascular tissues on splitting the twigs lengthwise.
The gummosis is found more prominent during winter after rainy season.
This pathogen also attacks ripe fruit in storage at the base of pedicle (stem end rot) and the circular brown area near the stem end further develops towards the lower portion of the fruit. Later entire fruit surface is covered with the dark brown to black area and complete fruit rots in 2 to 3 days. (PH symptom)
The disease may also start from injured portion on the fruit surface
Roots also show splitting of root epidermis and blackening of tissue and on digging roots emit bad odour. Growth of fungal mass is also observed on the root tissue.
Epidemiology:

High summer temperatures predispose the mango plants to the attack of the disease by reducing the vitality of the plant.

RH of about 80%; maximum and minimum temperatures of 31.5°C and 25.9°C, respectively, and rains favour disease development.

The optimum temperature for spore germination was 30°C.

Management

Harvesting of mangoes on clear dry day, preventing snapping of the pedicel and avoiding injury to the fruits at all stages of handling helps to reduce the disease incidence.

Preventive measures such as selection of scion from healthy trees, sterilization of the budding knife, keeping the grafted plants in a relatively dry environment and gradual exposure to full sunlight are effective.

Pruning of diseased twigs 7-8 cm below the infection site followed by pasting of Bordeaux paste (5-5-50) or 0.3% COC before the onset of monsoon was effective in controlling the disease.

Mango cv. Siroli was found resistant.

Application of fresh cow dung to the cut surfaces of twigs and pruned branches kept mango plants completely free from the incidence of gummosis and bark splitting up to 6 months due to the presence of high population of Actinomycetes which are reported effective against anthracnose, stem end rot diseases and gummosis of mango.

DISEASES OF CITRUS

1. Gummosis	:	*Phytophthora nicotianae* var. *parisitica* *P. palmivora, P. citrophthora* and other sps.
2. Diplodia gummosis	:	*Diplodia natalensis* *Physalospora rhodina* (Perfect stage)
3. Twig blight	:	*Diplodia natalensis,* *Colletotrichum gloeosporioides*
4. Pink/Cob webb/Rubellosis	:	*Pellicularia salmonicolar* *Corticium salmonicolar*
5. Felt	:	*Septobasidium pseudopedicillatum* + Scales
6. Powdery mildew	:	*Oidium tingitanium* (=*Acrosporium tingitanium*)
7. Scab	:	*Sphaeceloma fawcettii* (IPS) *Elsinoe fawcettii* (PS)
8. Dry root rot	:	*Macrophomina phaseolina* *Rhizoctonia bataticola*
9. Ganoderma root rot	:	*Ganoderma lucidum*
10. Sooty mold	:	*Capnodium citri, C. ramosum*
11. Canker	:	*Xanthomonas axonopodis* pv. *citri*
12. Greening	:	Candidatus Liberobacter asiaticum in Asian citrus Candidatus Liberobacter africanum in African citrus (FVB or Ricketsia like bacteriums)
13. Yellow Mosaic	:	*Citrus yellow Mosaic Virus*
14. Tristeza	:	*Citrus Tristeza Virus* (Closterovirus)
15. Psorosis	:	*Citrus Psorosis virus*
16. Exocortis	:	*Citrus Exocortis Viroid*
17. Post harvest diseases	:	
i) Penicillium fruit rot	:	*Penicillium italicum* (Blue mold) *P. digitatum, P. expansum* (Green mold)
ii) Sour rot	:	*Geotrichum candidum*
iii) Brown rot	:	*Phytophthora nicotianae*

1. Gummosis:

Caused by *Phytophthora nicotianae* var. *parisitica, P. palmivora, P. citrophthora* and other sps. Also known as collar rot, crown rot, foot rot, fruit rot etc. More in high rainfall areas.

Economic importance: Gummosis is widespread in Punjab and Assam. Lemons are highly susceptible compared to grapefruit, rough lemons and sweet orange. In South India, it is common in the sweet orange.

Symptoms

This is soil borne fungi. Primary colonization is on roots as discolouration, root decay, bark regradation at collar region and leaf falling

Disease starts as water soaked large patches on the basal portions of the stem near the ground level

First symptoms are dark staining of bark which progresses into the wood.

Bark in such parts dries, shrinks and cracks and shreds in lengthwise vertical strips.

Bark at the base is destroyed resulting in girdling and finally death of the tree.

Later profuse exudation of gum from the bark of the trunk occurs. There may be a considerable amount of gum formation in sweet oranges, but relatively little in grapefruit.

Infection extends to crown roots.

Prior to death, the plant usually blossoms heavily and dies before the fruits mature**.**

Disease cycle:

The pathogen survives on fallen leaves, twigs, leaves and in cracks on trees as **dormant mycelium**. It also survives as **oospores** in soil.

Secondary spread of the disease is mainly by moist wind/water disseminated sporangia and zoospores. Insects also spreads the fungus to some extent.

Favourable conditions:

- Varietal susceptibility: Sweet oranges, acid lime and lemons are highly susceptible.
- Heavy and ill drained soils with high water table.
- Constant wetting of the trunk base through excessive or improper irrigation leading to high soil moisture.
- Application of FYM close to trunk.
- Low grafting and deep planting of plants.
- Injuries to the crown roots.
- Soil temperatures of 25 – 28 ^{0}C and PH 5.4 – 7.5.

Management:

- Planting pits may be dusted with a mixture of $ZnSO_4$ + $CuSO_4$ + Quick lime (5:1:4) prior to planting.
- Avoidance of excessive irrigation and constant wetting of trunk. To avoid this double ring method of irrigation is practiced. Inner ring of about 45 cm around the tree trunk to prevent the moist soil and irrigation water coming in contact with trunk and heaping earth around plant base.
- Avoidance of application of FYM close to the trunk.
- Avoidance of injuries to the trunk.
- Avoidance of flow of irrigation water from one trunk to another.
- Selection of high budded plants. The bud joint should be 30-45 cm above the ground level.
- Application of bordeaux paste above the ground level 50 to 75 cm.
- In the initial stages of infection, affected portion should be removed with a knife and application of bordeaux paste or $ZnSO_4$ + $CuSO_4$ + Quick lime (0.6:0.2:0.5 kg/100 litres) followed by spraying of BM 1% or Aureofungin 3gm+30 ml liquid soap in 136 litres of water.
- Soil drenching with Fosetyl Al (Aliette) 0.3% or Ridomil MZ 0.2%.
- *Trichoderma viride* formulation 1 kg multiplied under moist conditions and in shade on 90 kg FYM and 10 kg neem cake and applied at 10 kg per plant.

2. Diplodia gummosis: Caused by *Diplodia natalensis*. Perfect stage is *Physalospora rhodina.*

Infection is mostly confined to branches and on trunks seen on upper portions.

Economic Importance: Occurs commonly in Andhra Pradesh, Tamil Nadu and Uttar Pradesh. It is common on Sathgudi and Batavian oranges, mandarinsand lemons in A.P. and Tamil Nadu.

Symptoms:

- Profuse gumming of higher portions of the plant on twigs, branches and trunks.
- Disease starts at cracks or crotches of branches.
- Diseased portions are cracked, black, dry and come out as flakes.
- From bark, infection spreads to inside.
- The gum is black in colour.
- Large limbs are killed and if left unchecked the whole tree may be killed in course of time.
- Sometimes branches break at the infected portion

Favourable factors:

The disease is favoured by rains, high humidity levels, reduced vigour, insect damage, poor nutrition and old age of the plants.

Mode of spread and survival
Black pycnidial bodies appear on the diseased bark which gets removed in flakes.
The disease spreads through dissemination of pycnidiospores or conidia by air, rain and insects
Management:

- Pruning affected branches 25-35 cm below the lowest point of infection prior to monsoon and protecting the cut surface with Bordeaux paste.
- Spraying Bordeaux mixture 1% or any copper fungicide prior to monsoon.
- Spraying of Carbendazim 0.1% or Chlorothalonil 0.2% or Iprodione (Rovral) 0.2% during Feb-March.
- Cleaning wounds and covering with bordeaux paste.

3. Twig blight : Caused by pathogens like *Diplodia natalensis, Colletotrichum sps.*
Symptoms:

- Twigs and branches die back.
- Drying of foliage associated with gumming and bark shreading.

Disease cycle:
PSI: Diseased twigs.
SSI: Wind/splashing rain disseminates the conidia.

Management:

- Pruning and burning of affected plant parts before monsoon and application of 0.1% Carbendazim or Kavach or Rovral 0.2%.

4. Pink/Cob webb/Rubellosis:
Caused by *Pellicularia salmonicolar* or *Corticium salmonicolar.*

Symptoms:

- Produces characteristic silvery white mycelium at branching points.
- With the progress of the disease pinkish pustules appear.
- Longitudinal cracking and gumming of the branches noticed.
- Bark dries and gets shredded.
- Branches dry out and leaves turn yellow and dry.

Disease cycle:
PSI: Dormant mycelium on dried branches or basidiospores on branches.
SSI: Wind or rain disseminated conidia.

Management:

- Pruning branches 40-45 cm below the lowest point of infection and burning.
- Protecting the cut ends with bordeaux paste.
- Spraying Bordeaux mixture 1% or any copper fungicide prior to monsoon.
- Spraying of Carbendazim 0.1% or Chlorothalonil 0.2% or Iprodione (Rovral) 0.2%.

5. Felt
Caused by *Septobasidium pseudopedicillatum* + Scales

Symptoms:

- Light brown to grey leathery growth on twigs and branches which encircles them.
- Sometimes petioles, leaf blades and fruit stalks are also attacked.
- Disease occurs due to symbiotic association of the fungus and scales insects.
- Twigs wilt and die back due to sucking of the sap by scale insects.

Disease cycle:
PSI: Fungus surviving on infected branches.
SSI: Wind or rain disseminated conidia.

Management:

- Pruning all dried twigs, removal of felt like growth and burning.
- Spraying of the plants with Captan 0.2% or Zineb 0.25% + Monocrotophos @ 1.5 ml/litre during monsoon season.

6. Powdery mildew:

Caused by *Oidium tingitanium* (IPS) *Acrosporium tingitanium* (PS).

Symptoms:

- More severe during Dec-March.
- Leaves consist of profuse powdery growth on the upper side of tender leaves and twigs. Leaves turn yellow, distorted and fall down.
- When fruitlets are attacked they drop out.
- Twigs show die back symptoms.

Disease cycle:

PSI: Fungus surviving on diseased bud, twigs and leaves. Some times by Ascospores produced on infected parts.

SSI: Wind disseminated conidia (Oidia).

Management:

- Dusting fine sulphur powder (200-250 mesh) early in the morning when dew is present at 8 days intervals.
- Spraying with wettable sulphur 0.2% or Cosan 0.1% or Thiovit 0.05%.
- Spraying of Tridemorph 0.1% or dinocap 0.1% or Carbendazim 0.1% or Bayleton 0.1%.

7. Scab:

Economic Importance: In India this disease is reported from Bengal, Punjab, Assam, Karnataka, Uttar Pradesh and Andhra Pradesh. It is one of the common diseases of sour oranges, rough lemon, tangelos and Pummelo. In Andhra Pradesh it is severe on Rangpur lime. The sweet oranges are mostly resistant to the disease

i) Sour orange scab	:	*Elsinoe fawcettii* (Similar to Indian scab)
ii) Australian or Tyron's Scab	:	*Sphaceloma fawcetti var scabiosa*
iii) Sweet orange fruit scab	:	*Elsinoe australis*

Symptoms:

- On leaves, small, semi translucent dot like areas develop on the lower surface and later become pustular.
- Leaves distorted, wrinkled, reduced in size and misshaped.
- The opposite surface of the leaf corresponding to warty growth shows a circular depression with a pink or red centre.
- Affected twigs are killed.
- On fruits, scabby or corky lesions appear.
- Scab symptoms resemble with those of canker in being warty and differs from canker by wartiness being confined to one side of the leaf and distortion of leaves (No distortion of leaves of leaves in canker)

Disease cycle:

PSI: Survives in off season as ascospores.

SSI: Conidia spread by air or rain.

Management:

- Collection and destruction of diseased material.
- Spraying of 1% Bordeaux mixture or 0.3% COC or 0.2% Chlorothalonil at 15 days gap.
- Using resistant root stocks like Cleopatra, Citrange, Citrumelo, Savage etc.
- Resistant varieties like Sweet lime, Kagzi lime, lemon, Gajanimma, etc.
- Kumquats, Khasimandarin and *Poncirus trifoliata* are immune.

8. Dry root rot: Caused by *Macrophomina phaseolina* and *Rhizoctonia bataticola.*

Symptoms:

- Moist decay of the bark in the early stages and later drying and shredding.
- Foliage becomes yellow, defoliation occurs rendering the twigs bare.
- Affected roots emit foul smell and peeling occurs.
- Black sclerotia are observed on the root surface.
- Affected trees die before fruits mature.

Disease cycle:
PSI: Scletotia developed on the infected plant parts i.e. roots.
SSI: Secondary spread of the disease is by irrigation water and root contact.

Management:

- Avoid injury to roots.
- Diseased roots to be cut and apply Bordeaux paste.
- Conservation of soil moisture in the basins of trees by means of mulching to avoid disease due to drought.
- Apply neem cake, *T. viride, T. harzianum* and other biocontrol agents.
- Drenching in the basins with Carbendazim 0.1% + Mancozeb 0.2%.

9. Ganoderma root rot: Caused by *Ganoderma lucidum.*
Ganoderma are also called as "Bracket fungi"

Symptoms:

- White strands of mycelium spreading along the surface of the bark of roots.
- Infection spreads from lateral roots to main root and then to trunk,
- Wood becomes soft and decomposed.
- Yellowing of leaves and drying and later the plant dies.
- During rainy season, brackets, the fungal fructifications will be formed at the base of the trunk.

Disease cycle:
PSI: The fungus is soil borne. It survives as basidiospores and also spreads by root contact.
SSI: Wind or water disseminated basidiospores.

Management:

- Removal of dead and decaying old stumps, brackets from the orchard.
- Treating the basal portions of stakes used for support to young seedlings with fungicides.
- Avoid injury to the roots.
- Green manuring crops should be grown and incorporated in the soil.
- In the initial stages of infection, infected roots should be removed and burnt. Cut surfaces should be protected by applying Bordeaux paste.
- **Isolation of severely infected trees by digging trenches around the tree to avoid infection by root contact.**
- Drenching in the basin with a mixture of Bavistin 10 gm + Vitavax 5 gm in 50 litres of water around the tree trunk once before monsoon and once after monsoon.

10. Sooty mold: As in Mango

11. Canker:
Caused by *Xanthomonas axonopodis* pv. *citri.*
Economic Importance: In India, citrus canker is endemic and occurs in all the citrus growing areas. It is reported from Punjab, Tamil Nadu, Andhra Pradesh, Karnataka, Rajasthan, Madhya Pradesh, Assam, and Uttar Pradesh. Acid lime is highly susceptible to canker
Symptoms:

- Small, brown, raised corky or warty out growths formed on leaves, twigs, thorns, petioles and fruits.
- On the leaves symptoms are seen on either surface and they are more prominent on lower surface. (In scab the warty growth is on one side only).
- No distortion of leaves as in the case of Scab.

- On Fruits, cankerous growth is limited to the rind portion and will not go deep into the flesh.
- Fruits look ugly and fetch less value in the market.
- Cankerous growth is also observed on petioles, thorns and twigs.
- Twigs die back and leaves fall.

Disease cycle:
PSI: Diseased plants and planting material-Cankerous leaves, twigs and branches.
SSI: Splashing rain water disseminates the bacterial cells.
The disease is also spread by leaf miners *Phyllacnistis citrella.*

Management:
- Pruning infected twigs and burning before the onset of monsoon.
- Procuring disease free seedlings.
- Spraying of BM 3-4 times at 15 days gap after pruning of twigs.
- Spraying with Streptomycin/Streptomycin sulphate 500-1000 ppm at 15 days gap.
- Spraying of COC 0.3% + an insecticide.
- In nursery, spraying of neem cake decoction (1kg in 20 litre water, kept 7-10 days and then filtered) through out rainy season.
- Variety Balaji, Tenali selection is important tolerant variety in A.P.
- Other improved varieties tolerant are Saisarbati, Pramalini etc.

12**. Greening:**
Economic importance: Greening disease is known to occur in Andhra Pradesh, Karnataka, Punjab, Uttar Pradesh, Himachal Pradesh, Rajasthan, Maharashtra, Jammu, Bihar, Bengal and Sikkim. The disease is more severe on sweet oranges than on acid lime, mandarin and grapefruits. This disease affects almost all citrus varieties irrespective of root stock. In India Mosambi sweet orange and Darjeeling orange (*Citrus reticulata*) are good indicator plants for greening.
The disease is first thought to be caused by virus, later MLO and now it is established that it is caused by phloem restricted Gram –ve obligate bacterium. Ricketsia like bacterium.
Candidatus Liberobacter asiaticum in Asian citrus.
Candidatus Liberobacter africanum in African citrus.
More severe in sweet orange.

Symptoms:
- Leaves develop striking chlorotic patterns resembling 'Zn' deficiency which cannot be corrected by 'Zn' sprays.
- Leaf size is reduced, they are thick and erect and gradually become chlorotic.
- Characteristic green islands appear against yellow background on leaves.
- Defoliation, twigs are short upright and become bare.
- Twigs show multiple bud formation, off season blooming, shortened internodes and die back symptoms.
- Fruits on the side exposed to the sun show yellowish area while those in the shade remain green even during ripening.
- Fruits are lopsided with columella curved/oblique.
- Diseased fruits fetch less price due to small size, distortion, low juice and insipid taste.

Disease cycle:
Transmitted by bud wood/grafting and through insects.
Citrus psylla *Diaohorina citri* and *Trioza erythreae* transmits the disease.

Management:
- Removal and destruction of infected trees.
- Control of insect vectors by spraying Dimethoate (Rogor) 0.2% or Metasystax 0.05%.
- Use of certified bud wood.
- Spraying of tetracycline 500 ppm for inhibition of pathogen.
- Bud sticks should be dipped in 500 ppm Ledermycin or penicillin extract in combination with 0.5% Bavistin for one hour before grafting.
- Using mild strains for cross protection.
- Heat therapy-Bud wood exposed to 47 ^{0}C for 4 hrs.
- Injecting tetracycline hydrochloride 6-10 grams as tree injections for mature plant.

- Injecting chemotherapeutant tree injection BP-101 or systemic anti fungal anti ricketsial chemotherapeutant TC 2298.

14. Tristeza or Quick decline: Caused by ***Citrus Tristeza Virus*** (Closterovirus)
In Brazil, "Tristeza" means 'sad appearance'.
Economic Importance: This disease was first reported in *Citrus aurantifolia* and *C. sinensis* from Italy and Florida in the U.S.A. In India, tristeza is present in Andhra Pradesh, Tamil Nadu, Karnataka, Madhya Pradesh, Maharashtra, Punjab, Bihar, West Bengal and Sikkim.
It affects all kinds of citrus plants but primarily orange, grapefruit and lime. Tristeza symptoms consisting of a quick or chronic tree decline are particularly common and severe on trees propagated on sour orangeroot stocks. The name "Tristeza" was suggested to describe the sad appearance of the diseased citrus trees. Kagzi lime and Nasnaran are indicator plants for CTV detection.

Symptoms:
- Trees look chlorotic and sick and exhibit deficiency symptoms.
- Trees defoliate and show die back symptoms and gradually die.
- Trees become open and subjected to sun burn.
- Phloem tube necrosis hinders the transportation of carbohydrates from top to roots and roots exhibit varying degrees of rotting of fibrous and bigger roots.
- In acid limes, leaves show intermittent vein flecking.
- In sweet orange, the specific symptom of tristeza is honey combing, a fine pitting of inner face of bark in the portion of trunk below the bud union. In acid lime which is highly susceptible to the disease, vermiform or linear pits appear in the woody cylinder.
- Tristeza infected citrus trees on sour orange rootstocks cause phloem necrosis at the graft union
- Diseased trees usually blossom heavily. Trees with stem pitting are stunted and set less fruits. The fruits are of smaller size and of poor quality (insipid fruits). As the fruits develop, the tree wilts partly or completely
- Sweet oranges are symptomless carriers.
- Kagzi like (*Citrus aurantifolia*) is an ideal indicator plant.

Disease cycle:
PSI: The disease is both vector and bud transmissible. Major transmission is by bud wood.
SSI: Transmitted by aphids.
Toxoptera citricidus is widely distributed in south India.
Toxoptera aurantii - Black citrus aphid transmits mild strains.
Other aphids involved are *Myzus persicae, Aphis crassivora, A. gossypii.*

Management:
- Destruction of infected trees and replacement with virus free plants.
- Using of certified bud wood.
- Cross protection (Pre immunization).
- Use of resistant root stocks- Rough lemon, Sweet orange, Rangapur lime, Trifoliate orange.
- CTV is inactivated by heat therapy by exposing the bud wood at 35-45 ^{0} c for a period of 7 days.
- Periodic spray of insecticides like Monocrotophos @ 0.05% to reduce the secondary spread.

15 Psorosis: Caused by *Citrus Psorosis virus.*

Symptoms:
- Different strains of viruses are involved Psorosis A, Psorosis B, Blind pocket psorosis, Concave gum psorosis.
- Vein clearing and flecking on young leaves.
- Typical scaling and browning of the bark.
- Symptoms on bark noticed 6-12 years age of the plantsas small pimple like eruptions followed by bark scaling.
- Exudation of gum, bark rupturing near the diseased portion.
- Wood turns brown to reddish brown and dies and defoliation occurs.
- Vascular vessels are blocked by gum.

Disease cycle:

The disease is probably spread by root grafts. Bud wood and Cuscutta also transmits the disease.

Management:

- Scrapping and disinfection of damaged bark using Potassium permanganate 1%.
- Heat therapy for young plants and grafts.
- Use bud wood from high performing trees of more than 25 years of age, which do not show any symptom till that time.

16. Citrus Yellow Mosaic

The disease is found on satgudi and Mosambi sweet oranges in South India and in Maharashtra. An incidence of the disease upto 70% was observed on Satgudi in Andhra Pradesh.

Symptoms: The most characteristic symptom of the disease is yellow mosaic on leaves of infected trees

The trees are systematically infected. The trees over 10 years of age show decline. Most of the commercial cultivars are susceptible to virus infection.

Temperature does not appear to effect the symptom development.

Virus -*Citrus yellow mosaic disease*

Pathogen: The causal virus has bacilliform particles measuring 130 x 30 nm in size. The genome of the virus is double stranded DNA andthe virus belongs to Badnavirus group. The virus is named as citrus yellow mosaic badnavirus (CYMBV). The vector of the virus is a mealybug, *Planococcus citri*. The virus is also transmitted by grafted.

Disease Cycle: The primary spread of the virus is through the use of infected budwood for propagation. Secondary spread may be by the mealybug vector but the spread by the vector is comparatively low.

Control: Use of healthy planting material and control of mealy bug fromcitrus and neighbouring crops such as sugarcane are the methods to control the disease.

17. Exocortis: Caused by *Citrus Exocortis Viroid*. The disease is also called as **scaly butt.**

Symptoms:

- Cracking and scaling of the bark below the bud union.
- Produces narrow, vertical thin strips on the outer portion of the bark near the soil and extends upwards and down wards.
- Bark of the green twigs may show chlorotic blotching.
- Trifoliate orange root stock over grows scion in three or more years but if infected with exocortis, the over growth is missing or absent.

Disease cycle:

Infected bud wood, contaminated tools and sap. **Not transmitted by insects and seeds.**

Management:

- Removal of unproductive trees and replanting viroid free trees on tolerant root stocks such as Rough lemon, Cleopatra, Sweet orange and Sour orange.
- Proper cleaning and sterilization of tools before moving from one black of trees to another with trisodium orthophosphate and soap water emulsion.

DISEASES OF GRAPEVINE

1.	Anthracnose	:	*Gloeosporium ampelophagum* *Elsinoe ampelina* (PS) *Sphaceloma ampellophagum*(IPS)
2.	Downy mildew	:	*Plasmopara viticola*
3.	Powdery mildew	:	*Uncinula necator* (PS) *Oidium tuckeri* (IPS)
4.	Alternaria leaf spot	:	*Alternarla vitis*
5.	Rust	:	*Physopella ampelopsides*, *Phakopsora vitis*
6.	Bacterial leaf spot	:	*Xanthomonas ampelina*
7.	Dead Arm	:	*Phomopsis viticola, Pestalotiopsis viticola*

1. Anthracnose:- *Gloeosporium ampelophagum* (Anamorph/Imperfect stage)

Elsinoe ampelina (Perfect stage/Telomorph). Anthracnose is a Greek word meaning **Ulcer.**
It is especially serious on new sprouts during rainy season. Among various foliar diseases of grapevine in India, anthracnose has **longest spell spread over the p**eriod from June to October

Symptoms:

- Visible on leaves, stem, tendrils and berries.
- Young shoots and fruits are more susceptible than leaves.
- Circular, greyish black spots or red spots with yellow halo appear.
- Later the centre of the spot becomes grey, sunken and fall off resulting in a symptom called '**shot hole'**.
- Black, sunken lesions appear on young shoots.
- Cankerous lesions on older shoots. Girdling and death of shoots occur.
- Infection on the stalk of bunches and berries result in the shedding of bunches and berries respectively.
- Sunken spots with ashy grey centre and dark margin on fruits (**Birds eye symptom**). In warm and wet weather pinkish spore mass develop in the centre of the spots
- Mummification and shedding of berries.

Diseases cycle:

The mycelium of the fungus over winters in the canes and give rise to conidia or rarely Ascospores in spring.
Secondary spread by wind borne conidia.

Management:

- All canes showing cankers / symptoms should be removed while pruning and burnt.
- One spray of Bordeaux mixture @ 1% should be given after pruning.
- Spray vineyards at the time of leaf emergence with Carbendazim (0.1%) or Hexaconazole (0.1%) or Bordeaux mixture (1%) or Thiophenate methyl (0.1%), Bitertenol (Baycor) @ 0.1%, Propineb (Anthracol) and Mancozeb @ 0.2%.
- Bharath Early, Hussaini, Banglore Blue, Gulabi, Khali are resistant.

2. Downy Mildew:- *Plasmopara viticola.*

Economic importance

Since 1875, this disease caused heavy losses in France to wine industry.
It led to discovery of B.M. by Prof. Millardet in 1885.

Symptoms:

- Symptoms appear on all aerial and tender parts of the vine. Symptoms are more pronounced on leaves, young shoots **and immature** berries.
- Greenish yellow translucent indefinite oily spots on upper surface.
- The lower side of the leaf is covered by whitish downy fungal growth.
- Infected tissues gradually killed and the necrotic lesions become light to dark brown.
- Whole shoot including tendril, young leaves, often become covered with white fruiting structures of the fungus.
- Infected inflorescence turns to oily yellow to brown, rot or blight.
- Infected clusters develop downy growth under high humidity.
- Berries become leathery, drop easily or mummified remaining attached to bunch.

Disease cycle:

Primary source of inoculum is "Oospores" which are embedded in the dead leaves and dormant mycelium in the host tissue.
Secondary spread is by Sporangia and Zoospores and they are spread by drops of water splashed in rains.

Pathogen: *P. viticola* is a biotroph. The intercellular mycelium of the fungus is coenocytic, thin-walled, hyaline, and produce spherical or pear shaped haustoria

Management:

- Sanitation of orchard to prevent the primary infection from Oospores.
- Free circulation of air by proper spacing and pruning, facilitating rapid drying reduces secondary infection.
- Protective application of fungicides such as Bordeaux mixture (1%), Copper oxy chloride (0.3%), Chlorothalonil (0.2%), Dithane Z 78 (Zineb) (0.2%), Ziram (Cuman L) (0.2%)
- Under heavy disease risk situations due to favorable weather conditions give sprays of systemic fungicides such as Metaloxyl + Mancozeb (Ridomil MZ) 2-3g/litre, Dimethomorph + Mancozeb, Curzate + Mancozeb, Fosetyl Al etc. at 10-12 days interval.

- Grow resistant varieties like Amber Queen, Cardinal, Champa, Champion, Dogridge and Red Sultana, Banglore Blue
- The disease can be effectively managed by giving 3-5 prophylactic sprays with 1% B.M or Fosetyl -Al (Aliette) 0.2% or metalaxyl + mancozeb 0.3 to 0.4% or Azoxystrobin or Dimethomorph
- Chemical -5 sprays with 1% B.M.
  1 – Immediately after pruning of vines
  2 - When new flush formed (3-4 weeks after pruning)
  3 – Before buds open
  4 - When bunches or berries have formed
  5 – During growth of shoots

3. Powdery mildew:- *Uncinula necator* (Perfect stage/ Telomorph/ Sexual stage)
Oidum tuckeri (Imperfect stage/ Anamorph/ Asexual stage)

More severe during Oct –Nov in North India and Feb –June in South India

Symptoms:
- The disease attacks the vines at any stage of their growth. All the aerial parts of the plant are attacked. Cluster and berry infections usually appear first.
- Whitish powdery patches appear on entire upper surface of the leaf has dusty coating.
- Powdery patches are also produced on stem, tendril and young fruit branches.
- If blossom clusters are attacked flowers may wither and drop i.e. Blossom blight.
- Young fruits when attacked develop powdery growth exposing pulp.
- Later the berries turn dark in colour, **may develop cracks and** don't develop further.
- Late in the season there may be numerous minute specks (Sexual fruiting bodies --Cleistothecia) on the surfaces infected leaves.

Disease cycle:
Primary source of inoculum – Ascospores released from the Cleitothecium or the mycelium surviving or in sheltered places on the shoots.
Secondary spread is by wind borne oidia.
Pathogen: The mycelium is ectophytic and produces bilobate or multilobate appressoria. The conidiophores are simple, multiseptate and erect bearing a chain of 3-4 conidia. Under Indian conditions, perfect stage of the fungus is not found. When the mating types are present cleistothecia canform on all infected tissues during later part of the growing season.

Management:
- Use training system that reduces dense growth and allows proper air circulation.
- Dustings of vines with 300 mesh Sulphur ($1^{st}$when new shoots are 2 weeks old, 2nd prior to blossoming, 3rd when the fruits are half ripe).
- Prophylactic spray with B.M. 1% or **Lime sulphur at dormant stage delays** development of disease by decreasing initial inoculum.
- Fungicides like Wettable sulphur (above 30^0 C avoid sulphur because it becomes toxic), Dinocap (0.05-0.1 %), Triademefon (0.1%), Mycobutanil (0.1%), Penconazole (0.1%), Carbendazim (0.1%) Hexaconazole (0.1%) etc are effective in checking the disease.
- Avoid excessive use of nitrogenous fertilizers as it promotes succulent growth and increases disease severity
- Bordeaux mixture is also effective against powdery mildew.
- Biological control-*Ampelomyces quisqualis* and *Verticillium lecanii* (which are mycoparasites) can be used for control of powdery mildew in combination with chemicals).
- Resistant Varieties: Red sultana , Saint George.

4. Alternaria leaf spot: Caused by *Alternaria vitis.*

Symptoms:
- Appears **more on old leaves** than on young growth.
- Appearance of patches mostly along the margin of leaves
- Individual spots appear rarely in the middle of the leaves
- In the initial stage, minute, yellow spots appear on the upper surface of leaves

- Later, spots enlarge and form brownish spots with **concentric rings in them**
- In severe cases of attack, leaves dry completely and defoliation occurs
- Water soaked minute lesions appear on the lower side.
- Gradually become necrotic brown angular spots.
- Spots coalesce to form large patches

Disease Cycle:
PSI – Conidia in crop residue. Survives in infected leaves & debris
SSI - Wind borne conidia.

Management :
- Spraying of Cuman L, Bitertenol, Iprodine, Mancozeb (2mg/litre).

5. Rust:
Caused by *Physopella ampelopsides* or *Phakopsora vitis*.

Symptoms:
- Small yellowish uredial pustules are formed on the lower surface.
- Similar pustules are also formed on shoots, rachis and petioles.
- Corresponding upper surface appears brown and necrotic.
- Telia develop as brown to dark brown eruptions.
- The disease eventually lead to weakening of the vine due to poor shoot growth
- Reduction in quantity and quality of fruit.

Disease cycle:
Fungus over winters as uredospores on the vine.
Fungus produces pycnial or aecial stages on ***Meliosma myriantha.***
Secondary spread – wind borne- uredospores

Management:
- Spraying of 1% Bordeaux mixture, 0.25% Zineb or Mancozeb or Dithane Z-78 or Indofil Z-78.
- Spraying of Hexaconazole -0.1% or Mycobutanil 0.1% or Tridemorph (Calixin) 0.05 to 0.1% or Chlorothlonil 0.2% and repeat after 10-15 days .
- Resistant Variety-Red sultana.

6. Dead Arm: Caused by *Phomopsis viticola, (Diaporthae)*
Dead arm or die back or wilt is a very serious disorder of grapes.

Symptoms:
- Vines fail to sprout after pruning.
- Even if they sprout, they soon wither and dry.
- Death of the bark on the canes, arms and even main stems.
- Dead bark dries, becomes shrunken.
- Wood beneath the bark may even crack longitudinallyand plants die.

Disease cycle:
Distributed mainly by windborne ascospores.

Disease Management:
- Cutting and destroying all the parts of the vine beyond the point of infection.
- Cut ends should be protected by Bordeaux paste or Copper oxychloride 0.3%.
- Spraying of Captan (0.25%) or Benomyl (0.25%) or Chlorothalonil 0.2% or Iprodione 0.2%.
- Spraying of Bordeaux mixture 1% to protect unfolding leaves in spring and repeating at 7-10 days interval.
- Spraying of Mancozeb 0.2% twice during bloom helps in control of fruit and rachis infection.

7. Bacterial leaf spot: Caused by *Xanthomonas compestris pv vitis- vinifera.*

Symptoms:
- Minute translucent water soaked lesions on the lower surface of the leaf along veins & veinlets.
- On the corresponding upper surface pale brown spots surrounded by yellow halo.
- Central portion of the spot is raised.

- Leaves give blighted appearance.
- Lesions on shoots are elongated, cankerous and later lead to cracking of stem.
- On berries – lesions are brown/black cankerous and shrivelled.

Disease cycle:
PSI: In buds, leaves, and berries
SSI: Bacterial cells spread by pruning tools.

M**anagement:**
- Spraying of Bordeaux mixture 1%.
- Avoid contamination by pruning tools.
- Spraying of Copper Oxy Chloride @ 3g/litre.
- Spraying of Streptomycin 100 g active ingredient/acre after pruning.

DISEASES OF GUAVA

1. Canker	:	*Pestalotia psidii*
2. Wilt	:	*Fusariumoxysporum* f. sp. *psidii, F. solani,* *Macrophomina phaseolina,* *Cephalosporium* sps.
3. Anthracnose	:	*Colletotrichum psidii* (=*Gloeosporiumpsidii*)(IPS) *Glomerella cingulata* (PS)
4. Red rust	:	*Cephaleuros virescens*
PHD		
1. Phytophthora fruit rot	:	*Phytophthora nicotianae* var. *parasitica*
2. Botryodiplodia fruit rot	:	*Botryodiplodia theobromae*
3. Phoma fruit rot	:	*Phoma psidii*

Wilt : *Fusarium oxysporum* f. sp. ***psidii, F. solani.***
Other pathogens reported are *Macrophomina phaseolina, Cephalosporium* spp.etc.

Economic importance: It was first reported in 1935 from Allahabad. Jhoty *et al*., in 1984 reported that seven thousand acres of land in A.P under guava cultivation was reduced to half the land value by the presence of the disease.
Favourable conditions:
pH 6.0is optimum for disease development. Both pH 4.0 and 8.0 reduces the disease.
Disease is more **in clay loam and sandy loam** compared to heavy soil.
Alkaline soils favours the pathogen
Higher disease incidence in monsoon period.
Disease appears from august and increases sharply during September-October.
The presence of spiral nematode, *Helicotylenchus dihystera*
Mode of spread and survival: The fungus first colonizes on the surface of the roots and enters the stem tissues at the basal portions near the ground level. It multiplies in vascular region and affects the cortical cells.
Disease Cycle:
PSI: Chlamydospores in dead trees.
SSI: Micro and macroconidia spread by irrigation water, implements and root contact.
Pathogen: *Fusarium oxysporum* f.sp. *psidii*
Mycelium is white or pink with a purple tinge. Microconidia are borne on simple phialides arising laterally on the hyphae. Microconidia are oval to ellipsoid, cylindrical, straight to curved and 7 to 10 x 2 to 3 pm. Macroconidia are 3 to 4 septate and 32 to 50 x 3 to 7 um in size. They are fusoid to subulate and pointed at both ends. Chlamydospores may be intercalary or terminal.
Asexual spores : Micro & macro conidia
Vegetative spores :Chlamydospores (Resting spores)

Symptoms:
The affected plants show yellow colouration with slight leaf curling at the terminal branches, becoming reddish at the later stage and subsequently premature shedding of leaves takes place.
Twigs become bare and fail to bring forth new leaves or flowers and eventually dry up. Fruits of all the affected branches remain underdeveloped, become hard, black and stony.

The entire plant becomes defoliated and dies. A few plants also show partial wilting, which is very common symptom of wilt in guava.
The finer roots show black streaks which become prominent on removing the bark.
The roots also show rotting at the basal region and the bark is easily detachable from the cortex.
The cortical regions of the stem and root show distinct discolouration and damage.
Light brown discoloration is noticed in vascular bundles. Bark splitting can be seen in wilted plants in later stages.
The disease can be catagorized into slow wilt and sudden wilt. In slow wilt, plant takes several months or even a year, to wilt after the appearance of initial symptoms and in sudden wilt, infected plant wilts in 15 days to one month.

Management:
Cultural:
Proper sanitation of orchard.
Wilted plants should be uprooted, burnt and a trench of 1.0-1.5m should be dug around the tree trunk. Treat the pits with formalin and cover the pit for three days and then transplant the seedlings after two weeks.
While transplanting seedlings avoid damage to the roots.
Maintain proper tree vigour by timely and adequate manuring, inter-culture and irrigation.
Intercropping with turmeric or marigld.
Soil solariztion with transparent polythene sheet during summer months.
Application of oil cakes like neem cake, mahua cake, kusum cake 5-10 kg/treesupplemented with urea.
Apply 6kg neem cake + 2kg gypsum per plant.
Judicious amendments of N and Zn.
Host Plant Resistance:
Resistant variety: Apple guava
Guava species, *Psidiumcattleianum* var. *lucidum* and *Syzigium cumini*(Jamun) are resistant to wilt.
Psidium cattleianum (*P. molle*), *P. quianense*, Chinese guava *(P. friedshthalianum*) and Phillippine guava are recommended as resistant root stocks
Cross of *Psidium malle* x *P. guajava*is free from wilt and this material can be used as resistant root stock
Varieties Safeda, Banarasi **(Andhra strain),**,Dholka, Sindh and Nasik are tolerant while Allahabad Safeda is resistant.
Biological:
Aspergillus niger strain AN 17, *Trichoderma viride*, *Trichoderma harzianum* and *Penicillium citrinum* can used as biocontrol agents
Chemical:
Stem injection with 0.1% water soluble 8-Quinolinol sulphate
Drench with 0.2% Benomyl or Carbendazim, four times in a year and spray twice with Metasystox and Zinc sulphate.
Disinfestation of soil with Metam-sodium at 250 ml/10m^2area to control nematodes

Fruit canker/Scab/Grey blight: *Pestalotiopsis psidii*
Symptoms:
- Disease occurs on green fruits and rarely on leaves.
- Minute, brown or rust- coloured, unbroken, circular, scabby lesions of 2 to 4 mm dia. appear on the fruit which later tear the epidermis open in a circinate manner.
- In severe cases raised cankerous spots develop in great numbers and fruits break open exposing the seeds.
- Infected fruits remain under developed, become hard, malformed and mummified and drop in great numbers.
- Sometimes on leaves small rusty brown angular spots appear on leaves.
- The scab disfigures the fruits and their market value is highly reduced.

Primary source of inoculum: Dormant mycelia. *Helopeltis antonii*, a kajji bug which punctures the young fruit sucking juice and that damage exposes the fruit to infection by the pathogen.
Secondary source of inoculum: Air borne conidia
Mode of spread: spread is through the wind-borne conidia.
Epidemiology:
The fungus is capable of growing at temperature between 20 and 25°C.

Mycelial growth with intensive sporulation takes place at 5.5°C.
Wounding results in quick attack by the fungus.

Management:

Since the wound by insect predisposes the fruit to infection, spray the young fruits after pollination with a suitable systemic insecticide (Dimethoate – 2ml/l) will take care of the infection.

Spread of the disease can be checked by three or four spraying with Bordeaux mixture1.0 percent or copper oxy chloride 0.2 per centeror lime sulphur (1:25) at 15 days interval.

Summer irrigation +Nutritional management reduces the disease

Resistant varieties like Sind, Nasik, Safeda, Apple colour.

Post-harvest washing of fruits with 200 ppm aureofungin can protect the fruits for five days

Guava cv. Nasik was almost immune ; Safeda and Apple Colour were highly resistant; Sind was resistant and Dholka was moderately resistant.

Stem Canker – *Physalospora psidii*

Symptoms:

Cracks and lesions are formed along the stem, arresting translocation of nutrients.

Infected fruits turn dark brown to black and dries up resulting in die-back symptoms.

Affected twigs show wilting and death.

Fruit rotting takes place, blighting of leaves to enlargement

Fungus: *Physalospora psidii*

Perithecia is glabrous with a fleshy wall and are formed on the infected tissue.

Ascospores are hyaline, narrow, ellipsoid and one celled.

Conidia are single celled, ovoid with a rough wall and measure 20 to 26 x 9 to 12 cm.

On the stems and fruits pycnidia are formed in stroma.

Mode of spread and survival: The pathogen remains in the infected tissues beneath the bark and become active under favorable conditions.

Disease Cycle: The pathogen remains inside the infected tissues beneath the bark and becomes active whenever weather conditions are favourable. The pathogen spreads from plant to plant by air-borne spores.

Management

The infected branches should be cut and destroyed. The cut ends should be pasted with fungicidal paste.

Pruned ends should be painted with fungicidal paste such as Bordeauxpaste. Insecticidal sprays after pruning will help in preventing the disease

Anthracnose: *Colletotrichum psidii*

Symptoms:

- On leaves fungus causes necrotic lesions at the tip or margins.
- Lesions are usually ashy grey and bear fruiting bodies.
- Tender twigs die from tip giving rise to die back symptoms.
- Dried twigs without leaves are seen in the orchard and under humid conditions, they bear black spots (Acervuli).
- Infection on flowers and buds result in blossom blight.
- **On fruits** pin head spots are seen on unripe fruits which gradually enlarge.
- Dark brown to black coloured sunken circular spots appear on ripe fruits.
- Produces creamy spore masses in moist weather.
- Infected unripe fruits become corky, harder and develop cracks under severe infections.

Pathogen

Conidia are hyaline, aseptate, oval to elliptical condiophore is cylindrical. Acervulli are dark brown to black

Mode of spread and survival:

The pathogen remains dormant for about three months in the young infected fruits.

It becomes active and incites rot when the fruit begins to ripe. In moist weather, acervuli appear as black dots scattered throughout the dead parts of the twigs.

From the twigs, the fungus penetrates the petioles and attacks the young leaves, which become distorted with dead areas at margins or tips. The conidia are spread by wind or rain.

Disease cycle:

PSI: Infected fruits, twigs and leaves which contain Acervuli.

SSI: Conidia dispersed by wind and rain.

Epidemiology:

The cool season (Jan - Mar) and the hot, dry weather (Apr-Jun) prevent the spread of infection. In moist weather, acervuli are produced in abundance on dead twigs and pinkish spore masses are seen. Conidia initiate fresh infection.
The temperature for disease development on fruits ranges from 30 to35°C.

Management:

- Fungicidal sprays of Benlate, Carbendazim or Thiobendazole at 7-10 days interval.
- Spraying of Bordeaux mixture 1% or Copper oxychloride 0.3%.
- Fumigation of fruits with sodium metabishulphite or bleaching powder for long distance transport.
- 'Apple guava' (**light red fleshed) is moderately resistant to anthracnose disease.**
- Surekha hybrid between Allahabad safeda X Benarasi is resistant.

Red rust: *Cephaleuros virescens*

This disease is exceptionally severe in guava.

Symptoms:

The alga produces specks to big patches on the leaves. They may be crowded or scattered.
The pathogen extends between cuticle and epidermis and penetrates the epidermal cells.
Fruit infection by alga is not common on fruits. Fruit lesions are usually smaller than leaf spots.
They are dark green to brown or black in colour.
Primary source of inoculum: Dormant mycelia
Secondary source of inoculum: Zoospores

Green algae: *Cephaleuros virescens*

Mode of spread and survival:

The disease is more common on closely planted mother plants. The zoospores cause the initial infection.
High moist condition favours the development of fruiting bodies of the alga.

Management: This algal disease is controlled by spraying with Bordeaux mixture 1.0 per cent or copper oxychloride 0.3 per cent.

PHD

Phytophthora fruit rot : *Phytophthora nicotianae* var. *parasitica*

Symptoms:

- Attacks unripe fruits at blossom end as small dark brown circular spots.
- Fruits are mummified.
- Riped fruits rot partly or completely either on tree or during storage and emit bad odour.

Management:

1. Spraying of Bordeaux mixture or Copper oxychloride 0.3% in the field.
2. Careful handling of the produce.
3. Provide proper storage conditions.

Botryodiplodia fruit rot : *Botryodiplodia theobromae*

Symptoms:

- Brownish discolouration at the stem end which proceeds downwards irregularly.
- Rot is soft and watery.
- Numerous pycnidia are produced on the diseased areas.

Management:

Careful handling and storage at 15 ^{0}C.

Phoma fruit rot: *Phoma psidii*

Symptoms:

- Brown coloured spots appear all over the fruits.
- Centre of the spot is depressed.
- Minute black dot like pycnidia appear on the spots.
- Under moist conditions, oozing of cream coloured spores from the pycnidia is observed.

Management:

1. Careful handling of the fruits.
2. Fruit dip treatment with Tecto-40 (TBZ), *i.e.*, Thiobendazole solution 0.1%.

Phomopsis fruit rot: *Phomopsis psidii*

The symptoms appear on unripe fruits at the blossom-end.
Infected fruits show small, conspicuous, white or light brown and circular spots. Some of

the infected fruits are shed prematurely.
As the fruits ripen, the spots extend and cover the fruit surface. The infected tissues become softer and emit an undesirable odour.
Weekly sprays with Bordeaux mixture 1.0 per cent or copper oxychloride 0.3 per cent are required for the control of fruit rot.
Six monthly sprays with Mancozeb 0.2 per cent during fruiting stage are helpful in controlling fruit rot.

PHD management

- Fruit rotting in transit and storage can be effectively controlled by thio urea (5% for 2 minutes)
- Good protection can be obtained by hot water treatment at 50°C for 5 minutes
- Irradiation of fruits with gamma rays (100 KV)
- Harvested fruits can be fumigated with sodium metabisulphate or bleaching powder.
- Thiram and Cuman were good protectants as pre-inoculation sprays.
- Application of bio agent viz *Streptosporangium pseudovulgare* on fruits before emergence of symptom

DISEASES OF SAPOTA

1. Phaeopleospora leaf spot	:	*Phaeopleospora indica*
2. Pestolotiopsis leaf spot	:	*Pestolotiopsis versicolor, P. sapoticola*
3. Flat limb	:	*Botryodiplodia theobromae*
4. Fruit rots	:	
i) Botryodiplodia rot	:	*Botryodiplodia theobromae*
ii) Phytophthora rot	:	*Phytophthora palmivora*
iii) Pestalotiopsis rot	:	*Pestolotiopsis versicolor*

1. Phaeopleospora leaf spot: - *Phaeopleospora indica.*

Symptoms:

- Numerous, minute, circular pinkish or reddish brown spots with whitish centres.
- Later such spots unite to form large necrotic patches leading to defoliation.
- As a result, there will be less number of fruits and reduced size of fruits.

Disease cycle:
PSI: The pathogen survives on leaf debris.
SSI: Air borne conidia.
Favourable conditions
Severe during Oct-Dec
Fungus grows best at 25^0C and 90% RH
Susceptible varieties: Cricket ball, Kirthabarthi, *Mimusops hexandra* (root stock)
Management:

- Removal and destruction of infected plant debris.
- Orchard sanitation.
- Spraying of Mancozeb 0.2% or Copper oxychloride 0.3%.
- Varieties *viz.*, PKM1, PKM2, PKM3, showed very less disease severity.
- Growing of susceptible sapota cvs Cricket ball, Kirthabarthi, *Mimusops hexandra* (root stock) should be avoided in new plantations

2. Pestolotiopsis leaf spot **:-***Pestolotiopsis versicolor*or *P. sapoticola.*

Symptoms:

- Occurs as minute reddish brown specks with greyish centre.
- In the centre of the spots black dot like fruiting bodies (Acervuli) are visible.
- Defoliation of leaves is observed under severe infection.

Disease cycle:
PSI: The pathogen survives in infected leaves and plant debris.
SSI: Air borne conidia.
Management:

- Removal and destruction of infected plant debris.
- Orchard sanitation.
- Spraying of Mancozeb 0.2% or Copper oxychloride 0.3%.

3. Flat limb:-*Botryodiplodia theobromae.* Also called as Fasciation.
First reported from Maharashtra and Gujarat by Khurana and Singh in 1972. Now prevalent in all sapota growing areas of the country. The disease is a problem in orchards above 10 years age
Symptoms of flat limb:

- Transformation of normal cylindrical part of the branch/shoot in to a broad flat structure with crowding of branches at its tip twisted showing rough marks.
- Affected branches are flattened, twisted and show rough appearance.
- Leaves are reduced in size, pale green in colour, born on limbs, clustered.
- Shortening of internodes and leaves appear bushy.
- Fruits are small sized, hard and mummify.
- Fruits and leaves fall prematurely

Symptoms of branch drying

- Symptoms of branch drying are predominantly observed on shoots and branches having girth of 3-10 cm.
- Initially, there is development of minute slightly raised streaks of reddish brown colour on the surface of the branch.
- These streaks break open to from longitudinal cracks.
- Gradually, bark at this point splits and get disintegrated partially to expose the internal hard wood.
- There is browning of the surface of internal hard wood followed by oozing of reddish brown coloured cell sap which down on the surface of the branch.
- When complete destruction of conducting vessels is accomplished, sudden drooping of leaves above the point of infection is noticed followed by complete drying of the branch within a period of two weeks.
- Fruits of all stages on such dried branch also dry and fall down.

Disease cycle:
PSI: The pathogen survives in unpruned infected shoots and plant debris.
SSI: Air borne conidia.

Management:

- Cut the branches showing symptoms till healthy portion.
- Apply Bordeaux paste to the cut ends.
- Spraying COC @ 3g/l or Carbendazim 0.1% or Thiophanate Methyl 0.1% or Chlorothalonil 0.2%.

- Sapota varieties KritiBharathi and DHS-2 were found to be least affected by flat limb while DHS-l and cricket ball were found most susceptible

Fruit rots:

i) Botryodiplodia rot	:	*Botryodiplodia theobromae*
ii) Phytophthora rot	:	*Phytophthora palmivora*
iii) Pestalotiopsis rot	:	*Pestolotiopsis versicolor*

i) Botryodiplodiarot : - *Botryodiplodia theobromae.*

Symptoms:

- Affected fruits show brownish discolouration mostly at the stem end and proceeds downwards in an irregular wavy manner.
- In advanced stages, numerous small pycnidia are produced on the entire surface of the fruit.
- The rot is soft and watery and maximum disease occurs at about 30 ^{0}C.

ii) Phytophthora rot *Phytophthora palmivora*

Symptoms:

- Initial symptoms are seen as water soaked, irregular spots which soon spread up involving major portion of the fruit.
- Decay is deep and pulp eventually disintegrates.
- Decayed portions are seen over grown with snowy white fungal colonies.
- Such infected fruits at later stages become soft, pulpy and rotted, emitting bad odour.

iii) Pestalotiopsis rot: *Pestolotiopsis versicolor* and other *Pestolotiopsis* spp.

Symptoms:

- Diseased fruits exhibit water soaked lesions, which become brown within 2-3 days. Subsequently whole fruit is covered with small tufts of mycelium.
- Soon mycelium is seen studded with black acervuli.

PHD Management:

1. Spraying of Bordeaux mixture or Copper oxy chloride 0.3% in the field.
2. Careful handling of the produce.
3. Provide proper storage conditions.
4. Treating of the fruits with Copper sulphate.

DISEASES OF BANANA

1. Panama wilt	:	*Fusariumoxysporum* f. sp. *cubense*
2. Anthracnose	:	*Colletotrichum gloeosporioides (musae)* (=*Gloeosporium musarium*) *Glomerella cingulata* (P.S.)
3. Sigatoka leaf spot	:	*Mycosphaerella musicola* (P.S.) *M. fijiensis* (P.S.)
4. Cigar end rot	:	*Verticillium theobromae* *Trachysphaera fructigera*
5. Freckle	:	*Phyllosticta musarum* *Guignardia musae* (P.S.)
6. Moko/bacterial wilt	:	*Ralstonia solanacearum*
7. Bacterial soft rot/head rot	:	*Pectobacterium caratovora* pv. *caratovora*
8. Bunchy top	:	*Banana Bunchy Top Virus* (BBTV)
9. Mosaic/heart rot/ Infectious chlorosis	:	*Cucumber Mosaic Virus (CMV)*
10. Bract mosaic	:	*Banana Bract Mosaic Virus*. (*BBrMV*)
11. Banana streak	:	*Banana Streak Virus (BSV)*

1. Panama wilt :- *Fusarium oxysporum* f. sp. *cubense.*

- First reported from Australia in 1874

- The popular variety Gros Michel (AAA), mostly grown for export quality fruits, had to be replaced with Cavendish bananas
- It first became epidemic in Panama in 1890
- Cultivars Rasthali (Amritapani), Gros Michel, Karpooravalli cultivars are susceptible

Favourable conditions

- Mono-cropping, soil factors (heavy soils)
- Cultivation of banana in heavy soils with poor drainage
- Cultivation of susceptible cultivars like Amritapani, Gros Michel, Karpooravalli
- Infection by burrowing nematode, *Radopholus similis* and *Meloidogyne incognita*

Symptoms:

- Conspicuous symptoms usually appear on 3 to 5 months old plants, although 2-3 months old plants are also killed under highly favourable conditions
- The earliest symptoms are faint yellow streaks on the petiole of oldest, lower most leaves ; Affected leaves show progressive yellowing
- Yellowing, withering and breaking of leaves at petiole and hang down along pseudostem.
- Longitudinal splitting of pseudostem at the base.
- Individual vascular strands appear yellow besides red or brown dots and streaks.
- Diseased plants when cut exhibit rotten fish smell.
- When diseased rhizomes are cut transversely, usually the light yellow to dark brown or reddish brown discolouration appears first in the outer or oldest leaf sheath and extends to the centre of rhizome or pseudostem.
- Presence of burrowing nematode *Radopholus similis* aggravates the disease by providing wounds on roots through which pathogen enters.

Disease cycle:- Spores germinate in the soil and grow towards the nearby roots of banana plants in response to chemical compounds exuded from the roots. Infection takes place on the secondary and finer roots and proceeds into the larger, primary roots through the xylem vessels before entering the rhizome.

The primary roots and the rhizome do not appear to be infected directly by the pathogen. The xylem network consists of a series of individual vessels with perforated end walls through which the sap flows. Movement of the spores with the sap flow is blocked temporarily when they become lodged at the end walls.

The spores then germinate and hyphae grow through the perforations into adjoining vessels where further spores are produced. The plant is often able to prevent infection from successfully travelling to and entering the rhizome by the production of gels and tyloses (a resistance mechanism) to seal off the infection.

PSI: The disease is soil borne. Survives saprophytically in rhizomes and spreads through suckers. The pathogen survives as chlamydospores, micro and macro conidia.

SSI: Root contact, irrigation water/implements spread conidia.

- *Fusarium oxysporum* f.sp. *cubense*
- Four races of the pathogen are known to exist but race 1 is mostly prevalent in India attacking silky group (AAB) of cultivars (Rasthali or Amritapani)
- Race –2 (ABB)
- Race -3 *Heliconia*
- Race -4 (AAA) or Tropical race

Management:

1. Selection of disease free suckers.
2. Growing rice prior to banana planting or flood fallowing of fields at least for 6 months.
3. Soil fumigation with Formaldehyde 5%.
4. Dipping suckers in 0.2% carbendazim for 30 minutes.
5. Application of lime (1-2 kg/pit) to the infected pits after chopping of the plants parts
6. Avoid root injury during tillage operations and provide good drainage.
7. Drenching with **Carbendazim 0.2%** starting from 6 months after planting at bi monthly intervals.
8. Corm injection of 3ml of 0.2% Carbendazim at 5th, 7th and 9th month after planting.
9. Incorporation of *Trichoderma viride* and *Pseudomonas fluorescence* @ 15 gm/plant 3 times starting from 2 months after planting onwards.
10. Cavendish and Poovan cultivars are resistant in India. Basrai, Moongil, Vamanakeli (potti patcha arati), Kovvur bontha etc are resistant. Robusta is immune (pedda patcha arati).

2. Sigatoka leaf spot:-

Yellow Sigatoka – *Mycosphaerella musicola* (I.S: *Pseudocercospora musae*)

Black sigatoka or black leaf streak – *Mycosphaerella fijiensis* (I.S: *Paracercospora fijiensis*)

- First observed in Java in 1902
- Epidemic in 1913 in Sigatoka valley in Fiji (Sigatoka – Name of the valley in FIJI)
- Yellow Sigatoka: Worldwide in distribution
- Black sigatoka is not prevalent in India.
- In India, yellow sigatoka is a serious threat to banana production in the states of Assam, T.N, Karnataka and A.P.

Favourable conditions

- Rapid spread and heavy infection is favoured by high humidity, heavy dew and rainy weather with temp above 21° C
- Prolonged leaf wetness periods
- High humidity due to poor drainage, **closer planting** and more suckers in a mat.
- Low soil fertility particularly of potassium
- Cultivation of susceptible cultivars like Grand Naine, Dwarf Cavendish and Giant Cavendish

Symptoms:

- Early symptoms appear on the third or fourth leaf from the top, i.e., on young leaves
- Appears as pale yellow or greenish yellow streaks on both sides of leaves running parallel to leaf veins.
- Fully developed spots are spindle or eye shaped with grayish centers and dark brown or black borders and chlorotic halo around.
- Spots are more concentrated towards the leaf edges.
- Spots coalesces and whole leaf blade dries up. On the upper surface of the spots, fructifications of the fungus appear as black specks
- Leaves scorched and hang down along pseudostem.
- Fruits in bunches of infected plants are under developed and may ripen prematurely

Disease cycle: Infection takes place through ascospores in the inflected plant debris. Infection takes place through stomata on the lower surface of young leaves. Disease spreads through wind borne conidia Surface moisture is necessary for release of both conidia and ascospores

PSI:- Both ascospores and conidia spread by wind and rain drops.

SSI: Conidia spread by rain drops.

Management:

- Resistant Varieties – Karpoora chakrakeli, Amruthapani
- Selection of healthy suckers and dripping in fungicide solution.
- Planting at recommended density (1000 plants/acre)
- Potash can be applied 10 to 20% more
- Removal and destruction of affected leaves followed by spraying with BM (0.1% + linseed oil 2% or Benomyl 0.1%)
- Spraying with Chlorothalonil 0.2% / Thiophenate methyl/Tridemorph/Propiconazole 0.1% 3-4 times at 20 days starting from October.
- Wetting agent such as teepol or sandovit @ 1ml/lit of water mixed with fungicide solution.

3. Anthracnose:-*Colletotrichum musae* (=*Gloeosporium musarium*) (IPS)
Glomerella cingulata (P.S.)

Severe during June- September when high temperatures prevail coupled with rains.

Favourable conditions

- High temperature (30-35^{0}C) and humidity
- Severe from June to September
- Ripe fruits in storage are more susceptible than unripe fruits
- Cavendish varieties more susceptible.
- *C. gloeosporioides* has also been reported to be associated with banana anthracnose

Symptoms:

- On green fruits, flowers and distal ends of hands develop circular black, sunken spots with yellow halo.
- Ripe fruits develop symptoms at tips as minute, dark brown, circular, sunken areas which subsequently cover entire fruit tip and is black in colour.

- Characteristic orange to salmon pink acervuli and conidia on the spots; Finally the whole finger and bunch are affected; Premature ripening and shriveling of fruits.
- Sometimes the main stalk of the bunch may become diseased. Infected fruits become black and rotten

Disease cycle:
PSI: Survive in crop residue as conidia. Leaves used for packing.
SSI: Wind borne conidia, fruit contact, insects.
Management:
- Spraying Chlorothalonil @ 2g/l at 15 days gap to reduce field infections/latent infections.
- Careful harvesting of bunches – harvested at right stage *i.e*. ¾ th maturity, dehanding, delatexing.
- Careful handling to avoid injuries.
- Dipping of fruits in fungicidal solution of TBZ/Carbendaizim/Benomyl 400 ppm or aureofungin solution 400 ppm.
- Clean packing, refrigeration, at 7-10 ^{0}C after harvest reduces fruit spots after harvest.
- Hot water treatment of fruits 5 min for 50°c

4. Cigar end rot: *Verticillium theobromae* (Deuteromycota)
Trachysphaera fructigera (Oomycota fungi)
The rotted portion of the banana finger is dry and tends to adhere to fruits (appears similar to the ash (grey) of a cigar

Symptoms:
- Black necrotic rot at the tip of the fingers up to 2 cm.
- Infection starts from the floral remnants and proceeds upwards.
- Necrotic area corrugated and covered with powdery fungal growth (conidia and conidiophores) looking like grayish ash of cigar end, hence the name.

Disease Cycle
PSI: Air borne spores in the debris of Banana crop. Also harbored by other *Musa* and *Heliconia spp.*
SSI: wind borne conidia
Pathogen The conidiophores are slender, erect, hyaline or slightly pigmented and simple or branched. The genus *Verticilliom* characterized by the production of balls of amerospores on verticillately arranged phialides. Conidia are small, hyaline, oval to ellipsoid, one-celled, borne singly or in small clusters apically. Chlamydospores, aleuriospores and microsclerotia are also produced by some species
Management:
- Removal of floral remnants by hand soon after fingers emerge.
- Spraying Mancozeb 0.25 % or Carbendazim 0.1% or copper oxy chloride 0.25% + wetting agent @ 0.5-1 ml of spray fluid.
- Covering bunches with polythene sleeves in field prior to bunch emergence.

5. Freckle: *Phyllosticta musarum* (IPS) *Guignardia musae* (P.S.)

Symptoms:
- Leaves and fruits affected.
- Numerous minute, grey, brown or dark brown, raised, more or less round spots of 1 mm size with black dots in the center with an indefinite margin. They are more abundant on the lower surface
- The raised structures are the fructifications, pycnidia of the fungus.
- Individual reddish-brown spots occur, 0.5-1mm diameter, later turning dark brown to black, usually on the upper surface of leaves
- Diseased portions are black and leaves turn yellow and die.
- Fruits also develop numerous spots giving ugly look.

Disease Cycle
PSI: Ascospores and conidia from crop residue.
SSI: Conidia spread by rain/ wind.

Management:
- Cavendish varieties have resistance to the disease
- Fungicides used are: Protectant fungicides: Dithiocarbamates (e.g., mancozeb).

- Systemic fungicides: Triazoles (e.g., propiconazole and flusilazole). Strobilurins (azoxystrobin)
- Spraying of 1% Bordeaux mixture or Zineb 0.2% effective.

6. Moko/bacterial wilt: *Ralstonia solanacearum.*

- First recorded in Guyana in 1840 in *Moko* plantain and reduction of yield up to 74%.
- Moko is endemic to Central and South America
- In India, first reported from West Bengal in 1968
- In India, the disease is reported on banana cvs. Robusta and Poovan from Tamil Nadu causing 75-100% yield losses.

Symptoms:

- Yellowing of young leaves which break at petiole, wilt rapidly and collapse.
- Vascular discolouration (pale yellow to dark brown or bluish black) is concentrated near the centre of the pseudostem, becoming less apparent on the periphery
- The most characteristic symptoms appear on the young suckers that have been cut once and begin re growth
- Suckers also wilt and turn black, stunted and twisted.
- When pseudostem and rhizomes are cut, characteristic bacterial ooze as shiny drops can be noticed.
- Unlike in panama wilt infected rhizomes will not emit rotten fish smell.
- Reddish or black colour dots present in vascular tissues of rhizomes in case of panama wilt are absent in Moko disease.

Disease cycle:

PSI: Soil borne bacterium and use of diseased suckers.

SSI: Irrigation water, implements, pruning knives, insects which feed especially on male flower bracts.

The bacterium *Ralstonia solanacearum* **Race 2, biovar 1** cause's disease in *Musa* and *Heliconia* while **Race-1 Tobacco: Race-3 Potato, Tomato: Race-4 Ginger: Race -5 Mulbery**

Management:

- Selection of healthy planting material.
- Deep summer ploughing, crop rotation & providing good drainage (3 years rotation with sugarcane or rice)
- Eradicate infected plants and suckers by rouging or killing *in situ* by application of herbicides
- Eradication of *Heliconia* sp which act as collateral host.
- Disinfection of tools and pruning machetes.
- Soil drenching with bleaching powder @ 25 g/litre at plant base.
- Soil application of *Pseudomonas fluorescens.*
- Removal of male buds when hands begin to turn upwards, since flower visiting insects transmit the disease.
- Growing of french marigold around the planation
- Grow relatively resistant varieties like poovan
- Soil solarisation (Katan) Exposure of soil to sunlight during dry hot weather for 40-45 days using 400 gauge transparent LDPE which increased soil temp. by 5-7°C
- Bio fumication with crucifers

7. Bacterial soft rot/Head rot/Rhizome rot/Tip-over :-*Pectobacterium caratovorum* sub.sp. *caratovorum* (formerly *Erwinia caratovora* pv. *caratovora*)

First recorded in Honduras on Gros Michel in 1949

Symptoms:

- Usually young plants are attacked. In newly planted young sprouting or sucker infection to pseudostem leading to toppling of the germinating sucker during the summer season.
- In established plants water soaked spots develop on outer leaf sheaths on base of pseudostem near the soil line. If affected plants are pulled, it comes out from the collar region leaving the corm with their roots in the soil.
- Massive soft odorous rot of the centre or a portion of rhizome.
- Rot progresses up the pseudostem destroying the growing point.
- Yellowing and wilting of the leaves are the characteristic symptoms.
- Yellowish to reddish ooze seen when affected plants are cut open at collar region.
- Infection may result in the production of cavities which resemble root borer tunnels.
- In infected rhizomes, pockets of dark water soaked areas develop. Soft rotting of pseudostem tissues and rhizomes .

- Scanty roots with necrotic tips are noticed.
- Due to poor root formation and root decay, the plants fall even with the slightest winds. (tipover of plants).

Disease cycle:
PSI: Soil borne bacterium and diseased planting material enters the host through wounds.
SSI: Bacteria spread by irrigation water.
Favourable conditions

- Juvenile stage of the crop is highly susceptible (3 to 4 months after planting)
- Prevalence of high temperature (37ºC) in early stages of crop growth
- Water logging and high temperatures
- Growing susceptible cultivars like Cavendish and Tella chekkarakeli

Management:

- Use of certified planting material.
- Avoid planting susceptible varieties during periods of high rainfall or in water logged soils in problematic areas
- Remove and destroy infected plants along with roots
- Crop rotation with rice or sugarcane for three to four years
- Maintain proper soil aeration by inter cultivation
- Soil application of bleaching powder at 8g/ plant at 10 to 15 days interval for 2-3 times.
- Spraying of Streptocycline 300 ppm + Copper Oxy chloride 4 gms/carbendazim 1 gm during rainy season.
- Grow sunhemp or cowpea (2-3 times) in interspaces till 5 months after planting
- Dip the suckers in copper oxychloride (5g/lt) + monocrotophos (2.5ml/l) solution.
- Dip the suckers with *Bacillus subtilis/ Pseudomonas fluorescens* followed by soil drenching at 15 days interval starting @1-2 l/plant (50g/lt of water at 0, 2nd, 4th and 6th months after planting)

8. Bunchy Top/ Strangles/Curly Rot/ Cabbage Top:- *Banana Bunchy Top Virus* (*BBTV*)

First reported from Fiji in 1889 in Cavendish varieties

- Around 1940, introduced into India from Srilanka
- The BBT is a **domestic quarantine disease in India** (Kerala, Assam, Bihar, West Bengal and Orissa).
- Introduced into India from Srilanka in 1943.

Symptoms

- **Chlorotic mottling** on leaf sheaths, bracts etc.
- Presence of interrupted dark green streaks along the mid rib and secondary veins **(Moorse codes)**
- Severe stunting of the plants, non elongation of leaf stalks, more erect leaves.
- Leaves are reduced in size with narrow and shortened petiole, become erect, brittle and crowded at the top of the pseudostem hence the name **Bunchy Top.**
- Margins of leaf lamina become wavy and slightly roll upwards **(Wavy margins)**
- Edges of the leaves turn yellow in colour *i.e*. **yellow fringes**
- Veins at the point of attachment to the midrib curl in the form of a hook *i.e*. **Green hooks**
- Non production of bunches or production of **deformed bunches.**
- Tips of inflorescence bracts of infected plants remain green and do not turn to normal pink or purple
- Infection of young plants leads to failure of bunch emergence

Spread

PSI: Infected suckers and also spread by banana aphid – *Pentalomia nigronervosa*, *Colocasia esculenta*, ***Cucumis sativus, C. melo*** carries BBTV symptomlessly. *Heliconia* sps. harbour the virus in the field. *BBTV* also infects ornamental *Canna* sps.

- **Family:** Nanoviridae : Genus *Babuvirus*
  Banana bunchy top virus : (ss DNA)

Management

- Application of strict quarantine measures.
- Diseased plants along with rhizomes should be destroyed as soon as they are detected.

- Diseased rhizomes should be dug out, cut into small sections and sprayed with kerosene or herbicides.
- 2, 4-D pseudostem injection @4 ml prepared in the ratio of 1:8 *i.e*. 50 g in 400 ml water. 2, 4, 5 T can also be used for killing the infected plants.
- Use of certified banana suckers. Use of tissue culture plants is recommended.
- Application of Carbofuran @40g in pits/plant at plant base and leaf axil.
- Spray of Methyldematon@1.5 ml or Dimethoate @0.05% thrice at 21 days gap for control of vector.

9. Mosaic/Infectious chlorosis/Heart rot:- *Cucumber Mosaic Virus* (CMV)

- This disease is under domestic quarantine in India (Gujarat and Maharashtra).
- *CMV* causes great losses in vegetables, ornamentals and fruit crops
- Poovan is highly susceptible
- **Prevalence of infected reservoir hosts (*Cucucmis sativus*)**

Symptoms:

- Most characteristic symptoms is light green streaks running parallel to veins imparting a striped appearance to leaves.
- Streaks extend from margin to midrib.
- In young leaves, loss of chlorophyll noticed in patches giving a mosaic look.
- Outer leaf sheath gets separated from pseudostem.
- Heart leaf is necrotic (heart rot).
- Bunches not formed generally and if formed, only a few fruits produced.

Spread:

PSI: Use of diseased suckers and by aphid vectors – *Aphis gossyppi*, *Aphis craccivora* , *Aphis maydis*, ***Rhapalosiphum maydis***, *M. persicae*.

Can be mechanically transmitted to cucurbits and some other indicator hosts such as *Chenopodium amaranticolar*, *C. quinoa*, Cow pea *etc.*,

- Family *Bromoviridae*. Genus- *Cucumovirus*
- *Cucumber mosaic virus* (+ssRNA)

Management:

- Adoption of quarantine measures.
- Systematic destruction of diseased plants, keeping fields clean, avoiding cucurbits.
- Use of disease free suckers (tissue cultured plants).
- Sucker treatment of 40 ^{0}C using dry heat for 14 days prior to meristem tip culture.
- Heat treatment at 40 ^{0}C for 1 day followed by aureofungin treatment 120 ppm.
- Spraying of Methyl demeton or Dimethoate @ 2ml/lit at 3-4 weeks interval.
- Rhizome dip treatment with a mixture of Monocrotophos 2.5 ml/litre + Copper oxy chloride 0.5% (5gm/litre) for 15 minutes prior to planting.

11. Bract mosaic: *Banana bract mosaic virus*.

First reported from Philippines in 1979

In India, this virus disease was first reported in Kerala as Kokkan disease

Symptom expression is clearly seen in cooking banana and Karpura Chakkerakeli cultivars

Symptoms:

- Development of black streaks on petiole, yellow or pink discolouration on pseudostem.
- Mosaic like purple or greenish spindle shaped streaks on the bracts.
- Suckers show reddish brown streaks.
- Clustering of leaves and ‘traveller’s palm’ like appearance.
- Elongated peduncle and half filled hands are other symptoms.

Spread

PSI: Spread by planting material.

SSI: Aphids are the vectors for transmitting the viruses.

Family *Potyviridae*	Genus- *Potyvirus*
Banana bract mosaic virus (*BBrMV*)	+ssRNA

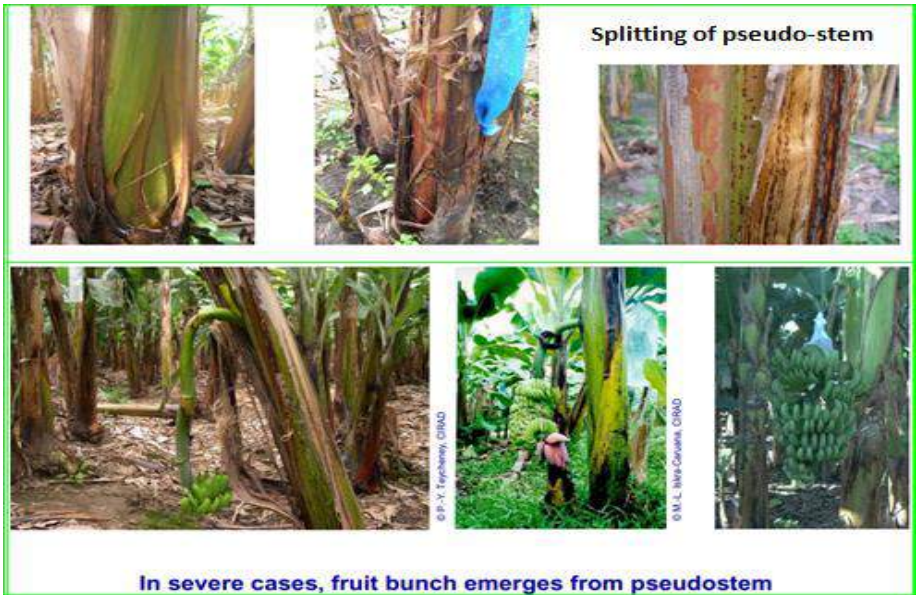
Management:-

- Use of only certified banana suckers or tissue culture plants for planting
- Periodical monitoring and rouging of infected plants with all suckers in the plant by rouging or killing by injecting herbicide, 2, 4-D (*BBrMV)*
- Dry heat treatment of suckers at 40^0 C for 1 day (Mosaic and Bract mosaic)
- Avoid growing other hosts of viruses as intercrops
- Apply more amount of organic manures especially oil cakes
- Raise barrier crops like sunhemp in three to four rows on the field boundaries to check aphids from entering the fields from neighbouring infected fields
- Vector control with systemic insecticides, *viz*., Phosphomidon @ 1ml/l or Methyl demeton or Dimethoate @ 2 ml/l

12. Banana Streak: *Banana streak virus*.

Symptoms:

- The primary symptoms of disease formation of continuous or broken yellow streaks that run from leaf midrid to the margin
- In later stages of the disease, these streaks may become necrotic, and the heart of the pseudostem may rot, ultimately leading to death of the plant
- Splitting of pseudostem. In some cases, spindle or eye-shaped pattern or blotches are noticed



Spread:

Spread by planting material also by mealy bugs *Planococcus citri, Sacharomyces sachari.*

- *Family:Caulimoviridae* *Genus:Badnavirus*
- *Banana streak virus* *dsDNA-RT*

Management:

- Use of certified planting material,
- Field sanitation and weed free fields.
- Eradication of infected plants.
- Control of vectors by spraying insecticides.

DISEASES OF PAPAYA

1. Damping off : *Pythium aphanidermatum, P. butleri, Rhizoctonia solani*.
2. Powdery mildew : *Oidium caricae papayae* (Anamorph) *Sphaerotheca fuliginea*
3. Anthracnose : *Colletotrichum gloeosporioides*
4. Phytophthora root rot : *Phytophthora nicotianae* var. *parasitica*
5. Alternaria leaf spot : *Alternaria alternata*
6. Ring spot : *Papaya Ring Spot Virus* type P and W
7. Mosaic : *Papaya Mosaic Virus*.
8. Leaf Curl : *Papaya Leaf Curl Virus*
9. Post harvest diseases :
   - i) Rhizopus rot : *Rhizopus stolonifer*
   - ii) Fusarium fruit rot : *Fusarium equiseti, F. solani*

1. Damping-Off/ Stem Rot/Collar Rot/Foot Rot- *Pythium aphanidermatum***,** *P. butleri.* Sometimes caused by *Rhizoctonia solani.*

- This pathogen causes damping off in nursery in the form of root rot or collar rot.
- Complete failure of the crop if it appears in the early stages of plant growth.

Disease cycle:

PSI: Oospores and chlamydospores surviving in the soil (*Pythium*), Sclerotia (*Rhizoctonia*). Seedlings raised in the infested soils.

SSI: Through irrigation water and implements Zoospores (*Pythium*) and conidia (*Rhizoctonia*) are dispersed in the field.

Mycelium: Well-developed, branched, coenocytic, hyaline, intracellular, without haustoria. Zoospore differention takes place in vesicle, which emerged out of sporangium.Oospores smooth, thick walled, round and light brown.

Symptoms:

- The disease is observed both in nursery and in the main field.
- **The typical stem rot is most common in 2-3 year old trees,**
- Water soaked patches appear on the stem at the ground level.
- Which enlarges girdle the base of stem and later turn dark brown or black and rot.
- Terminal leaves turn yellow, droop and wilt. Entire plant topples over and dies
- Fruits shrivel and drop.
- Disintegration of parenchymatous tissue and internal tissues are honey comb like finally leading to toppling of plants.

Management:

- Use of disease free seedlings raised in healthy soils.
- Avoid nurseries and planting of papaya in ill drained soils.
- Crop sanitation, up rooting and burning of diseased plants.
- Application of FYM and biocontrol agents will reduce the disease incidence.
- Seed dressing with Captan or Chlorothalonil or Apron (Metalaxyl) 2-3 gm/kg seed.
- Drenching the pits or base of the plants with Bordeaux mixture 1% or Captan 2gm/litre or Copper oxychloride 3 gm/litre or Metalaxyl 2 gm/litre or Chlorothalonil 2 gm/litre.
- Avoid movement of irrigation from one plant to the other and go for drip irrigation.

2. Powdery mildew: *Oidium caricae papaya* (Anamorph) *Sphaerotheca fuliginea* (Telomorph).

Disease is more in cold and dry season.

Disease cycle:

PSI: Pathogen survives in the form of mycelium on the infected plant parts or Ascospores or sometimes the oidia from the near by fields where the pathogen survives.

SSI: Wind borne conidia (Conidia)

Symptoms:

- Initially appears on the lower surface and later on upper surface as powdery patches.
- Later entire leaf surface is covered with powdery patches.
- Leaves become yellow leading to defoliation.
- In severe cases, the disease is also observed on flower stalks and young fruits.
- Fruits do not develop properly and may drop out.
- If seedlings are attacked may lead to death of seedlings.

Management:

- Orchard sanitation, removal and destruction of infected plant parts.
- Spraying of Wettable sulphur 0.3% or Carbendazim 0.1% or Tridemafon 0.1% or Dinocap 0.1% in the field.

3. Anthracnose: *Colletotrichum gloeosporioides.*

Disease cycle:

PSI: Crop residue harbouring the fungus.

SSI: Conidia spread by wind/rain or fruit contact.

Pathogen The mycelium is branched, septate and hyaline at first becoming dark with age. The asexual fruiting body (acervulus) contains a layer of 3 to 50 conidiophores. Conidia are borne successively and acrogenously on short conidiophores pinkish spore masses appear on the surface and setae are produced sparingly. Conidia are hyaline, cylindrical having rounded ends or some what pointed at one end and without vacuole. Sexual fruiting body (Perithecia) contain hyaline and filiform periphyses and asci. Each ascus contains eight allantoid ascospores, which are ejected from the tip of ascus.

.**Symptoms:**

- On leaves, stem and immature fruits small necrotic lesions appear.

- Necrotic lesions will also be found on petioles and flowers.
- Young fruits if infected will not develop properly and are mummified.
- Most characteristic symptoms appear on fruits at ripening stage.
- Initially water soaked discoloured brown spots appear on the fruit surface which are depressed or sunken with black centres.
- Black centres contain lot of dot like fruiting bodies called 'Acervuli'.
- The spots on fruits are sometimes very big and look like a 'bull's eye'.
- Under favourable conditions production pink spore masses is observed from the infected parts.
- Latent infections of the fruits are common.

Management:

- Spraying of Carbendazim 0.1% or Chlorothalonil 0.2% or Captan 0.2% at regular intervals reducing inoculum coming from field up to 10 days prior to harvest.
- Post harvest damage can be minimized by fumigating fruits with Benzyl isothiocyanate and coating with groundnut oil controls post harvest spots and rots..
- Hot water treatment of fruits at 45°C for 30 min.
-

Alternaria fruit spot- *A. alternata*

Fruit spots are characterized by depressed, circular to oval, lesions that eventually become black as a result of mass sporulation.

Lesions are restricted to the surface of the fruit and do not cause extensive rotting of the flesh.

Lesions from multiple infection sites can coalesce as they expand and eventually cover the entire fruit surface.

4. Phytophthora root rot:- *Phytophthora nicotianae* var. *parasitica*

Phytophthora root rot also called soft foot rot, stem canker and soft fruit rot

Disease cycle:

PSI: Oospores and chlamydospores but formation of oospores is rare. Mummified fruits serve as source of reservoir for further spread of the disease.

SSI: Zoospores liberated from sporangia spread through irrigation water in soil and rain splashes in aerial spread. African snails act as vector the pathogen.

Symptoms:

- Water soaked areas on roots and characteristic fungus growth.
- Roots rot completely and plants get killed.
- On the stem lesions occur as water soaked lesions.
- Latex exudation from the lesions is common.
- Lesions girdle the stem and completely kill the plant.
- Older fruits at the bottom of the fruit column are infected.
- Infected fruits become hard, shrivelled and mummified.
- Mature fruits get infected and there will be whitish mycelium with numerous sporangia.

Management:

- Avoid stagnation of water around the stem by mounding up to 1 ft.
- Spraying of basic copper sulphate 1-2 kg/500 litres during wet seasons.
- Spraying of Dithane M-45 0.2% at fortnightly intervals.
- Spraying of Ridomil MZ 0.2% or Aliette 0.2% is also effective.
- Seedbeds in nurseries should be steamed or fumigated prior to planting.

5. Ring spot: *Papaya Ring Spot Virus*

- It is a reported and devastating virus disease in Hawaii
- Khurana first reported from Gorakhpur.
- *Papaya ring spot virus*
- Occurs as 2 strains
  - PRSV- P infects both papaya and cucurbits such as squash, pumpkin, cucumber, and watermelon
  - PRSV-W (watermelon) infects cucurbits but does not infect papaya.

Spread:-Ring spot virus is transmitted mechanically by sap and also by grafting but not seed borne. Major transmission is by aphid vectors like *Aphis gossypii, A. malvae, Myzus persicae, Rhopalosiphum maydis.*

- ***Family Potyviridae*** Genus- ***Potyvirus***
- **Long, flexuous rod-shaped +ssRNA**

Symptoms:

- Severe mosaic mottling, puckering, blistering and deformation of affected leaves.
- The virus induces mosaic, mottling, dark green blisters, necrosis of chlorotic areas (green islands)
- Infected leaves malformed into tendril like or filiform structures called '**Shoe string appearance or Rat tailing'.**
- Tapering of stem with only few leaves at the top.
- **Mosaic or mottle symptoms, dark green spots and oily or water-soaked streaks are seen on stems and petioles of young plants.**
- Circular, coalescent, concentric ring spots on the fruits.
- Diseased fruits contain 40% less sugar content and deterioration in latex quality. Trees infected at a young stage almost never bear fruit, and the fruit of trees that do bear show characteristics ringspots, can be deformed, and generally have lower sugar contents and fall prematurely.

Management:

- Avoid infection in the field by using healthy seedlings.
- Seedlings should be protected from aphids using insect proof nets.
- Prophylactic sprays on community basis with Dimethoate or Methyl Demeton one week before aphid occurrence (Nov-Feb and June-Sept).
- Using mild strains of PRSV in cross protection.
- Manipulating the transplanting time and organic amendments was found most effective in improving the yield and quality inspite of presence of virus.
- Transplanting in September and November and supplemented by 5 kg neemcake, 100 g urea, 100 g P_2O_5 as SSP and 200 g K_2O.
- Spraying of 0.5% $ZnSO_4$ and 0.1% boric acid at monthly interval improved fruit quality and yield.
- Rows of barrier crops like maize and sorghum
- Papaya cv. Rainbow is resistant to ring spot virus.

6. Mosaic: *Papaya Mosaic Virus*

In India 1st reported from Mumbai and Pune in 1947

Survival and spread

- P.I: Affected plant parts (Mechanical)
- S.I: Aphids (*Myzus persicae*)
- Other aphids: *A. gossypii, A. medicaginis, Aphis malvae, Rhopalosiphum maidis, Microsiphum solonifolii*
- Most of the cultivated species of *Carica papaya* are susceptible
- Other hosts: *Cucumis sativus, Cucurbita pepo, C. maxima, Citrullus vilgaris, Luffa acutangula, Lagenaria siceraria,* Safflower and some ornamental plants.
- ***Family- Alphaflexiviridae*.** **Genus -*Potexvirus***
- **Species -*Papaya mosaic virus* (*PapMV*)** ***PapMV* is a filamentous, flexuous rod**
- **Monopartite, (+) ssRNA**

Symptoms:

- Vein clearing and vein banding.
- Mosaic mottling and puckering of leaves, especially the young leaves.
- Older leaves show marginal curling down and with chlorotic spots.
- Older leaves fall down and a small tuft of younger leaves is left at the top with upright position**.**
- Plant growth reduced, leaf lamina reduced and modified into filiform like structures.
- Symptoms in the form of water soaked spots, rings or streaks present in case of ring spot virus are usually absent on petioles, stem and fruits.

Management:

- Spread of the disease can be controlled by elimination of infected plants.
- Mechanical spread by implements and tools should be avoided.
- *Carica california* is immune.
- Disease free seedlings
- Rouging and destruction of diseased plants
- Vector control
- Weekly sprays with 1% groundnut oil

7. Leaf Curl: *Papaya Leaf Curl Virus*.(***PaLCuV*)**
Papaya leaf curl was first reported in Tamil Nadu in 1939.

Symptoms:

- **Downward curling of foliage** and inward rolling of leaf lamina.
- Crinkling and distortion of leaves.
- Severely affected **leaves** exhibit **inverted cup like** appearance.
- Thickening of leaf veins and midrib, become brittle and leathery.
- The inter-veinal areas are raised on the upper surface **due to hypertrophy** which gives rugosity to the leaves.
- **Petioles** get reduced in length, twisted in **zig-zag** manner.
- Affected plants fail to flower and produce no fruits.
- Even if **fruits** are produced they are **malformed**.

Spread:- White fly -*Bemesia tabaci* is the vector transmitting the leaf curl virus.
Collateral hosts like Tobacco, Chillies, tomato, Datura, Petunia, Zinnia and some other weeds act as reservoirs for the virus.

- ***Family Geminiviridae*.** Genus ***Begomovirus***
- ssDNA

Management:

- Procure healthy seedlings from nursery.
- Selection of site-avoid fields nearer to tomato, tobacco etc.
- Rouging of diseased plants.
- Spraying of Monocrotophos 1 ml/litre or Dimethoate 2 ml/lit or Metasystax 0.1% at an interval of 10-12 days.

8. Post harvest diseases:

i) Rhizopus rot : *Rhizopus stolonifer*
ii) Fusarium fruit rot : *Fusarium equiseti, F. solani*

i) Rhizopus rot/ Watery Fruit Rot- *Rhizopus stolonifer.*

Symptoms:

- Water soaked spots on the fruit skin involving major part of fruit.
- Fruit flesh becomes soft and watery **and emits a foul odour.**
- The fungus produces characteristic strands of coarse white/grey moldy growth consisting of sporangiophores and sporangia covering the fruit.

ii) Fusarium fruit rot:*Fusarium equiseti, F. solani*

Symptoms:

- Larger depressed lesions of water soaked nature towards stem end.
- Lesions are soft with scanty fungus growth.
- Fruits also exhibit dry rot in some cases.
- The water soaked lesion enlarges rapidly and turns light brown, later centre of lesion gets depressed, white mycelial growth profuse, appear on rotted surface .

Management:

1. Storing fruits at 10 ^{0} C.
2. Treating with Aureofungin 500-1000 ppm.
3. Treating with hot water (HWT) at 49 ^{0}C $\pm$ 1 ^{0}C for 20 min.
4. Fruit dip in DCNA (2,6 Dichloro-4 nitroalanine) at 1000 to 2000 ppm.

DISEASES OF POMEGRANATE

1. Cercospora leaf spot : *Cercospora punicae*
2. Anthracnose : *Colletotrichum gloeosporioides*
3. Bacterial leaf spot : *Xanthomonas axanopodis* pv. *punicae*

4. Post harvest diseases :
   i. Aspergillus rot : *Aspergillus niger*
   ii. Fusarium rot : *Fusarium sps.*
   iii. Cladosporium rot : *Cladosporium* spp.

1. Cercospora leaf spot :- *Cercospora punicae*

Symptoms:
- Minute brown to black circular spots on leaves.
- Sporulation on the under surface with stroma.
- Minute black circular spots on flowers.
- Elliptical spots on twigs.
- Black circular spots on fruit rind which enlarge and become irregular and ugly.

Disease Cycle:
Survives as mycelium in shoots/fruit/leaves.
Secondary spread is by wind disseminated conidia.
Management:
- Spraying with Mancozeb 0.2%, Thiophenate Methyl 0.1%, Carbendazim 0.1% at 15 days interval.

2. Anthracnose : *Colletotrichum gloeosporioides*

Symptoms:
- Small irregular dull violet to black spots on leaves with yellow halo.
- Leaves turn yellow and fall.
- Black fruiting bodies i.e. Acervuli are produced.
- Circular brown to black sunken spots appear on fruits.

Disease Cycle:
Survives as Acervuli or mycelium in infected plant parts.
Spread by wind borne conidia.
Management:
- Spraying if Carbendazim 0.1% or Thiophenate methyl 0.1% or Zineb 0.2% at 20 days intervals.

3**. Bacterial leaf spot : *Xanthomonas axanopodis* pv. *punicae***
Symptoms:
- Minute irregular water soaked brown to dark brown spots on leaves.
- Brown to black elongated oily spots on twigs and flowers.
- Exudation of bacterial ooze.
- Oily spots on fruits.
- Pericarp of the fruits exhibit characteristic L or Y shaped cracks.

Disease Cycle:
Pathogen survives on fallen fruits, leaves, twigs and also on plants.
Spread by rain splashed bacterial cells.

Management:
- Obtain cuttings from healthy plants.
- Before pruning, spray 1% Bordeaux mixture on diseased leaves. Then spray with ethrel solution (2.0-2.5 ml/litre) to defoliate the diseased leaves. Collect and destroy the fallen leaves.
- Disinfect the pruning knife in Sodium hypochlorite solution (25 ml/litre).
- Dust the tree basins with bleaching powder at 20-25 kg/ha to kill the bacteria on left over leaves.
- After pruning, paste the diseased stems with copper oxychloride (3 g/litre) mixed with red sandy loam soil.
- Spraying of Bordeaux mixture 1% or COC 0.3% alone or in combination with antibiotics like Streptomycin or Streptomycin sulphate or Cyclomycin 200 ppm.
- At the beginning of disease incidence, give 5 to 6 sprays with streptomycin+copper oxychloride (3 g/litre) at 10 days interval.
- After each spraying of bactericides, give mineral spray (1 g each of Cu SO_4, Mg SO_4, Ca SO_4 and Boron in 1 lit of water) to boost the plant growth.

DISEASES OF BER

1. Powdery mildew : *Oidium erysiphoides* f. sp. *zizyphi*
2. Sooty mold/black leaf spot : *Capnodium ramosum*
3. Alternaria leaf spot : *Alternaria chartarum, A. passeriniana*

1. Powdery mildew : *Oidium erysiphoides* f. sp. *zizyphi*

This disease was first reported from Allahabad

Symptoms:- The disease is distributed in almost all ber growing regions of India. The disease is noticed generally at the end of October and prevails from November to April

Powdery mass appears on young leaves and fruits

- The disease first appears on young leaves in the form of white floury patches and later spreads to the young shoots and developing fruits.
- With the passage of time, the infected area becomes slightly raised and rough.
- Severely affected leaves shrink and defoliate.
- The initial symptoms are seen on flower buds as powdery growth of the fungus.
- As the disease advances, the powdery growth will be seen on small fruits of pinhead stage.
- In severe cases, the entire fruit is covered with white powdery growth of the fungus.
- The spots change into light brown to dark brown discolouration,
- Infected area becomes slightly raised having cankers and fruits crack.
- The infected fruit often becomes malformed and may shed from the tree.

Disease cycle:

P.S.I -The pathogen survives on the branches, twigs and bud wood during the absence of flowers and fruits in the form of dormant mycelium and also on alternate hosts including weeds.

S.S.I- The secondary spread of the disease takes place by the air borne conidia.

The decrease in night temperature and increase in day temperature favour the disease spread.

Management:

- Spraying of Dinocap 0.1% or Wettable sulphur 0.1% during 1st week and 3 rd week of November.
- Spraying of Carbendazim 0.1% or Hexaconazole 0.1%.
- Resistant varieties like Gola, Safed Rohtak, Seb, Mehrun, Glory etc.

2. Sooty mold/black leaf spot : ***Isariopsis indica* var. *zizyphi***

Symptoms:

- Circular to black /sooty spots develop on lower side of leaves.
- Corresponding upper surface turn brown and lead to defoliation.
- Unlike other sooty molds the pathogen is pathogenic.

Disease cycle:

PSI: The pathogen survives in soil, plant debris and leaves sticking to crotches of branches.

SSI: Air borne spores/conidia.

Management:

- Spraying of Carbendazim or Captafol 0.2% or Thiophenate methyl 0.1% or Chlorothalonil 0.2% during rainy season.

3. Alternaria leaf spot : *Alternaria chartarum, A. passeriniana*

Symptoms:

- Irregular brown spots on upper surface of the leaves.
- Lower side spots are dark brown to black in color.
- Defoliation of leaves.
- Fruits show brown depressed and elongated spots.

Disease cycle:

PSI: Crop debris

SSI: Wind disseminated conidia.

Management:

- Spraying of Zineb, Captafol, Rovral or Chlorothalonil 0.2%.
- Resistant Varieties Gola Gurgaon, Popular gola, Bahadur Garhia etc

DISEASES OF FIG

1.	Rust	:	*Cerotelium fici* (***Physopella fici***)
2.	Anthracnose	:	*Sphaceloma fici-caricae*
3.	Cylindrocladium leaf spot	:	*Cylindrocladium scaparium*
4.	Mosaic	:	*Fig Mosaic virus*

1. Rust : *Cerotelium fici.* Rust is an important disease of Fig.

Symptoms :

- Rust is an important disease of fig in all the fig growing areas.
- Symptoms generally develops late in the summer.
- Initially, symptoms are small, yellowish spots on the upper surface of the leaves. As these spots (or lesions) grow larger, they turn a reddish -brown colour but remain relatively smooth.
- On the lower surface of the leaf, the lesions are a reddish-brown color and have a slightly raised, blister-like appearance. Heavily infected leaves often turn yellow or brown, particularly around the edges
- Later, the rust-infected areas form angular brown spots on the upper leaf surface
- The angular spots then coalesce and form necrotic regions.
- The premature shedding of leaves.
- The teliosori on leaves are less powdery than uredosori in appearance.
- In severe, it can cause the trees to defoliate in a matter of a few weeks.
- The raised, blister-like pustules appear on fruits.

Fig. 1. Symptoms of fig rust on the upper leaf surface

Fig. 2. Fig rust – pustules on the underside of the leaf

Fig. 3. Fig rust symptoms – necrosis and curling of diseased leaves

Fig. 4. Fig rust symptoms – seared appearance of a diseased tree

Raised spots on fruits

Disease cycle:

PSI: Teliospores surviving in diseased plants.

SSI: wind borne uredospores.

Management:

- Dusting sulphur powder 250 – 500 g per plant.
- Spraying wettable sulphur 0.2% or Zineb / Bitertenol (Baycor) 1g / litre during December.
- **Tiger' slightly more resistant.**

2. Anthracnose: *Sphaceloma fici-caricae*.
Symptoms:

- Anthracnose symptoms tend to be most conspicuous on leaves and ripe fruits.
- Anthracnose first appears on leaves as small black, yellow or brown spots.
- As the anthracnose disease progresses, these spots enlarge and merge to affect entire areas.
- Mature spots circular, oval or elongate or irregular, light brown or reddish in colour.
- When such spots coalesce present a patchy look on the leaves.Defoliation of the leaves.
- Disease can also cause cankers to form on stems and petioles, resulting in severe defoliation and root rot.
- Infected fruit develop sunken, circular spots that might emit pink spores.

Disease cycle:
PSI: Conidia surviving in infected plant parts.
SSI: Wind borne conidia.

Management:
Spraying Aureofungin 40 ppm+ Copper sulphate 20 ppm in soap solution is effective.

3. Cylindrocladium leaf spot: *Cylindrocladium scoparium*

Symptoms:
- Initially minute brown spots appear on the leaves which enlarge into uniform or zonate, prominent, reddish brown lesions with dark brown margins.
- These lesions later coalesce to form irregular patches and affected leaves sheds earlier.
- The centre of the leaf become papery and drops off. Cobweb like mycelium spread over the lower surface of leaves and later on ultimately become powdery.

Disease cycle:
PSI: Mycelium surviving in infected plant parts.
SSI: Wind borne conidia.

Management:
Spraying of mancozed 0.2% or captan 0.3%.

4. Mosaic: *Fig Mosaic Virus*.
Symptoms:
- **Fig mosaic can cause symptoms on both leaves and fruits.**
- On wild varieties masked mosaic symptoms with blister like patches appear. Such leaves later become pale yellow and wither away
- On cultivated varieties, initially yellow green spots scattered over the lamina, which sometimes coalesce to form bigger spots of various shapes and sizes. Occasionally such infected leaves develop white mottle and are deformed.
- Mosaic spots or lesions may be uniformly scattered over the surface of the leaves or may appear as irregular patches of light green diffused widely throughout the leaf blade.
- Later in the season, a rust-colored band develops along the border of the mosaic spots, apparently caused by the death of epidermal or subepidermal cells.
- Deformed leaves may occur on the same twig as normal leaves. Mosaic spots on fruits are very similar to those on leaves, but less conspicuous.
- Premature fruit drop may also occur in certain cultivars. Infected trees seem to produce smaller and fewer fruit.

Transmission:
Transmitted through grafting, seed and sap and eriophid mite *Aceria ficus.*

- *Family-Bunyaviridae*
- *Genus-Emaravirus*
- Species- *Fig mosaic virus*
- (- ssRNA)

Management:

- Spraying of insecticides like Dicofol for vector management.
- Use of healthy planting materials.

DISEASES OF PHALSA

1.	Rust	:	***Uredopeltis* (*Dasturella or Phakopsora) grewiae*.**
2.	Cercospora leaf spot	:	*Cersospora grewiae*
3.	Powdery mildew	:	*Oidium grewiae*

1. Cercospora leaf spot: *Cercospora grewiae*.
Symptoms:

- This disease is common in rainy season
- Minute black spots develop on both sides of the leaves.
- Up on maturity, the diseased areas turn black in the centre consisting cushion like sporulation.
- This results in early defoliation of leaf.

Disease cycle:
PSI: Pathogen survives in the crop residue as stroma or dormant mycelium.
SSI: Wind / rain spread conidia.

Management:

1. Spray of Aureofungin 30 ppm or Benomyl 0.1%
2. Spraying of Carbendazim (0.1%), Thiophenate methyl (0.1%), Kavach (0.2%) etc.

2. Rust: *Dasturella grewiae.*

Symptoms:

- Light brown spot develop on the lower side of leaves as a result of **uredospore** infection
- The corresponding upper surface is necrotic.
- **The severely affected leaves start drying followed by defoliation**

Disease cycle:
PSI: Diseased plant serve as the primary source of infection.
SSI: Wind brone uredospores.

Management:
- Spraying of Wettable sulphur 0.2%, Chlorothalonil 0.2%, Triademefon 0.1% during rainy season and in November.

3. Powdery mildew : *Oidium grewiae*
- The disease manifest by powdery patches on leaves, tendrils, flowers, young fruits.
- On leaves the powdery patches enlarge and the upper surface becomes dusty, whereas on lower surface the mealy submerged growth is observed.
- The affected leaves may curl upward in dry weather.
- They may be malformed and discoloured.
- Later infection of berries results in soft rot symptoms.
- In case of a severe attack, the fruits shrivel extensively.

Management
- Clean cultivation of vines or removal and destruction of all diseased parts
- Dustings of vines with 300 mesh sulphur
- Spray wettable sulphur @ 0.3% or Dinocap or Tridemorph @0.1%
- Biocontrol agent- *Ampelomyces quisqualis*

DISEASES OF AMLA

1. Rust- *Ravenelia emblicae*
2. Anthracnose- *Colletotrichum gloeosporioides*

1. Rust- *Ravenelia emblicae*

Symptoms- Symptoms appears both on leaves and fruits,

Leaves- circular or semi circular Pinkish brown pustules develop which may appear singly or in groups

Interestingly, the plants with a severe attack on fruits show no symptoms on leaves and *vice versa.*

Conspicuous brown rust pustules are formed on fruits and finally these pustules become dark brown to black in colour and which later develop into a ring.

The fruit gives a dirty appearance.

Severe infection of fruits results in premature fruit drop.

Fig 2. Rust on leaves

Fig. 1 Rust on fruits

Management

1. The spread of disease can be checked by giving monthly sprays of wettable sulphur (0.5 %) or Zineb (0.2 %) during July-September.

2.Three sprayings of 0.1 % Bayleton at 15 days interval

3. Resistant –Banarasi, Chakaiya

2. Anthracnose- *Glomerella cingulata*

Fig. 10. Anthracnose on fruits

Symptoms appears on leaflets and fruits during August-September

Initially minute, circular or spindle shaped brown to grey spots with yellowish margin appear on leaflets, which later bear dot like dark fruiting bodies in their centre.

On fruits, depressed lesions develop which later turn dark in colour bearing dot like fruiting bodies arranged in rings.

Consequently the infected fruits become shriveled and rot

DISEASES OF CUSTARD APPLE

1. Anthracnose / Glomerella fruit rot : ***Glomerella cingulata***
Colletotrichum annonicola

Symptoms:

- Light brown to black spots on leaves.
- Spots coalesce to form patches on the leaves.
- Leaves defoliate, causing die back of plants.
- On the fruits, appears at blossom end as brownish black, dry spots.
- Fruits are transformed into a shrivelled mass (Mummification)

Plate 5. Leaf symptoms of anthracnose more commonly seen

Disease cycle:
PSI: Survive in Plant debris or on collateral hosts.
SSI: Wind / rain disseminated conidia.

Management:
- Orchard sanitation, removal and burning of infected twigs and fruits.
- Spraying of benzimidazoles like Carbendazin or Benomyl 0.1%
- Spraying of Mancozeb 0.2% or Bordeaux mixture 1% is also effective.

DISEASES OF APPLE

1. Scab	:	*Venturia inaequalis* [*Spilocaea pomi* (IPS)]
2. Powdery mildew	:	*Podosphaera leucotricha*
3. Fire blight	:	*Erwinia amylovora*
4. Crown gall	:	*Agrobacterium tumefaciens*
5. Mosaic	:	*Apple Mosaic Virus*

Post Harvest Diseases
1.Soft rot- *Pencillium expansum*
2.Bitter rot- *Glomerella cingulata*
3.Pink mold- *Trichothecium roseum*

1. Scab caused by : *Venturia inaequalis*
- This disease endemic in Himachal Pradesh

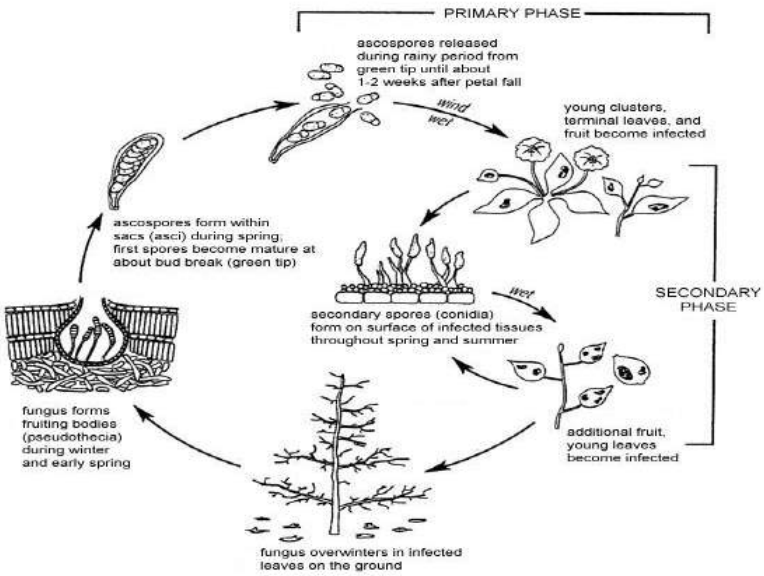
Losses
- Losses from scab result from midseason defoliation of trees, reduction in fruit production, devaluation in fruit grade, weakening of the trees, failure of fruit bud formation and increased expenses on spray operations.
- The first epidemic of scab in Kashmir valley in 1973 completely ruined the apple crop worth US$ 540,000.
- During 1983, it rendered 10% apple crop (30,000 MT) unfit for consumption in Himachal Pradesh which has to be destroyed and as a result the state suffered a loss of Rs. 1.5 crores.

Symptoms:
- Leaves- Large brown olivaceous spot on lower surface of leaves. Ultimately got wavy margin Lower surface is convex Later lesion on upper surface of leaves . Lesion become velvety with slight depression on upper surface
- Leaf become curled and fall off.
- Dwarf and distorted leaves.
- Fruits show small, warty rough, black or brown, circular lesions on their skin with white halo surrounds The centre of the spots become corky and on mature fruits, yellow halo around the lesions. Cracks developed in the lesion area.
- During storage diseased areas depressed, shrink and rot due to secondary infection.

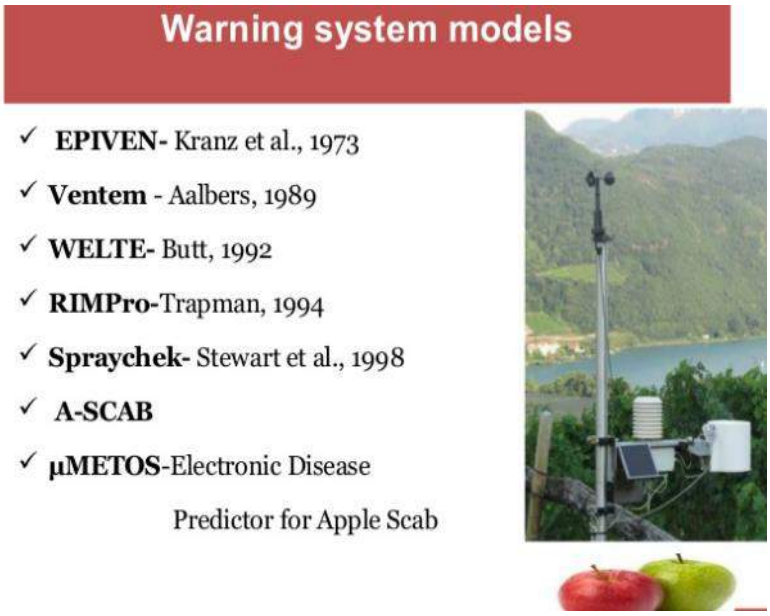
Disease cycle:
This disease, caused by the fungus *Venturia inaequalis* (anamorph *Spilocaea pomi*), may be quite severe when rainy, cool weather occurs in the spring. Fungal spores are produced in early spring on dead, fallen apple leaves about the time buds begin to develop. These spores are splashed by rain and blown by wind to land on developing plant tissue and initiate infections. After spots appear on the newly formed leaves, more spores are produced that spread infection to other parts of the tree. Again, rainy weather greatly encourages spore spread and infection during the secondary phase of spore production. The fungus over winters on fallen leaves.

PSI: Ascospores released from pseudothecia on fallen leaves in spring.
SSI: Wind borne conidia.
Favourable conditions

- High relative humidity, temperatures between 16° and 20°C and leaf wetness favour the fast development of the disease.



MILLS rules



Management:

- Breaking of the overwintering cycle and reducing the production of ascospores
- Collection and destruction of fallen leaves pruned material.
- Spraying of 5% urea after harvesting in Oct-Nov to accelerate decomposition of leaves.
- Spraying of 2% urea solution at pre bud break stage to prevent ascospore release in spring.
- Spraying of Urea 5% or Benlate 0.2% once in autumn prior to leaf fall and again just before bud break with 2% urea followed by two sprays of Captan 0.2% or Mancozeb 0.25% at 10-15 days after petal fall.
- Spraying of 0.2% captan or 0.25% dodine at short intervals after petal fall or a single application of 0.3% difolatan at green bud stage followed by 0.2% captan at petal fall controls the disease.
- Apple Cvs. Emra, Red Free, Prima, Priscilla, Macfree, Liberty, and hybrids Ambstarking, Ambroyal, Ambrich and Ambred are resistant.
- The biocontrol agent- *Athalia bombacina*

  Spray schedule

Stage picture	Chemical spray
Green Tip	Mancozeb 75WP@3 gm/L of water or Captan50WP @3 gm/L of water
Pink Bud	Dodine 65wp@ 0.5 gm/L of water, Mancozeb 63% + Carbendazim12% @2.5 gm/L of water
Petal Fall	Bitertanol10EC @ 0.5 ml/L of water or Kresoxim methyl 44.3SC @ 0.4 ml/L of water or Difenconazole 25EC @0.3 ml/L of water
Pea Shaped fruit	Protectant fungicides like Hexaconazole5%+ Captan70% @ 0.5 gm/L of water
Walnut shaped fruit	Mancozeb63%+ Carbendazim12% @2.5 gm/L of water or Myclobutanil10WP @0.5 gm/L of water

2. Powdery mildew: Caused by *Podosphaera leucotricha*

Cletothecia having dichotomously branched appendages. Single asci in the cletothecia. Eight ascospores are present in the ascus. Conidia are ellipsoidal, truncate and hyaline.

Symptoms:

- New growth is affected *i.e.* on buds, leaves and shoots.
- Buds killed before they develop into leaves.
- Whitish or grayish growth on the lower side of foliage or on both sides.
- Leaves curl and crinkled, hard, brittle and narrow.
- Affected fruit buds results in non bearing of flowers and fruits.
- Fruit remain small, deformed and cracked.
- Fungus produces Cleistothecial bodies on twigs.

Disease cycle:

PSI: Resting mycelium or as haustoria in the buds and fruits.
SSI: Wind borne conidia.

Management:

- Pruning, thinning and cutting back laterals to reduce the primary load.
- Spraying of Wettable sulphur 2-3gm/litre or Carbendazim or Thiophenate methyl 0.05% or Kerathane 0.05% at late dormancy, bud swell, petal fall and again 15 days later.
- Application of Lime sulphur
    - 1:15 dilution at green tip stage
    - 1:35 dilution at open cluster stage.
    - 1:60 dilution at full pink stage.
    - 1:100 dilution at half petal fall stage gives better control.
- Spraying of Difenconazole, Penconazole and Hexaconazole etc..
- Resistant varieties Golden delicious and Red delicious.

3. Fire blight : Caused by *Erwinia amylovora*

The bacterial nature of the disease was described by T.J Burril in 1882. **This was the first plant disease proven to be caused by a bacterium.**

Economic impact – Results from killing of flowers, fruit spurs, twigs and girdling of large branches and trunks that results in death of the trees

- Young trees in nursery or orchard can be killed in a single season.
- Hosts – over 75 rosaceous plant species are susceptible
- Pear is considerably more susceptible than apple

Symptoms- All above ground tissues including blossoms, fruits, shoots, branches and limbs, and in the rootstock near the graft union on the lower trunk.

Blossom blight- The initial symptom usually occurs on flowers, which become water soaked, then shrivel turn brownish to black in colour and fall or remain hanging in tree. The symptom spread to leaves, twigs. Terminal twigs wilt from tip to downward and also spread to branches fruits becomes water soaked, turns brown, shrivels and finally becomes black. Oozing may be seen in the affected area.

Shoot blight- The infected shoot becomes light to dark brown in contrast to pear which becomes black. Blighted shoots will often form the characteristic “shepherd's crook” at their tip and the dead leaves remain clinging to the affected twigs. From there the disease can progress to whole limbs and when infection is severe, the whole tree appears to be scorched by fire, hence the name fire blight.

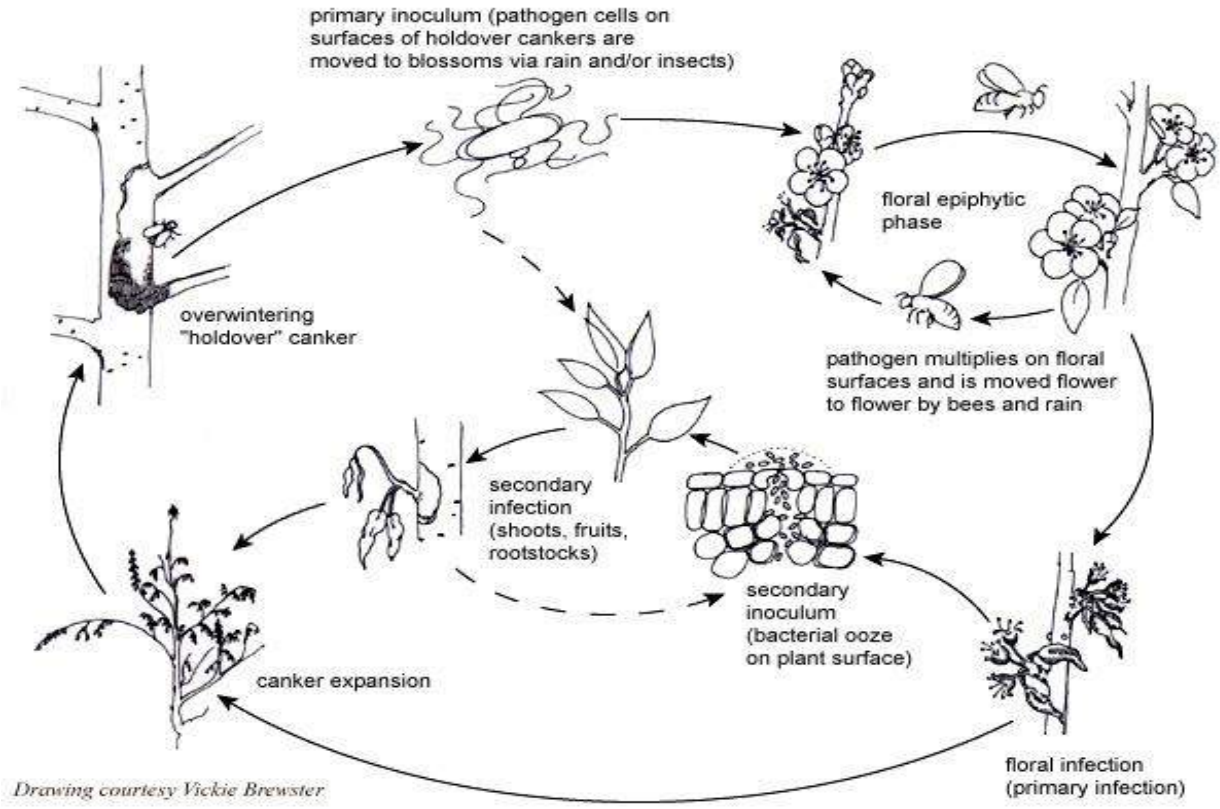
Canker blight- The canker itself is dead bark, phloem and cortex cells. Canker blight is also referred to as ‘limb’, ‘trunk’, or ‘body’ blight depending on where the infection occurs. Cankers form as result of the bacteria travelling systemically into the woody tissue of the tree

There are two types of cankers; determinate and indeterminate. The determinate cankers have a distinct margin with cracks that circle the infected area.The indeterminate cankers have indistinct margins that can be raised or blistered but a canker could simply be a water soaked. If the bark is peeled back from the margin area of both types of cankers, a distinct demarcation between healthy and diseased tissue is visible where the diseased tissue is a red-brown color.

Rootstock blight -Trees on M-26 and M-9, the most commonly used dwarfing rootstocks, are the most susceptible. Infected root stock turns dark brown to black and a contrast outline between the scion and rootstock at the graft union is often noticeable.

- Water soaked spots on leaves which later, shrivel, turn brown to black and fall.
- Leaves curl, shrivel and hang down
- Twigs gets blighted and dieback.
- Bark of the twigs turns brownish black, shrink and hardened.
- Cankerous growth formed on branches. Branches above the point of infection dies.
- Water soaked areas on fruits, turn brown, shrivel and mummified.
- Appearance of milky coloured, sticky droplets of bacterial exudation under humid conditions.

Disease cycle:



Overwintering- *Erwinia amylovora* overwinters in a small percentage of the annual cankers that were formed on branches diseased in the previous season. These overwintering sites are called "holdover cankers". As temperatures warm in spring, the pathogen becomes active in the margins of holdover cankers. Free bacterial cells are released onto the bark surface, sometimes as visible ooze. Insects attracted to the ooze (e.g., flies) or rain disseminate the bacteria from the canker to flowers.

Floral epiphytic phase- **Stigmas**, which are borne on the end of the styles, are the principal site of epiphytic colonization and growth by *E. amylovora*. During the floral epiphytic phase, the ultimate population size that the pathogen attains is influenced by temperature, which regulates the generation time of the pathogen, and by the number of blossoms in which the pathogen becomes established, which is facilitated by pollinating insects, honey bees in particular. Under ideal conditions, stigmas of each flower can support ~10^6 cells of the pathogen.

Primary infection in flowers- Blossom blight is initiated when cells of *E. amylovora* are washed externally from the stigma to the hypanthium (floral cup). On the hypanthium, *E. amylovora* gains entry to the plant through secretary cells (nectarthodes) located on the surface. In pear, the importance of blossom blight is expanded further by the tendency of this species to produce nuisance, secondary or "rattail" flowers during late spring and early summer, long after the period of primary bloom.

Secondary phase- This includes shoot, fruit, and rootstock blight. These phases are usually initiated by inoculum produced on tissues diseased as a result of blossom infection. Wounds are generally required by *E. amylovora* to initiate shoot and fruit blight. Insects, such as plant bugs and psylla, create wounds on succulent shoots during feeding. Strong winds, rain, and hail can create numerous, large wounds in host tissues. Infection events induced by severe weather are sometimes called "trauma blight." Rootstock blight of apple can result from shoot blight on water sprouts or from internal translocation of *E. amylovora* from infections higher on the tree.

Canker expansion- Both primary and secondary infections can expand throughout the summer, with the ultimate severity of an infection being dependent on the host species, cultivar, environment, and age and nutritional status of the host tissues. Rates of canker expansion also can be enhanced by a high water status in a tree caused by excessive or frequent irrigation or poorly drained soils. Canker expansion slows in late summer as temperatures cool and growth rates of trees and shoots decline.

Epiphytic growth of *E. amylovora* on stigmas combined with movement of the pathogen from flower to flower by pollinating insects .

Erwinia amylovora also can reside as an endophyte within apparently healthy plant tissue, such as branches, limbs, and bud wood. Migration of the pathogen through xylem is one mechanism by which floral infections of apple can lead to rootstock infections near the graft union

PSI: Bacterium over winters in the cankers of previous season formed on large branches.

SSI: Bacterium spread by rain splashes, flies, wasps, honey bees etc.

Management:

- Pruning all blighted and dried twigs and cankerous 10 cm below the lowest point of infection and burning.
- Disinfection of tools after each cut in 0.1% mercuric chloride solution.
- Control of insects to restrict spread of the bacterium using 1.6 ml of Monocrotophos per litre or 2ml of Dimethoate.
- Spray application of Streptomycin when the max temperatures are above 18 ^{0}C four times at 550-1000 ppm.
- Spraying of Copper oxy chloride 0.3% as blossom application.
- Insect control esp. aphids and plant bugs to prevent infections
- **Biocontrol agents** *Erwinia herbicola strain C 9-1 or Pseudomonas fluorescens strain A 506.*

4. Crown gall: Caused by *Agrobacterium tumefaciens.*

Symptoms:

- Small outgrowths on stems and roots.
- Initially galls are soft, spherical, white or fleshy varying 7-10 mm in diameter.
- Plants are stunted with chlorotic leaves.
- Crown Gall:- The characteristic symptom of the disease is swelling along the roots or on the trunk just above the soil line.
- The young galls are soft, composed of light coloured frosty tissue resembling fresh callus growth and globular to elongated.
- Galls vary in size from 0.6 to 10.0 cm in diameter and in extreme cases may go up to 30 cm or more.
- Mature galls are hard, corky, knobby on woody stem.
- With the enlargement of galls, the outer tissue becomes darker and often turns black. Infected trees are often stunted and may produce small, chlorotic leaves.

Disease cycle:
PSI: Soil borne and also through planting material.
SSI: Root contact, Implements.

Management:

- Field sanitation and use of healthy planting material.
- Avoiding wounding plants, disinfection of tools.
- Dipping healthy grafted plants in 1% $CuSO_4$ for 90 minutes.
- Rotation of nursery sites.
- Use of biocontrol agents such as *Agrobacterium radiobacter* K 84 or K-1026.

5. Mosaic : *Apple mosaic virus*.
Fy. *Bromoviridae*;
Genus-*Ilarvirus*; Ilarvirus subgroup 3
+ssRNA
Symptoms:

- Apple trees infected with apple mosaic virus develop pale to bright cream spots on spring leaves along with margins.
- Apple leaves tend to be mottled with white to creamy yellow flecks, spots, and patches

Transmission:
Transmitted by grafting & budding.
Not associated with any of insect vectors, sap or seed.

Management:

- Treating affected bud wood at 50° C in hot water for 15 minutes or at 70°C with warm vapour (AST) for 10 minutes inactivates the virus.
- In seedlings, the virus could be inactivated at 40° C exposed for 20 days.
- Rouging of infected trees

- Use of certified disease-free planting stock
- Virus-free mother' plants should be indexed for graft and bud wood.

Post harvest Diseases

Soft Rot- *Pencillium expansum*

- Young spots starts from stem end of the fruit. In the early stages, the rotted area is light in colour and soft and watery in texture. As the fruit ripens area of the rotting increases, skin becomes wrinkled. In humid atmosphere, grayish blue cushions of fruiting bodies of the pathogen appear on the diseased skin. A peculiar musty odour is emitted.

Bitter Rot- *Glomerella cingulata*

- Bitter rot is a fairly firm rot that starts as a small, circular, light brown spot which enlarges rapidly, darkens, and eventually turns almost black. Characteristic of this rot is the saucer-shaped depression in the centre and later the concentric rings of tan fungus fruiting structures that form inside the spot. Tiny, raised, black dots appear beneath the cuticle and give rise to acervuli which break the cuticle.
- In moist weather, sticky pink masses of spores are seen which may be arranged in clearly defined rings
- **Pink mold rot- *Trichothecium roseum (Cephalothecium roseum)***
- The rot affects all delicious varieties of apple. The attack is both external and internal but is sometimes only internal. The rot is usually found associated with scab following which the disease has two wall marked stages.
- In the first, rot appears as sunken, about 5 cm wide brown bands encircling the scab spots with fungal threads on the rotted area producing pink spore masses under moist warm conditions. The affected tissue firm corky, and bitter taste.
- In the second or final stage of rot is characterized by chocolate brown sunken area with an irregular outline which remains rather firm and dry. Superficially pink mold rot resembles fish-eye and bull's-eye spots.

MANAGEMENT

- Avoid the damage at the time of picking, grading and packing.
- The diseased and injured fruits should be discarded.
- Growth and spread of the rot is very slow at low temperature and may be almost entirely inhibited at 0 to 1.6°C. (cold storage after packing).
- Sulphur dioxide fumigation by sodium metabisulphite.
- Apple coated with paraffin oil and pungent mustard oil
- Fruits wrapped with SOPP (Sodium ortho phenyl phenate)
- Dipping fruits in 1000 ppm benomyl or diphenylamine for 5 min or in 500 ppm of aureofungin for 20 min.

DISEASES OF PEACH

1. Leaf curl : *Taphrina deformans*
2. Rust : *Puccinia pruni-spinosae*
3. Scab/Freckle/Blackspot : *Venturia carpophila* (PS)

Minor diseases

Brown rot of Fruits	:-	*Monilinia fructicola, M. fructigena, M. laxa, M. laxa* f.sp. *mali*
Powdery mildew	:-	*Sphaerotheca pannosa*

Bacterial diseases

Bacterial leaf spot	:-	*Pseudomonas syringae* pv. *persicae*
Peach- X	:-	Phytoplasma
Peach yellows	:-	Phytoplasma

Viral diseases

Necrotic leaf spot - *Necrotic leaf spot virus* (*NLSV*)

1. Leaf curl : *Taphrina deformans.*

- Dimorphic mycelium, Nacked asci, Ascocarps are absent

- Asexual reproduction is through small oval or spherical uni-nucleate haploid blastospores that bud from ascospores either with in the ascus or after their release.
- Asci are formed from special binucleate ascogenous cells.

Symptoms:

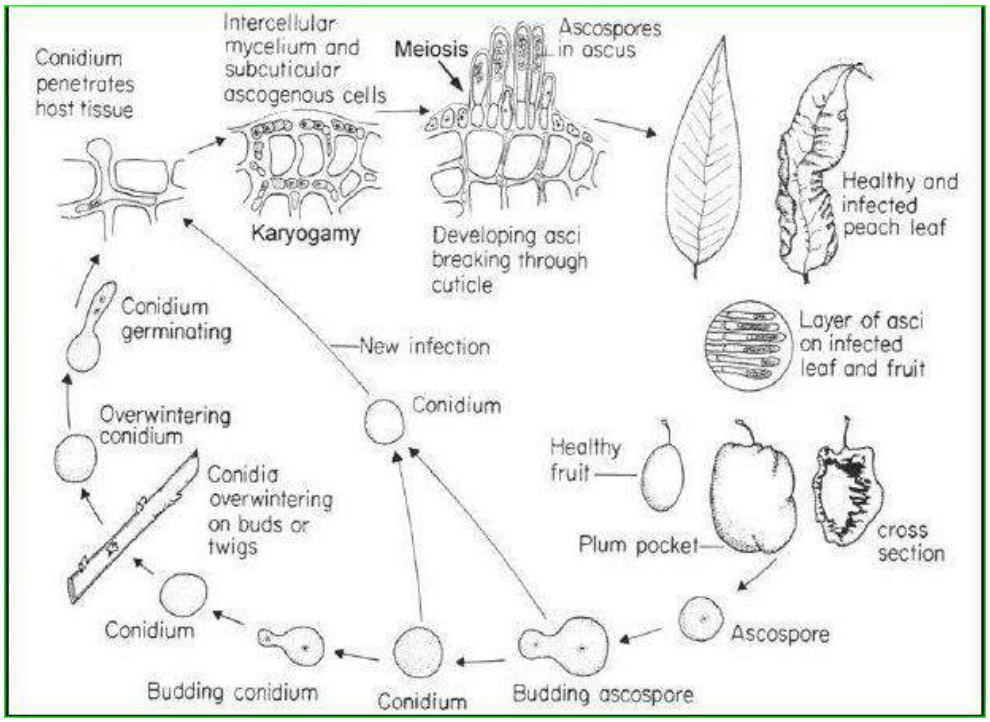
- Leaf lamina becomes thick, puckered along midrib and curl.
- Leaves 2-3 times longer than healthy and develop red or purple colour.
- Puckered parts of leaf may turn yellowish and then greyish white, as velvety spores are produced on the surface
- Finally infected leaf turns to reddish purple ting**. A**scospores, which give the leaf a greyish white, powdery or velvet like appearance
- Older leaves become necrotic, wither and fall out prematurely.
- Twigs swollen and die.
- Only rarely do reddish, wrinkled to distorted (or hypertrophied) areas develop on fruit surfaces. Later in the season these infected areas of fruit become corky and tend to crack.
- Flowers and fruits also drop off.

Disease cycle:

PSI: Fungus over winter as ascospores or as thick walled conidia on the bud scales or as perennial mycelium.

SSI: Air borne ascospores.

When the weather turns cool and wet in fall, the ascospores germinate to produce more bud-conidia. The new and old bud-conidia continue to increase in number by budding. Eventually a film of bud-conidia is formed on the tree's surface. In spring, the bud-conidia move by splashing water from irrigation or rain and can infect new leaves.



Management:

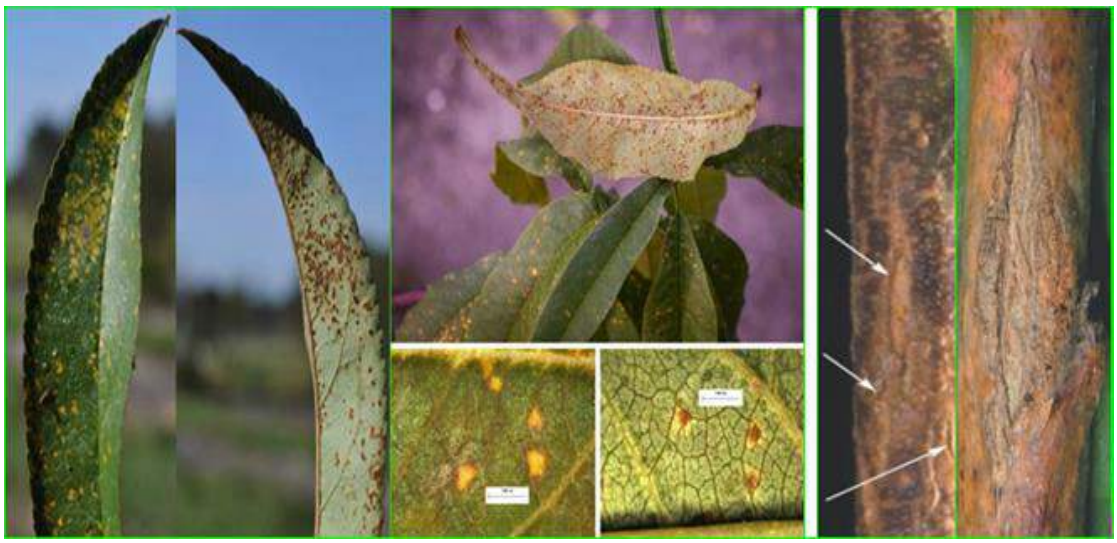
- Spraying of copper oxy chloride 0.3% or Carbendazim 0.1% or Chlorothalonil 0.1% either before bud swell in spraying or after general leaf fall in autumn.
- Resistant varieties red heaven, favorite, modeline pouyet.

2. Rust : Caused by *Puccinia pruni- spinosae*.

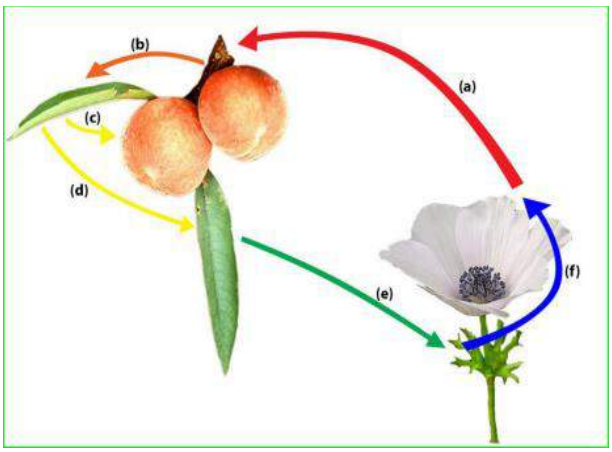
Symptoms:

- Leaf lesions are pale yellowish-green spots on upper and lower leaf surfaces that become bright yellow, angular spots from the top and blisters with rusty spores on the bottom.
- Leaf cankers turn necrotic, causing leaf drop and early bloom in winter.

- Affected leaves remaining on the tree contain overwintering structures that allow infection in the spring.
- Symptoms on young stems start as water-soaked lesions, which develop into splits with uredinia (flacks) formation. Twig cankers appear as blisters, with splits measuring up to ¼-inch long usually visible on the upper side of the twig. After the twig lesions emerge, rusty brown, powdery masses of spores (urediniospores) are produced in the cankers.
- On immature, yellow, or green fruit, lesions first develop as small, brownish spots (about 0.1 inch diameter) with green halos. When the fruit matures and develops darker skin color (blush), these lesion halos become greenish-yellow. The lesions are sunken and extend several millimeters into the fruit. Numerous infections may develop on each fruit, and these can lead to secondary infections by other fungal species and cause fruit decay



Disease cycle :



PSI: Survives as perennial mycelium in the underground stems of *Anemone coronaria* which serves as alternate host on which the fungus produces its aecial stage. Can also survive as uredosori on twigs.

SSI: Wind borne uredospores.

In a sexual life cycle, infected peach leaves can produce teliospores, which may overwinter on a peach tree until the next the growing season. If so, teliospores can germinate to produce basidiospores, which are capable of infecting alternate host, poppy anemone (*Anemone coronaria*) of peach rust where basiodiospores are infective and aecio spores are formed.

Then, aecio spores can infect peach trees and continue the cycle. Aeciospores can re-infect a peach tree and produce rust lesions with urediniospores in the spring

In the asexual cycle, infected leaf tissue erupts, producing a rust pustule under the leaf. A single rust pustule on the leaf can produce thousands of uredioniospores, which can reinfect peach leaves in favorable conditions.

In subtropical conditions, urediniospores overwinter in twig cankers and pustules under the leaves, producing the primary inoculum for infection in the next growing season.

Management :

1. Destruction of alternate host

2. Spraying with wettable sulphur/ Zineb 0.2%.
3. Spraying of ergosterol biosynthesis inhibitors like Propiconazole or Mycobutanil etc.

3. **Scab/ Freckle/ Black spot:-** *Cladosporium carpophilum*

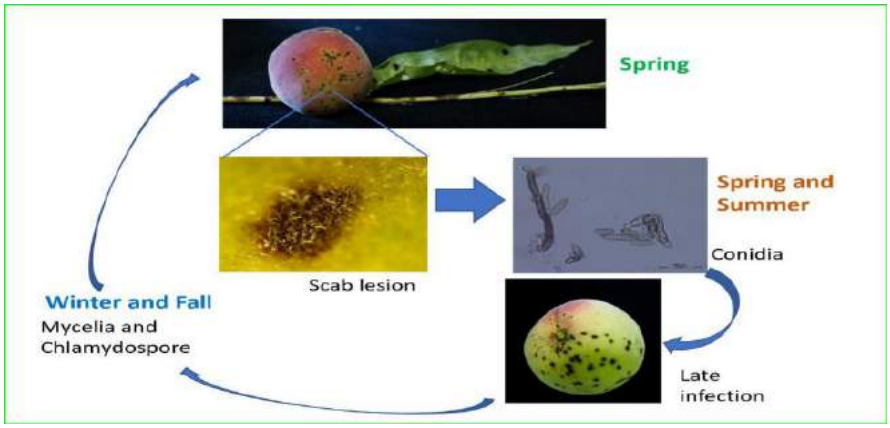
Symptoms :

- Leaves, twigs and fruits are affected.
- Leaf infections appear as inconspicuous chlorotic (yellowing) spots on the undersides of leaves. When the necrotic (dead) tissue falls out, it leaves shot holes in the leaves. Premature defoliation (leaf drop) can occur in some instances
- Infection in young, green shoots commonly begins with small, slightly raised, reddish-grey oval or circular lesions approximately 0.08 in (2 mm) in diameter. As shoots mature, the lesions expand and develop dark brown borders.
- The fungus overwinters in these twig lesions and produces tufts of olive spores in the spring of the following season, triggering a new infection cycle.
- Small, green to olive circular spots on surface of fruit, usually close to stem end, which enlarge, darken and develop a green or yellow halo; Lesions may coalesce and cause fruit surface to crack.
- When mature spots coalesce form velvety blotches.
- Fruits become abnormal in shape, crack and lose quality

Disease cycle:

PSI: Fungus overwinters in lesions on twigs. Conidia, mycelium and chlamydospsores for its dispersal and survival structures.

SSI: Conidia spread by air and water splashes.



Peach scab can overwinter as mycelia (filamentous part of the fungi) in lesions or as chlamydospores (large, thick-walled structures) on vegetative tissue or in the bark of 1-year-old shoots. Chlamydospores are the primary source of inoculum in an orchard. During the spring and summer, conidia are produced when relative humidity is at least 100% for 24 hours, and temperatures exceed 16°C. The conidia (spores) are spread by wind or by rain splash. They can also be spread by irrigation systems such as those used for overhead frost protection during the early spring. Wind dispersal is relatively minor compared to rain/ irrigation splash, the major means by which fungal spores are spread.

Management :

- Pruning helps in circulation of air and hence the chances of infection are less.
- Avoiding selection of low lying areas for planting.
- Spraying Bitertenol, a week prior to bloom and repeated 15-20 days later to boost yields.
- Spraying of 5% urea solution to the fallen leaves

Spraying schedule

- a) Silver tip to green tip- Mancozeb @ 4g/litre water
  b) Pink bud stage- Carbendazim @ 1g/ litre water.
  c) Petal form stage- Bittertenol
  d) Fruit stage-(pea nut stage) – Capton @ 3g/ litre water.
  e) Walnut stage or 40 days before harvest –Hexaconozole @1.5g /litre water.
- Biological agents like *Athelia bombacina*, *Chaetomium globosum*

- Alternaria leaf spot :- *Alternaria alternata*
- Anthracnose :- *Colletotrichum acutatum*
- Armillaria root rot :- *Armillaria mellea*
- Brown rot or stone fruit rot :- *Monilinia fructicola, M. laxa* or *Sclerotinia sclerotiarum*
- Ceratocystis canker :- *Ceratocystis fimbriata*
- Leaf curl :- *Taphrina deformans*
- Phytophthora root and crown Rot :- *Phytophthora cactorum*
- Powdery mildew :- *Podosphaera tridactyla* var. *tridactyla, Sphaerotheca pannosa*
- Scab :- *Venturia carpophila*
- Verticillium wilt :- *Verticillium dahliae*
- Rust or stone fruit rust :- *Tranzschelia discolor* .
- Bacterial canker :- *Pseudomonas syringae* pv. *syringae*
- Bacterial spot :- *Xanthomonas campestris* pv. *pruni*
- Crown gall :- *Agrobacterium tumefaciens*
- Almond leaf scorch :- *Xylella fastidiosa*

LIST OF DISEASES OF PLUM

Plum pocket :- *Taphrina maculans*.
Wilt :- *Verticillium albo-atrum*
Leaf curl :- *Taphrina deformans*
Bacterial canker :- *Pseudomonas syringae pv. syringae*
Bacterial leaf spot :- *Xanthomonas arboricola* pv.*pruni*
Line pattern :- *Plum American line pattern virus*.
Plum mosaic :- *Plum line pattern virus & ring spot virus*
Creamy-white spot :- *Creamy white spot virus*.

LIST OF DISEASES OF PEAR

- Leaf spot :- *Mycosphaerella chaubattiensis*
- Rust- :- *Gymnosporangium sabinae*
- Fruit decay :- *Botrytis cinerea, Pencillium expansum, Trichothecium roseum, Glomerella cingulata*

Bacterial Diseases

- Crown gall :- *Agrobacterium tumeficians*
- Bacterial leaf spot :- *Xanthomonas arboricola* pv.*pruni*
- Bacterial canker :- *Pseudomonas syringae* pv. *syringae*

Viral Diseases

- Ring spot :- *Prunus Necrotic Ring spot virus* (*PNRSV*), Ilarvirus
- Dwarf :- *Prunus Dwarf virus* (*PDV*)- Ilarvirus

LIST OF DISEASES OD STRAWBERRY

Septoria leaf spot :- *Septoria aciculosa*
Leaf and Stem blight :- *Mycosphaerella fragariae*
Rust : - *Puccinia fragariae*
Botrytis rot :- *Botrytis cinerea*
Anthracnose :- *Colletotrichum acutatum*
Angular leaf spot :- *Xanthomonas fragariae*

LIST OF DISEASES OF WALNUT

- Anthracnose :- *Marssonia juglandis*
- Phytophthora root rot :- *Phytophthora citricola, P. cinnamomi*
- Leaf spot :- *Mycosphaerella juglandis*
- Fusarium canker :- *Fusarium solani, F.oxysporium*
- Cylindrocladium root rot :- *Cylindrocladium scoparium*

Bacterial blight :- *Xanthomonas juglandis*

DISEASES OF JACKFRUIT

1. Die back : *Botryodiplodia theobromae*

2. Rhizopus rot : *Rhizopus atrocarpi*

1. Die back : *Botryodiplodia theobromae*

Symptoms:

- Shrivelling and curling of leaves.
- Discoloration of bark of the twigs.
- Exudation of gum from the infected twigs.
- Die back of twigs and branches.

Disease cycle:

PSI: Survives in infected plant parts.

SSI: Wind borne conidia.

Management:

- Pruning and burning of infected twigs and branches.
- Spraying of Carbendazim 0.1% or Benomyl 0.1% in rainy season.

2. Rhizopus rot: *Rhizopus atrocarpi, R. arrhizus*

Symptoms:

- Brown to black spots from the petiole end of the fruits.
- Infection spreads and covers the entire fruit.
- Fruits gets rotted and mummified.
- Post harvest rotting is also observed.
- Diseased fruits are covered with grayish mycelia growth consisting of sporangia and sporangiospores.

Disease cycle:

PSI: Survives as saprophyte in soil, compost heaps etc

SSI: Wind borne conidia.

Management:

- Removal and destruction of infected fruits.
- Orchard sanitation.
- Spraying of Carbendazim 0.1% or Benomyl during the fruit development.
- **The disease can be minimized by improved sonication near the plants.**

DISEASES OF PINEAPPLE

1. Heart or Stem Rot or top rot : *Phytophthora nicotianae* var. *parasitica*
2. Root rot : *Phythophthora cinnamomi*
3. Base (butt) rot/white spot or soft rot : *Ceratocystis paradoxa*
4. Wilt : *Pineapple mealybug wilt associated virus (PMWaV)*

Minor diseases

4. Anthracnose : *Colletotrichum ananas*
5. Leaf Blotch : *Pestalotia microspora*
6. Fruitlet core rot/Black spot/ leathery pocket and inter fruitlet corking : *Penicillium funiculosum, Fusarium subglutinans*
7. Fusariosis : *F. subglutinans (G. fujikuroi)*

Bacterial diseases

1. Pink disease : *Erwinia herbicola, Gluconobacter oxydans* and *Acetobacter aceti*
2. Fruit collapse/ Bacterial heart rot : *E. chrysanthemi*
3. Marbling : *Acetobacter peroxydans*

Viral Diseases

1. Yellow spot : *Tomato Spotted Wilt Virus*

1. Heart rot : *Phythophthora parasitica*

Heart or stem rot or top rot

More in high rainfall areas with cool climate.

Disease cycle:

PSI: Oospores surviving in soil and planting material

SSI: Zoospores spread through rains and irrigated water.

Symptoms:

- Plants of all ages are attacked, but three to four months old crown plantings are most susceptible. Fruiting plants or suckers on ratoon plants may be affected.
- The colour of the heart leaves turn yellowish green or light coppery brown from normal green colour.
- Later, the heart leaves wilt (causing the leaf edges to roll under), turn brown and eventually die.
- Once symptoms become visible, young leaves are easily pulled from the plant, and the basal white leaf tissue at the base of the leaves becomes water-soaked and rotten with a foul smell due to the invasion of secondary organisms
- Central leaves come out easily when pulled.
- Symptoms spread from crown to the fruits (stem) resulting in soft rot.
- The growing point of the stem becomes yellowish-brown with a dark line between healthy and diseased areas.
- Fruit development is arrested and entire plant wilts from the tip downward .
- The fruits exhibits spongy texture and salty to taste
- When the plant is pulled out the roots appear dark and in the process of decaying.

Management:

- Use of healthy planting material.
- Avoidance of deep planting.
- Dipping of suckers in 1% Bordeaux mixture helps in reducing the disease incidence.
- Planting of suckers in well drained soil can minimize the disease incidence
- Planting on raised beds at least 25 cm hight or more
- The addition of sulphur which results in reduced sporangial formation.
- Dipping plant material in Fosetyl Al 0.2%.
- Spraying or drenching with Fosetyl Al 0.2% or Bordeaux mixture 1% or metalaxyl (Ridomil).
- Fosetyl aluminium is used very effectively as a pre-plant dip.
- Metalaxyl has also been shown to be very effective for heart-rot control as a replant ‘seed-piece’ dip.
- The cultivar designated ‘53–323’ with resistance to *P. cinnamomi* was a cross between ‘Smooth Cayenne’ and F-236.

- ‘59–656’, is resistant to both *P. cinnamomi* and *P. parasitica*

2. Base rot : *Ceratocystis paradoxa* {*Thielaviopsis paradoxa* (IPS)}

Base (butt) rot/ watery blister/ or Black rot or Thielaviopsis fruit rot
Also called fruit and leaf rot.

- The disease is prevalent in warm with wet weather conditions.
- Serious losses occur when the suckers are kept in a damp place or when planting is followed by prolonged wet weather condition.

Disease cycle:

PSI: The fungus is a wound parasite. Survives as black spores in soil. Produces chlamydospores.
SSI: Secondary spread is by wind or rain splashed conidia.

Pathogen

C. paradoxa mycelium is dark grayish green growth consisting of septate.
Endoconidia were hyaline, cylindrical and formed endogenously.
Aleurioconidia were thick-walled ellipsoidal, pyriform, truncate at the base, golden-brown and borne singly or in chains and were intercalary, lateral or terminal on hyphae.
Perithecia were blackish to brown-blackish in colour, globose to subglobose with a characteristically long neck.

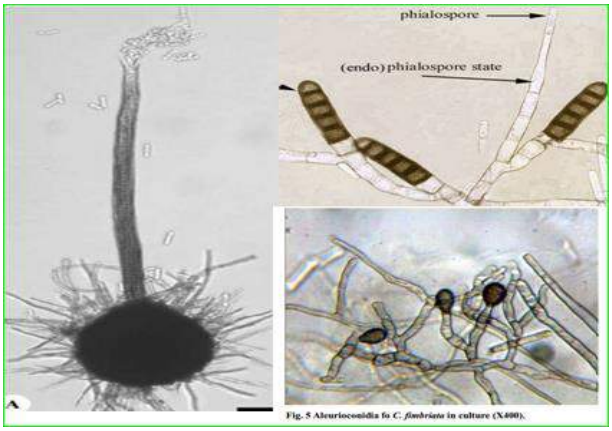


Fig. 5 Aleurioconidia fo C. fimbriata in culture (X400).

Symptoms:

- Symptoms are seen only on crowns, slips and suckers before or immediately after planting.
- The leaves show grey spots with dark margin and the spots turn olive brown or white.
- With advance of the diseases tissues dry and leaves become destroyed.
- A grey to black rot of the soft butt tissue develops, **only stringy fibers remain** decay of the butt is followed by wilting of the foliage and the diseased plant break off at low level and a cavity at the base of the stem.
- Finally skin flesh and core disintegrated.
- The fruit decay is accompanied by **a sweetish odor**, during such times the fungus is found on rotting tops and sucker left lying in heaps in damp situation.
- Unlike *Phytophthora* heart rot, the young leaves remain firmly attached to the top of the stem Infected plants can easily be broken off at ground level.

Management:

- Curing of plant material for 2-3 days in the sun prior to planting.

- Dipping of planting material in fungicidal solution of Captafol @13.5 kg/ha.
- Fruit dip after harvest in TBZ 0.1 % or 0.2% Benomyl.
- Careful handling of the fruits to avoid injuries.

3. Wilt : *Pineapple mealybug wilt associated virus*

The disease is also called as Mealy bug wilt.

Mealybug wilt of pineapple (MWP) was first described in Hawaii in 1910

Survival and spread

- This disease involves an association with virus particles, pink mealy bug (*Dysmicoccus brevipes*) and *Dysmicoccus neobrevipes* (gray mealy bug) vectors, and ants which spread the mealybug vectors.

Etiology

Family *Closteroviridae*

Genus *Ampelovirus*,

PMWaV

Long, flexuous, rod-shaped virus +ssRNA

Symptoms:

- The early symptoms are a slight reddening of inner leaves about halfway up the plant.
- Leaves develop characteristic bronzing starting from third or fourth whorl on words.
- The leaf colour then changes from red to bright pink and browning of the tips with downwards curling of the margin.
- The pink colour becomes more pronounced and leaves from the top dry downwards.
- Leaf tips curl tightly back.
- Central whorl of leaves is upright without any turgor.
- In quick wilt, the drying and wilting symptoms begin at the leaf tips which turn a reddish-yellow or pinkish color, This symptom does not develop in plants afflicted by “slow” wilt.
- Fruit that are markedly smaller than fruit from healthy plants.
- The bright pink colour becomes dull and the root system collapses.
-

Management:

- Use of disease free planting material
- Removal of infected planting material (disease incidence is more than 3%)
- Hot water treatment of planting material at 50-60 ^{o} C for 1 hour.
- Application of Disyston granules at plant base @ 1.5 kg.
- Treatment of planting materials by insecticide dips was found to be effective for the pre-planting control of mealybugs
- Spray applications of systemic insecticides, such as Diazinon and other contact insecticides
- Insecticidal baits are the most common and effective way to control ants in pineapple fields
- Biological controls for the pink and grey pineapple mealybugs

DISEASES OF COCONUT

1.	Bud rot	:	*Phytophthora palmivora*
2.	Ganoderma root rot	:	*Ganoderma applanatum, G. boninense, G. lucidum, G. tornatum*
3.	Stem bleeding	:	*Ceratocystis paradoxa*
4.	Grey blight	:	*Pestalotiopsis palmarum*
5.	Tatipaka	:	Phytoplasma

1. Bud rot : *Phytophthora palmivora*

- First reported by butler in 1906
- Coconut, arecanut and palmyrah palm are attacked
- Datepalm is immune to the disease
- Bud rot affects the palms at various stages of growth but it is more severe in young palms of 5-20 years.
- As the name indicates, the ultimate effect is rotting of terminal bud due to infection in or near the bud

Disease Cycle:

- **PSI:** Oospores, mycelium and chlamydospores in soil/or infected plant parts or debris
- **SSI:** Wind blown or rain splashed zoospores or rhinoceros beetle, *Oryctes rhinoceros*.

Favourable conditions

- Seedlings and tender palms
- More infected palmyrah palms in the vicinity
- Rhinoceros beetle infestation
- High relative humidity (>90%) and temperature of 18-20 ^{0}C and high and frequent rainfall

Symptoms:- Severe on seedlings in nursery and young palms in the main field, particularly during monsoon when the temperature is low and humidity is very high.

On seedlings

- The earlier symptom is the yellowing of one or two younger leaves.
- Black spots appear on spindle leaves.
- Basal tissues of the leaf rot quickly and can be easily separated from the crown.
- In seedlings, the spear leaf turns pale and comes off with a gentle pull
- Infection spreads to the older leaves, causing sunken spots covering the entire leaf blade.
- Internal blackening or browning of tissue

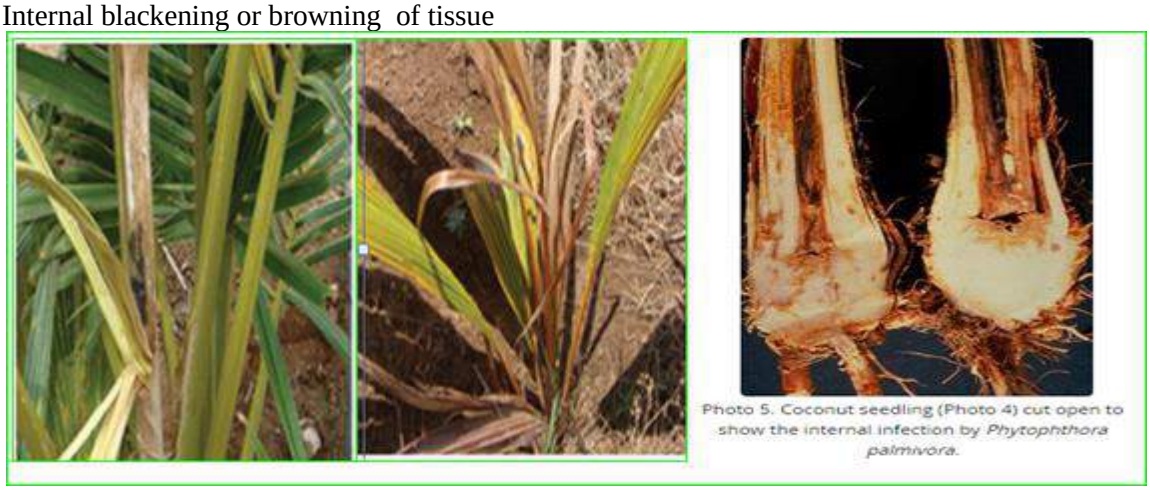


Photo 5. Coconut seedling (Photo 4) cut open to show the internal infection by *Phytophthora palmivora*.

On young palms

- The first indication of the diseases is seen on the central shoot of the tree (spindle).
- The heart leaf shows rotting at the base, which becomes brown instead of yellowish brown.
- This is followed by drooping and breaking off the heart leaf.
- With the progress of diseases, more number of leaves get affected with loss of lusture and turn pale yellow.
- Water soaked sunken lesions develop on leaf blade.

- The entire base of the crown may be rotten emitting a foul smell and can be pulled out easily from crown.
- The leaves fall in succession starting from the top of the crown.
- The leaf falling and bunch shedding continue until a few outer leaves are left unaffected.
- But within few months the infection leads to complete shedding of leaves, within subsequent wilt and death of the tree.
- The leaves and sheath in the central spindle fall off leaving an outer whorl of green leaves
- Fungus also attacks female buttons and causes button drop particularly in periods of high relative humidity and rainfall
- Fungus attacks tender nuts causing fruit rot.
- Infected nuts become heavier because of water soaked fibre and fall down

Management:

- Removal and burning of badly infected palms
- Infected trees can be saved, if the disease is detected early by removing the infected tissues (surgery) and protecting by smearing the growing head with Bordeaux paste
- Spray copper fungicides, Bordeaux mixture 1% or Copper oxychloride 0.3% three times at the beginning, middle and cessation of monsoon as a prophylactic measure
- Treated wound should be given protective covering (polythene covers) till normal shoot emerges.
- Placing a mixture of $CuSO_4$ + NaCl (1: 3-5 parts) in a cloth bag in the crown or above bud
- Stem injection or root infusion with systemic fungicides like Aliette (Fosetyl Al), Ridomil Mz.
- *Bacillus subtilis, B. macerans, B. amyloliquifciciens* and *Myrothecium roridum*
- The Hybrid PB 12 is tolerant (Malayan Dwarf x West African Tall)
- Manage rhinoceros beetle

2. **Ganoderma root rot** : *Ganoderma applanatum, G. boninense, G. lucidum, G. tornatum etc.*

The "Basal Stem Rot (BSR) disease" of coconut is called by various names in India viz. Thanjavur wilt (Tamil Nadu), Anab e Roga (Kamataka) and Ganoderma wilt (Andhra Pradesh)
Prevalent in light soils in Srikakulum, Visakhapatnam, Vizianagaram, East Godavari and West Godavari districts particularly by the side of river beds

- Dry soil and weather conditions

Disease cycle:

PSI: Survives as basidiospores and spreads through root contact or saprophytic mycelium in soil and infected debris or several host tree species like mango, jackfruit, citrus, arecanut, poplar and several forest trees.
SSI: Basisdiospores spread by irrigation water, air borne or rain splashed basidiospores, by bark eating caterpillars carrying basidiospores.

Pathogen
The fungus produces a semi circular basidiocarp (bracket), which is attached to the tree with a stalk. The bracket is very big about 10-12 cm diameter and woody. The upper surface is tough, shining, light to dark brown or almost black with concentric furrows. The lower surface is white and soft with numerous minute pores. These pores represent the opening of the hymenial tubes, which are lined with basidia and basidiospores. Basidiospores are oval, brown and thick walled.
Indicator plants- Red gram plants shows bark splitting symptom as the identification mark for basal stem rot disease.
Bengal gram plants shows withering, yellowing and drying of lower set of leaves followed by upper leaves as the identification mark of basal stem rot disease or Ganoderma wilt disease of coconut
Symptoms: Infected palms are predisposed to infestation by shot hole borer *Xyleborus testaceous* which bores infected trunks from which dry powder comes out

- All stages of the palm growth are attacked by the pathogen, The fungus is soil-borne and infects the roots.
- Roots are infected and gets rotted and later spread to the trunk.
- The most usual symptoms are yellowing, withering and drooping of the outer fronds which remain hanging around the trunk for several months before shedding.
- The new fronds produced become successively smaller and yellowish in colour which do not unfold properly
- More often the spindle is blown off leaving the decapitated stem.
- The wilting plants also show bleeding patches near the base of the trunk. As the infection advances, fresh bleeding patches appear above the old once, up to 3-5 meters height.
- A brown gummy liquid oozes out from the cracks in the tree which slowly result in the death of outer tissues.
- Production of fungal fruiting bodies (brackets) along the side of the basal trunk particularly during periods of heavy rains.
- The roots of wilting trees show discoloration and severe rotting.

Management:
- Avoid deep ploughing or digging
- Arrange separate irrigation channel to each palm basin.

- Digging trenches to isolate diseased palms with 50 cm wide, 1 m deep, 2-3 m away from the diseased plant base.
- Application of biocontrol agents.
- Green manuring with sunnhemp/sesbania to improve moisture retention capacity of soil and multiplication of antagonistic microflora.
- Dig out infected palms with entire root system and destroy by burning.
- Disinfect the pit formed due to removal of infected palms by burning trash and drenching with Bordeaux mixture 1%
- Replanting pits should be filled with FYM, 5 kg neem cake and *Trichoderma viride*
- Apply FYM (200 kg) and neem cake (5-10 kg) inoculated with *Trichoderma viride* to each palm during June – July months
- Also apply 2 kg superphosphate and 3 kg muriate of potash and 1 kg urea to each bearing palm, in two splits, one in July and other in November to maintain the vigour of the palm
- Treat basin of each palm with 40 liters of Bordeaux mixture 1%, once every year during August – September
- Apply Aureofunginsol 2g+Copper sulphate 1g in100 ml of water or Tridemorph 2ml/100 ml of water through stem injection or root feeding at quarterly intervals for one year.
- Bark eating caterpillars should be managed by mixing 5 ml/lit Chloropyriphos and applied on the infested area with a brush at 15 days gap.

3. **Stem bleeding** : *Ceratocystis paradoxa*
*Thielaviopsis paradoxa (*IPS)

Syn: *Ceratostomella paradoxa, Ophiostoma paradoxa, Chalara paradoxa*

Disease cycle:

PSI: *T. paradoxa* is a soil borne fungus and entry takes place through growth cracks and wounds where it multiplies, producing endoconidia and chlamydospores.

SSI: Conidia dispersed by irrigation water, insects, wind

Favourable conditions

- Heavy rainfall followed by drought
- High water table and highly alkaline soils
- Cracks and insects wounds and wounds caused during harvest and leaf removal
- Poor fertilization and poor maintenance of orchard

Pathogen

- *C. paradoxa* also produces hyaline perithecia with a long neck base is ornamented with knobbed appendages and ostiole is covered by numerous pale-brown, erect, tapering hyphae.
- Asci are clavate and ascospores are hyaline ad ellipsoid.
- The fungus produces two type of conidia.
- Endoconidia (Microconidia) are thinwalled, hyaline and cylindrical in form and are produced endogenously inside the long cells ruptures when mature and release the microcondia in long chain.
- Aleurioconidia (Macroconidia) are produced on conidiophores singly or in chains. They are spherical and dark green in colour.

Symptoms:

- Trunks of tender palms are more attacked, water soaked patch on trunks appears initially which later increases in size
- Reddish brown ooze from the cracks in the patch is a characteristic symptom. The fluid dries up to form black encrustations with brownish orange margins.
- The patch progresses linearly upward usually in a direction that is opposite to phyllotaxis and they are coalesce, involve larger areas.
- When the severely decayed patches are pierced a yellowish brown fermented fluid comes out, large cavities are formed in diseased trunk (except vascular bundles)
- Infected trunks are usually infested with *Diocalandron* beetle,Infected trunks are weakened because of decay of tissues and are prone to be felled by moderate winds

Common symptoms

- Leaf size reduces leading to reduced crown size
- The rate of leaf production slows down
- Bunch production is reduced
- Nuts and buttons shed heavily

- As a result of extensive damage to the stem, uniform yellowing and dropping of leaves leads to complete death of the palm

Management:

- Avoid any mechanical injury to the stem
- Provide proper drainage during rainy season
- Affected bark should be chipped off and paint with 5.0% calixin (5 ml Calixin in 95 ml water) and left to dry for one day.
- The wound should be covered with molten coal tar or or Bordeaux paste.
- In older trees, after draining the fluid, cavities should be filled with a mixture of tar coal, sand and cement to strengthen the trunks and to prevent secondary damage
- Coat the lower part of trunk with coal tar or Bordeaux paste to prevent the disease
- Application of 50 kg of organic manure+ 5 kg neem cake + *Trichoderma* to the basin during September
- Root feeding with Carbendazim 0.1% or Tridemorph 6ml in 25 ml of water.
- Soil application of phorate granules or Carbofuran granules @ 6-8 kg/ha.
- Coconut stem boring insects should be controlled by applying Carbaryl 50% WP on the trunk at 3g per litre
- West Coast Tall was found more resistant to stem bleeding

4. Grey blight/papery blight : *Pestalotiopsis palmarum*

Disease cycle:

PSI: Survives as conidia in crop residue/infected palms.

SSI: Wind/rain spread conidia.

Favourable conditions

Neglected gardens with palms suffering from **potassium** deficiency

Heavy weed infestation in the basins of palms

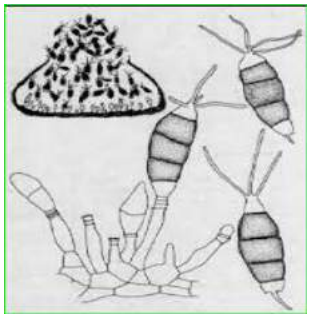
Pathogen

The fungus produces conidia inside the acervuli.

The acervuli are black in colour, cushion shaped and sub epidermal and break open to expose conidia and black sterile structures, setae.

The conidiophores are hyaline, short and simple, bear conidia at the tip singly.

The conidia are five celled, the middle three cells are dark coloured, while the end cells are hyaline with 3-5 slender, elongated appendages at the apex of the spore called as sepulae.



Symptoms:

- Disease symptoms develop in mature leaves
- Outer whorl of mature leaflets consist of minute grayish white spots with a brown band.
- Gradually these dark brown margins surrounded by yellow halo
- Spots coalesce (merge) to form irregular necrotic patches.
- Infected leaf blade completely dries up and shrivels giving a blighted or burnt appearance.

- Large number of globose or ovoid black acervuli appears on upper surface of affected leaves

Management:

- Remove older affected leaves and burn
- Correct potassium deficiency by applying recommended dose of potassium @ 3 kg /palm
- Keep basins of palms weed free
- Spray foliage with BM 1% or Copper Oxychloride or Captan 0.25%

5. Tatipaka : Phytoplasma

- The disease first appeared in Tatipaka village, 'Razole taluk' of East Godavari district, Andhra Pradesh, following a cyclone in 1949.
- It generally occurs in heavy black deltaic soils than in sandy, sandy loan and red loam soils.
- **Gangabondam** to be free of disease and the incidence was high on the **East Coast Tall**.

Disease cycle:

- Disease is spread by infected nuts and seedlings.

Symptoms:

- The disease-affected palms generally bear profusely for 2-3 years before the expression of visual symptoms
- With the onset of disease there is a reduction in both number and size of leaves. The leaves exhibit characteristic chlorotic water soaked spots and the fronds bend abnormally sometimes twisting in loops.
- Development of abnormally large crown with dark green inner leaves. Subsequently crown becomes smaller.
- Leaves of infected palms show chlorosis and newly emerging leaves become smaller leaflets of fronds in infected palms do not unclasp after emergence and hence leaves give a fascinated appearance due to improper unfolding of leaflets
- Leaves turn pale and bend abnormally in the centre resembling a bow and are small sized.
- In the advanced stage with the narrowing of leaflets and reduction in size of crown, the affected palm looks like a date palm in appearance.
- The spathes produced are very small with very few rachiliae
- Affected palms produce smaller bunches contain a mixture of normal and atrophied nuts or barren nuts or bear very few nuts that have soft pericarp
- The atrophied nuts are barren with thinner spongy mesocarp with or without shell, copra and nut water.
- The undersized nuts show longitudinal cracks with occasional gumming.
- Severely infected palms are barren with sickly crowns and slight tapering trunk (pencil point).Trunk below the crown becomes 'S' shaped.

Management:

- Destruction of affected palms.
- Avoid using nuts obtained from infected trees.
- Procure healthy seedlings.

DISEASES OF OIL PALM

A. Nursery diseases

Botryodiplodia leaf spot	:	*Botryodiplodia theobromae*
Glomerella leaf spot	:	*Glomerella cingulata*
Pestalotiopsis leaf spot	:	*Pestalotiopsis palmarum*
Curvularia leaf spot	:	*Curvularia lunata* *Curvularia geniculata*
Corticium leaf rot or leaf rot	:	*Corticium solani*
Alternaria leaf spot	:	*Alternaria alternata*
Cercospora leaf spot/Freckle	:	*Cercospora elacidis.*

B. Field diseases

Basal stem rot	:	*Ganoderma boninense*
Bunch rot	:	*Marasmius palmivora*
Bud rot	:	*Phytophthora palmivora*
Spear rot	:	***Pectobacterium*** *(Erwinia) chrysanthemi*

A. NURSERY DISEASES:

1. *Botryodiplodia* leaf spot : *Botryodiplodia theobromae*

Appearance of light brown spots on the distal portions of the leaves and the leaves become pale, extremely brittle and entire leaf gets killed.

2. *Glomerella* leaf spot : *Glomerella cingulata*

Yellow to brown water soaked lesions between veins which elongate and contain acervuli in the centre.

3. *Pestalotiopsis* leaf spot : *Pestalotiopsis palmarum*
Lesions with large irregular shape and orange red in colour which later become necrotic with dark grey margin and light brown centre.

4. *Curvularia* leaf spot : *Curvularia lunata, Curvularia geniculata*
Small circular translucent yellow spots which later turn dark brown with sunken centre and yellowish orange halo.
Persistence of primary lesions within the dead tissue is a diagnostic feature of this leaf spot.

5. *Corticium* leaf rot or leaf rot: *Corticium solani*
Irregular pale green patch at the base of the spear with brown zone which turns dark brown before drying.
Extensive dessication and fragmentation of dead tissues is characteristic symptom.

6. *Alternaria* leaf spot : *Alternaria alternata*
Leaves show distinct reddish brown localized spots which enlarge to become irregular. Later the spots become dark brown with blackish tinge.

7. *Cercospora* leaf spot : *Cercospora elacidis*.
Water soaked yellow lesions appear on the upper leaf surface and on the lower surface. These lesions are light brown in colour. The lesions on the upper surface enlarge and become irregular in shape.

Management of nursery diseases:

- Pruning of severely affected leaves.
- Destroy the badly affected seedlings.
- Prophylactic sprays with mancozeb 2 g/lit or captan or ziram 1 g/lit or thiram 2g /lit
- Spraying of carbendazim 1 g/litre.
- **Copper fungicides should not be used because of the extreme susceptibility of oil palm seedlings to copper burn (scorching)**

Bud rot : *Phytophthora palmivora*

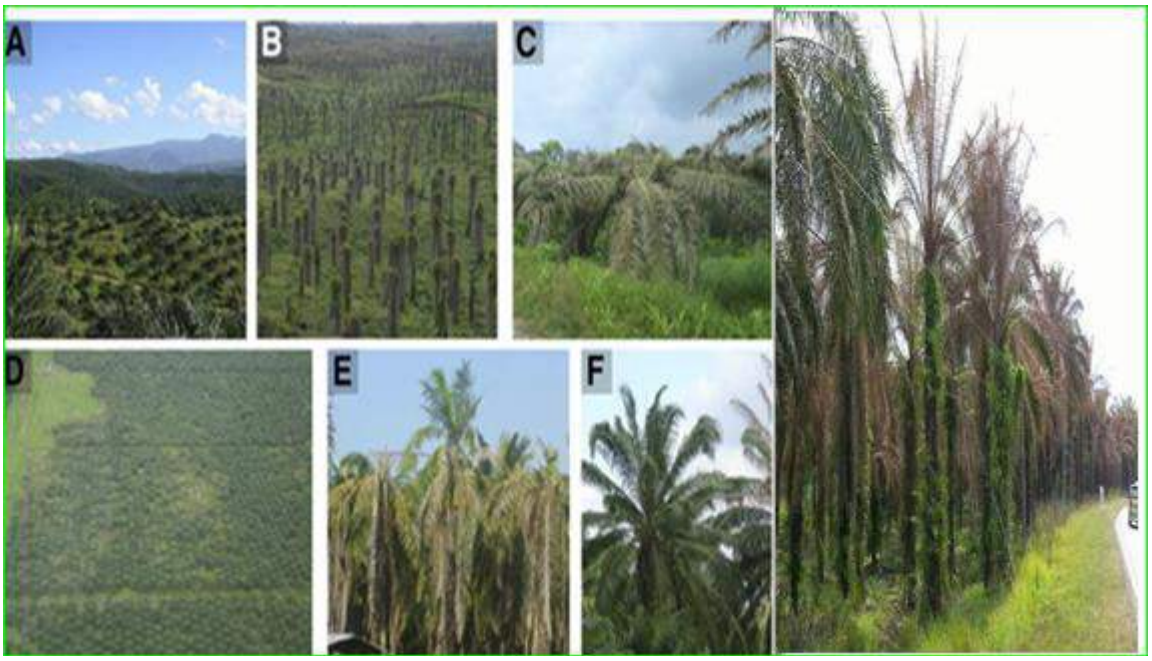
Survival and spread

PSI: Oospores or dormant mycelium in soil and infected debris

SSI: Sporangia and zoospores disseminated through rain splash, wind and rhinoceros beetle, *Oryctes rhinoceros*

Symptoms

- Palms of all ages are susceptible to the disease, but it is more severe in young palms of 5-20 years.
- The first indication of the diseases is seen on the central shoot of the tree (spindle)
- The heart leaf shows discolorations which become brown instead of yellowish brown.
- This is followed by drooping and breading off the heart leaf.
- With the progress of diseases, more number of leaves get affected with loss of lusture and turn pale yellow
- The entire base of the crown may be rotten emitting a foul smell, the central shoot comes off easily on slight pulling
- The leaves fall in succession starting from the top of the crown.
- The leaf falling and bunch shedding continue until a few outer leaves are left unaffected.
- But within few months the infection leads to complete shedding of leaves, within subsequent wilt and death of the tree



Management

- Removal and burning of badly infected palms
- Infected trees can be saved if the disease is detected early by removing the infected tissues (surgery) and protecting by smearing the growing cabbage with Bordeaux paste
- Spray copper fungicides, Bordeaux mixture 1% or Copper oxychloride 0.3% three times at the beginning, middle and cessation of monsoon as a prophylactic measure
- Keep a mixture of $CuSO_4$ + NaCl (1: 3-5 parts) in a cloth bag in the crown or above bud
- Manage rhinoceros beetle
- *Bacillus subtilis, B. macerans, B. amyloliquifciciens* and *Myrothecium roridum* have all been found to be checking the growth of the pathogen.

Basal stem rot : *Ganoderma boninense*

Disease cycle:

PSI: It is a soil borne fungus. Primary infection takes place through roots.

SSI: Secondary spread by Basidiospores, irrigation water and tools used for cultural operations.

Symptoms:

- Withering, yellowing and discolouration of the leaves followed by necrosis on one side of older fronds.
- Necrosis of the lower leaves and extends progressively to younger parts of the crown.
- Palm appears pale and produces excessive number of spears.
- Dessicated fronds droop or break at some point encircling the top of the trunk.
- Appearance of light brown lesions and rotting of the bole at the stem base.
- The disease produces dry rot of internal tissues.
- Formation of brackets is seen at the base of dead palm.

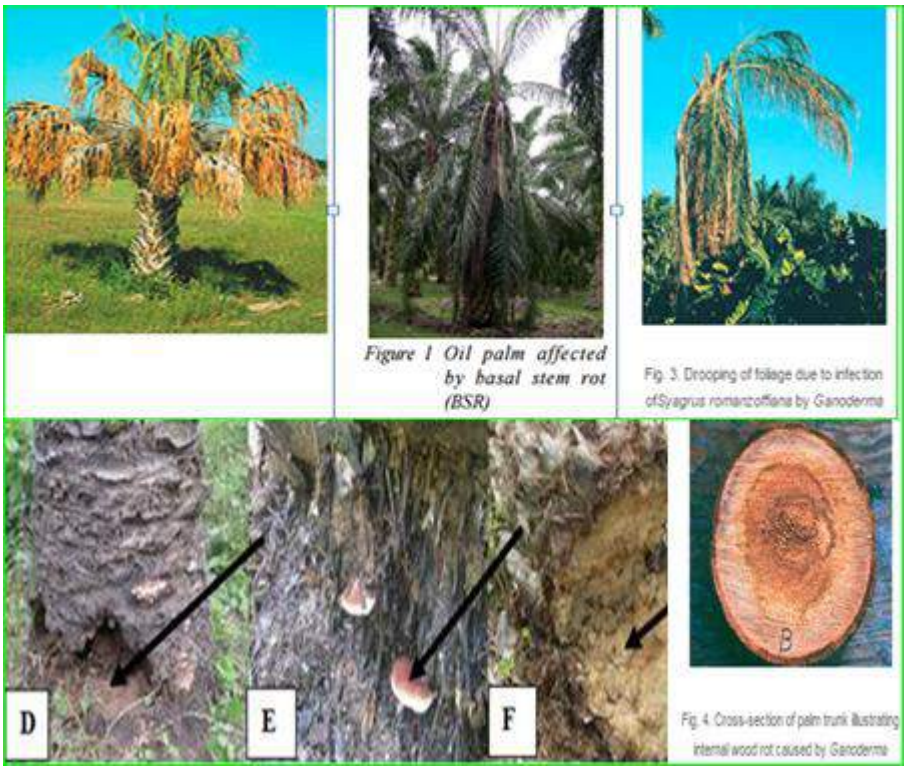


Figure 1 Oil palm affected by basal stem rot (BSR)

Fig. 3. Drooping of foliage due to infection of Syagrus romanzoffiana by Ganoderma

Fig. 4. Cross-section of palm trunk illustrating internal wood rot caused by Ganoderma

Management:

- Bleeding patchs should be chiseled and protected with Tridemorph (5 % Calixin) and subsequently with hot coal tar.
- *Ganoderma* resistant inter/mixed crops like banana.
- Digging of isolation trenches around the palm 2-3 meters away from the infected plant and 1 meter deep and 50 cm wide to save the healthy plants.
- Green manuring with Sunhemp or Sesbania to improve the soil texture.
- Application of bioagents like *Trichoderma spp.* that are antagonistic to the pathogen.
- Application of Neem cake at 5 kg/tree.

Bunch rot : *Marasmius palmivorus*

- Marasmius bunch rot disease of oil palm was first reported in Malaysia and Indonesia in 1924. In India by Reddy *et al* 1987
- This disease occur in where natural pollination is inadequate with poor sanitation, as well as during periods of wet weather and constant high humidity found in high density plantings
- Basidiocarps are capable of surviving in soil and plant tissues for several months
- Marasmioids and gymnopoids are basidiomycetes with small basidiocarps from the class of Agaricomycetes (Order: Agaricales). They are mainly white-spored saprophytic fungi.
- Apart from oil palm, basidiocarps of *M. palmivorus* have also been found on rotten wood and fruits of coconut, on leaves and stems of banana

Symptoms:

- In the early stages of infection, whitish or pinkish-white mycelial threads can be seen over the bunch surface, especially at the base particularly outer layer of the fruitlets.
- The fungus penetrates the mesocarp of the fruit and causes a soft, brown, wet rot which is sharply defined from healthy tissues.
- If affected fruits are left on the palm, the rot ultimately dries out, leaving the fibrous tissues of the mesocarp with abundant mycelial growth of the pathogen.
- During the late stage, the fruit bunch will eventually rot and fall and small, mushroom-like white basidiocarps will develop on the colonized bunches.
- The fungus is not confined to the fruit bunch of the oil palm, but is wide spread in the oil palm cultivation area, and can be found abundantly on old stems or debris of wood, fruits, and fruit bunches and also rachis of oil palm
- Shoot infection is usually characterized by the presence of whitish or pinkish white mycelial growth on the shoot and the edges of the scale leaves.

- As the disease progresses, fan-shaped mycelial mats become very conspicuous on the husk in the vicinity of the shoot.
- In the advanced stages of infection, fructifications of the fungus can be seen on the husk.
- The infection spreads, causing dry brown rot which eventually kills the shoot.

Management:

- Crown cleaning by removing aborted and dried bunches and inflorescences.
- Spraying of carbendazim 0.1% or Bordeaux mixture 1%.

Spear rot : *Pectobacterium caratovarum* (*Erwinia chrysanthemi)*

- First reported in Indonesia in the 1920s.
- Crown disease/common spear rot (CD/CSR)
- It is a lethal and infectious disease found in the oil palm plantations of Kerala (an endemic disease in Kerala state)

Symptoms: Two categories of spear rot can be recognised based on symptoms

1. Symptoms of spear rot with yellowing.

- This bacterial disease is noticed to affect oil palms of all ages.
- Yellowing of the youngest whorl of the unfolded leaves is the initial symptom.
- Yellowing starts from the tips of leaves and spreads mostly along the margins of leaflets.
- The chlorotic area later turns brown and dries up.
- Size of newly emerging leaves is reduced.
- Water soaked lesions are seen at the central portion of spear.

- These lesions extend downwards as well as upwards
- As the disease advances, new leaves become rudimentary (reduced leaf emission) and show rotting.
- As the disease advances the trunk gradually tapers
- Extensive rotting and foul smell are noticed in most of the cases. However, the rotting never extends to the meristem
- Inflorescence emergence arrested leading to loss in productivity.

2. Symptoms of spear rot without yellowing

- Water soaked lesions are seen at the central portion of spear.
- The lesions extend upwards and downwards, but only to a limited area.
- The spear on unfurling exhibits leaf rot symptoms.
- No leaf yellowing is seen in this case.

Spread:

The disease is infectious, vector borne and lethal.

Management:

- Avoid obtaining seedlings from infected areas to healthy areas.
- Tetracycline injection 500ppm to trunk reduces the disease.
- Rougueing of affected plants.
- Raising quick growing barrier trees in the border of the plantation to prevent vector movement from RWD/YLD source

DISEASES OF CACAO

1. Charcoal pod rot : *Botryodiplodia theobromae* (*Lasiodipoldia theobromae*)
2. Black pod rot : *Phytophthora palmivora*
3. Swollen shoot : *Cocoa Swollen Shoot Virus* (*CSSV*)

1. Charcoal pod rot: ***Botryodiplodia theobromae***

- Charcoal pod rot is found throughout the year. But the disease is severe during summer months
- The disease occurs on wounded pods or pods which are" under stress or overripe or weakened cocoa pods.

Symptoms:

- The infection appears as yellow spots and then become black colour on any place on the pod surface and spreads rapidly.
- The whole mass of the pod including the beans (internal tissue) becomes black in colour.
- If the infection occurs in the early stages of pod development, the beans may not develop fully and get mummified. The surface of the fruit is covered with large number of black spores giving the appearance of sooty.
- Leaves on the outer twigs yellow first and the damage may then extend along the whole branch, reaching the main trunk, eventually resulting in tree death.
- The twigs and branches of diseased trees show internal discolouration with brown streaks in the vascular tissue. White or yellowish exudates from trunks has also been reported.
- Desiccated leaves and mummified fruits remain attached to declining trees for several weeks.

Management:

- Spraying of 1 % Bordeaux mixture or 0.2% Rovral on fruits.
- Since the fungus spreads through wounds, control of insects and rodents also help in controlling the disease.

2. Black pod rot : *Phytophthora palmivora*

- The cacao pod rot caused by *P. palmivora* was first reported on the island of Luzon, the Philippines,by Reinking(1918).
- The disease may incur pod losses of upto 20 to 30% annually.
- Phytophthora is the most important pathogen of cocoa causing black pod, stem canker, seedling dieback, chupon blight and twig dieback

Disease cycle

PSI-Chlamydospores are produced asexually and can survive for months in soil or dead plant material in the absence of their host. The spores germinate when host plants are again present under favorable weather conditions.

SSI- Sporangia are produced on infected fruit, leaves, stems, or roots. They can germinate directly on the plant surface or in the soil and are capable of producing zoospores. Zoospores can swim in soil water or in water on a plant surface until they find entry into the plant.

Symptoms:

- The disease is common during rainy season
- This disease attacks several parts of the plant but the most severe damage occurs on the fruits, particularly on those that are nearly ripe.
- It produces a small, chocolate brown to dark brown circular lesions anywhere on the pod surface with a regular edge that spreads rapidly, covering the entire fruit within a few days.
- Obvious symptoms are the rotting or necrosis of pods. Internal tissues, including the beans, are colonized and shrivel to form a mummified pod. Internally it causes a brown rot.
- Infected beans quickly deteriorate and rot, rendering the pod useless.
- The pods then dry up and mummify on the tree, becoming a major source of inoculum to nearby pods, leaves, and stems.

Stem canker symptoms

- The pathogen can also infect flowers and can cause stem canker on young twigs, older branches, and the trunks of trees. The infection can spread from diseased pods towards the tree branches or trunk
- Appearance of dark brown, round to oval shaped discolored area on the bark.
- Dead and dying leaves are sometimes the first indication of branch die-back. The leaves die because the branch they are on is killed; the pathogen does not directly infect and kill the leaves.

- Cankers can form under the bark of infected stems and branches. There may be a dark spot on the bark that oozes reddish fluid. Tissue beneath the leision appeared as greyish brown water soaked lesion with a broad dark brown to black margin.
- The canker can continue to expand until it girdles and kills the branch.

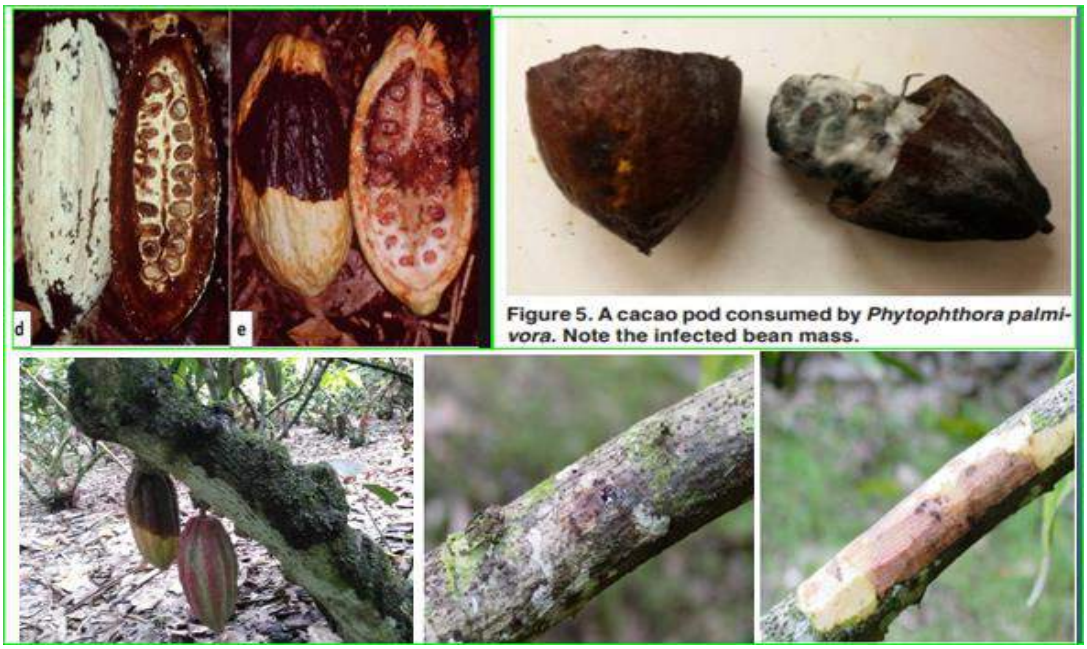


Figure 5. A cacao pod consumed by *Phytophthora palmivora*. Note the infected bean mass.

Management:

- Cultural practices like proper canopy pruning, shade reduction, weeding etc.
- Spraying of Metaloxyl or Ridomil MZ 0.2%.
- Spraying 1.0 % B.M at the onset of southwest monsoon and thereafter atleast twice during the monsoon season at an interval of 45 days.
- Avoid inter crop with areca
- Affected pods should be destroyed.
- Regular removal & destruction of infected pods at weekly intervals.
- Bio agent *Trichoderma viride*

3. Swollen shoot : *Cocoa swollen shoot virus*

- *CSSV* is a serious constraint to the production of cocoa in West Africa, particularly in Ghana where the disease was first recognized in 1936.
- Amelonado cocoa varieties are particularly susceptible to infection.

Pathogen- *Cacao Swollen Shoot Virus*

- *Caulimoviridae*, *Badnavirus*, dsDNA-RT
- Bacilliform in structure
- Disease spread: Spread by mealy bugs *Plannococcus citri* and *Pseudococcus longispinus*

Symptoms:

- 1.Reddening of primary veins or 'red vein banding' in young leaves (immature "flush" leaves)
- 2.Yellow banding along the main veins of leaves
- 3.Vein clearing in leaves, sometimes producing a 'fernlike' pattern
- 4.Chlorosis or flecking and mottling of mature leaves
- 5. Stem and root swellings-Stem swellings occur at the nodes, internodes or tips of the stem. Swellings are caused by the localized increase in secondary phloem and xylem.
- 6. Pod-Smaller, rounded to almost spherical pods. Occasional green mottling on the surface of the pods and their surface may be smoother than that of healthy pods

Management:

- Well fertilized gardens with good water relations show tolerance to the virus
- Eradication of infected plant material is the best method.
- Use of insecticides for control of insects.

DISEASES OF ARECA NUT

1. Fruit rot/Mahali/Kole roga : *Phytophthora meadii*
2. Foot rot/Basal stem rot/Anab e Roga : *Ganoderma lucidum*

1. Fruit rot/Mahali/Koleroga: *Phytophthora meadii.*

Fruit rot (commonly called kole roga in Kannada and Mahali in Malayalam. Kole raga occurs as an epidemic in the heavy rainfall of Karnataka and Kerala.

It is also called as Bud rot or Crown rot

Fruit rot of areca nut is caused by the fungus *Phytophthora arecae* and *P. heveae* were also reported to be involved in the disease.

Disease cycle:

PSI: Mycelium persisting on tree tops, Chlamydospores and Oospores formed in fallen fruits

or on dried nuts and mummified on the palm of infected inflorescence remaining in the crown..

SSI: Sporangia and sporangiospores (zoospores) spread by rain splashes and wind.

Favourable conditions

- The severity, persistence and spread of fruit rot are related to the pattern of rain.
- The disease appears usually 15 to 20 days after the onset of regular monsoon rains and may continue up to the end of the rainy season.
- Continuous heavy rainfall coupled with low temperature (20 to 23°C), high relative humidity (>90%) and intermittent rain and sunshine hours favour the occurrence of fruit rot.

Pathogen

- The fungus produces intercellur, non septate, hyaline mycelium.
- Sporangiophores are hyaline and simple or branched occasionally.
- The sporangia are hyaline, thin walled, pear shaped with a prominent papillae.
- Sporangia releases reniform, biflagellate zoospores upon germination.
- The fungus also produces thick walled, spherical oospores.
- In addition, thick walled, yellowish brown chlamydospores are also produced.

Symptoms:

- The disease first makes its appearance after monsoon period as water soaked lesions on the developing green nuts in the form of **dark green/yellowish**.
- Lesions first start at perianth region but slowly cover the entire fruit surface.
- Falling of the affected nuts covered by whitish mass of the fungal mycelium.
- Kernals of the infected fruits exhibit discoloration and dried nuts are lighter in weight.
- The infected nuts lose their natural green lusture
- Sometimes, the infected nuts may not be shed and remain mummified in the bunches. Such type of infection is known as 'dry mahali.
- Fruit stalk & inflorescence rachis are also affected. Affected nuts show vacuole inside and dark brown radial strands. Nuts of all ages are attacked and if unchecked, the pathogen invades crown causing the leaves and bunches to whither.

Management:

- Infected dried bunches, shed nuts, and dead palms should be removed and destroyed.
- Proper drainage should be provided to reduce soil inoculum.
- Dense planting should be avoided to regulate the light penetration, airflow and humidity inside the plantation.
- Developing fruit bunches should be covered with polythene bags (200 guage) before the starting of regular monsoons.
- Spraying of Bordeaux mixture 1% immediately after the first monsoon showers and again after 40-45 days.
- Metalaxyl (Ridomil) 0.5 % Fosetyl Al (Aliette) 0.2% are also effective.

- Remove the infected tissues from the crown and treat the wound/ cut end with 10% Bordeaux paste. Cover the treated bud with protective covering till new shoots emergence.
- Use Adhesive or vegetative oil as adhesive to ensure tenacity of the spray deposit on treated substrate

2. Foot rot/Basal stem rot/Anabe: G*anoderma lucidum, G. applanatum.*

Disease cycle:

PSI: The fungus is a soil borne and primary infection takes place through roots. Pathogen survives as mycelium or chlamydospores in the soil.

SSI: By Basidiospores, irrigation water and tools used for cultural operations.

Symptoms:

- The fungal invasion interrupts uptake of water and nutrients by the palm, leading to yellowing and wilting**.**
- Roots discolored, brittle, dry and give nasty smell.
- Infection spreads to the bole and internal conducting tissues.
- Brown patches appear at the base of the stem and as the disease progresses the patches increase in size and may extend up to one meter height.
- Oozing of brown colored liquid is observed.
- Discoloration and rotting of xylem and xylem parenchyma of the root and internal tissues of the basal stem is very characteristic.
- Leaves in the outer most whorl shows yellowish discolouration. In advance stage whole leaf become droop and dry.
- As the deterioration of the roots and stem increases, leaves in the inner whorl turn yellow and finally the whole crown wilts and droops and topples leaving a bare trunk.
- Tapering of the stem and reduction in inter nodal length. Then the stem also become brittle and easily broken by heavy wind.
- Fruiting bodies of the fungus (Bracket or Anabe) appears at the base of the stem immediately after the death of the palm.

Management:

- Dead palms along with root system have to be removed and destroyed by burning.
- Isolation of infected palms by digging trenches 30 cm wide and 60 cm deep.
- Root feeding with 125 ml of Tridemorph (Calixin) 1.5 percent concentration.
- Drenching the basin of the affected palms with 1.5 % Tridemorph at quarterly intervals or drench with 0.3% captan or calixin (0.1%) or copper oxychloride (0.3%).
- Drench the soil with 1% Bordeaux mixture before planting.
- Apply organic manure at 15-20 kg/palm and 15-20 kg green leaf along with 2 kg neem cake per palm
- General health of the palms should be maintained by giving recommended dose of NPK fertilizers.
- Damage to the root system due to repeated ploughings should be avoided.
- Water stagnation and close spacing of the orchards should be avoided.
- Discourage growing of collateral hosts of the fungus like *Delonix regia* and *Pongamia glabra.*

DISEASES OF BLACK PEPPER

1. Phytophthora foot rot	**:**	***Phytophthora capsici***
2. Anthracnose or Pollu	**:**	***Colletotrichum gloeosporioides***
		Colletotrichum necator
		Colletotrichum capsici

1. Phytophthora foot rot : *Phytophthora capsici*
***Phytophthora palmivora* var. *piperina* (Earlier name)**

Quick Wilt/Foot Rot- *Phytophthora capsici*
Slow decline – *Rhizoctonia bataticola* and/or *Fusarium* spp., and *Pythium* spp.,

- The disease was first reported in 1902 from Wynad district of Kerala
- The crop losses due to foot rot of black pepper have been reported to the tune of 119 to 905 tonnes per annum in Kerala

Disease cycle:

- PSI: Oospores in soil and infected plant debris.
- SSI: Zoospores spread by rain splashes, irrigation water.

Mode of spread and survival

- The inoculum survives in infected debris and soil, sporangia are produced in infected leaves, runner shoots and are carried to upper leaves by rain splashed, thereby spreading in a ladder like fashion.

Favourable conditions:

- The leaf infection is serious in irrigated areca nut -pepper mixed plantations.
- Collar infection is severe in pure plantations.
- Susceptible variety -Karimunda

Symptoms:

1. Leaf infection-

- The symptoms start as water soaked lesions on the lower surface of the leaves and these lesions later enlarge rapidly, involving 25-70 per cent of the leaf lamina

2. Die back:-

- The aerial branches get infected at any point. At the site of infection of branch, the discoloration occurs and rotting progress upwards and downwards resulting in die-back symptoms. The lateral branches of the affected vines break off at the nodes and fall off.

3. Foot rot or collar rot

- The stem near the ground level get infected shows water soaked spots followed by and the rotting and death of vine occurs with in 2-3 weeks leading to yellowing and drooping of the foliage.
- The affected portion emits bad odour. The necrosis progress down wards to the underground stem and to the root system.
- Infected vines wilt and die.
- The death of the plants is rather fast and hence called as "Quick wilt".

4. Root rot

- A dark patch appears in stem and spread to one foot from base of the plant in case of root rot. The infection starts at main root or at feeder root. Stem breaks off at the nodes, leaves turn yellow and droop the vine dies in 3-4 weeks. In leaves dark brown spots appear and enlarge causing a concentrate zonation with a grey centre. Leaves and spikes are shed.

Management:

- Avoid movement of infected soil through farm implements from diseased to healthy gardens.

- Pruning of the runner shoots lying on the ground.
- Soil around the basin may be covered with mulch and grass cover in the inter spaces to the prevent the spread of the inoculum.
- Soil amendment with neem cake or FYM or cotton seed meal or ground nut cake favours the growth of antagonistic organisms.
- Two sprays of Bordeaux mixture 1% or **0.3% potassium phosphonate** during the monsoon period reduce the aerial infection.
- Soil drenching with copper oxychloride 0.2% at 5-8 lit/vine minimizes the building up of the inoculum.
- Open the soil and apply 10% Bordeaux paste on the base of the stem up to 50cm.
- Sprays and soil drenching of Metaloxyl, Mancozeb and benomyl are also effective.
- Incorporation of biocontrol agents like *Trichoderma spp., Gliocladoim virens* etc.
- Resistance: *Piper colubrinum, P. arboretum, P. sarmentosum* are found to be resistant.
- Cultivars Narayakodi, Kallu valley, Uthirankotta and Balankotta are tolerant.

2. Anthracnose or Pollu disease : *Colletotrichum gloeosporioides*
Colletotrichum necator
Colletotrichum capsici

Pollu means hallow or chaffy

- *Diascorea triphylla* is a collateral host.
- The pathogen require high humidity in the form of dew, the maximum disease incidence occur during the months of August and September.
- Spike shedding due to this disease can go up to 10%.
- In early infection, there is up to 77 % weight loss of berries while late infection causes 56% infection.

Disease Cycle:

PSI: Infected plant parts/debris.

SSI: Wind borne conidia.

Pathogen

The mycelium is septate, intercellular as well as intra-cellular.

Acervuli are saucer shaped and elongated black coloured setae are scattered, 1-5 septate, rigid and swollen at base and acute at apex.

Conidiophores are short, hyaline to faintly brown, cylindrical, septate or aseptate.

Conidia are falcate, fusiform with acute apices and narrow truncate base, one celled, hyaline and uninucleate.

Symptoms:

- Young leaves and spikes show small, brown specks which are surrounded by the yellow halo. **Leaves show concentric zonations**
- Defoliation and spike shedding in severe cases.
- Small cracks develop on the berries and reduces their quality.
- Darkening of the pericarp of tender berries affecting their development.
- Ultimately results in chaffey berries and hence the name “POLLU” which means hollow berry.

Management:

1. Pre-monsoon and post-monsoon spray with Bordeaux mixture.
   i) Sprays of carbendazim or benomyl (0.1%) at 10 to 14 days interval
   ii) Three sprays of Bordeaux mixture (5:5:50) during June-July, late July and late August.
2. A combination spray of (0.1%) quinolphos and (0.2%) mancozeb during September controls both the "Pollu" beetle and disease.
3. Spraying of Bavistin or Benlate was effective which lessen the disease intensity by reducing the number of black spots ber berry and also increase in green berry yield.

DISEASES OF BETELVINE

1. Phytophthora root and foot rot	:	*Phytophthora spp.* *Phytophthora palmivora* *Phytophthora capsici* *Phytophthora nicotianae*
2. Sclerotium foot rot	:	*Sclerotium rolfsii*
3. Anthracnose or leaf spot	:	*Colletotrichum capsici*
4. Powdery mildew	:	*Oidium piperis*
5. Bacterial leaf spot	:	*Xanthomonas campestries* pv.*betlicola*
6. Mosaic	:	*Pipper Mosaic virus*

1. Phytophthora root and foot rot: Caused by various species of *Phytophthora*.
Also called as leaf rot or stem rot/ wilt

Disease cycle:

PSI: Survives in the soil and crop debris as oospores and chlamydospores
SSI: Sporangia and zoospores by irrigation water and rain splashes

Mode of spread:

The fungus is mainly soil-borne and is also carried through irrigation water and through collateral hosts.
The secondary spread is through sporangia produced on the leaves

Favourable conditions

- Low lying fields with poor drainage
- High soil moisture, Frequent rains
- Favourable period - September to February (winter), low temperature (23°C and below) and high relative humidity.

Symptoms:

Foot rot:

- Small brown water soaked lesions develop on the vine stem near soil line and progresses upward and downward.
- The lower portion of the stem near the soil level shows irregular black lesions up to second or third inter node.
- The affected vines show yellowing and drooping of the leaves from tip to downwards.
- The affected plant dry up completely within 2 or 3 days.
- The diseased inter nodes undergo 'wet rot' and the tissue become soft, slimy with a fishy odour
- Internal symptoms- Narrow brown streaks develop on in the regions of vascular bundles.The central core of the stem also becomes brown or black
- Decomposition of the underground portions of cuttings leaving only the fibrous parts
- Rotting of fine young roots first and later spreads to older roots and ultimately reaches the foot or collar region of the plant.

Leaf rot:

- Water soaked lesions on mature leaves near the soil under high moisture conditions and the rot extends to the petiole and stem.
- Type I symptoms: Spots are circular, necrotic, deep brown in colour with distinct gray brown zonation, when the relative humidity fluctuates greatly during day and night.
- Type II : Brown circular spots with no zonations when the weather is continuously wet.
- Affected leaves fall off prematurely.

Management:

- Provide ridges and furrows for good drainage.
- Collection, removal and destruction of dead and dying plants by burning.
- Removal of collateral hosts like *Colacasia* around the plantations.
- Cuttings should be selected from a height of 1.5 mt above the soil level.
- Treating the ridges with Bordeaux mixture 2:2:50 at monthly intervals.
- Selection of healthy cuttings for new plantings (more than one year old)
- Crop rotation with non host crops like rice or maize or sugarcane.
- Drenching bases of infected vines with Bordeaux mixture 1% at monthly intervals @ 500 ml/hill.
- Treatment of cuttings with 2:2:50 Bordeaux mixture before planting.
- Spraying Metalaxyl+Mancozeb 0.2% on infected and surrounding healthy vines
- Biocontrol -Neem cake + *Trchoderma viride*
- Application of neem cake or neemcake urea @ 150 kg N/ha/year near the root zone.
- Apply shade dried neem leaf or *Calotrophis* leaves at 2t/ha in 2 split doses and cover it with mud.

2. Sclerotium foot rot: *Sclerotium rolfsii*

The disease is found to occur during summer months *i.e* May to July

Disease cycle:

PSI: Sclerotia in infected debris.

SSI: Sclerotia spread by implements and irrigation water.

Symptoms:

- Dark lesions appear just below the ground level and reach up to 10 cm above the ground.
- Leaves turn yellow, become flaccid and droop and finally entire vine wilts and dies.
- White cottony mycelium strands will be observed on the rotting stem portion.
- Development of small, brown,mustard seed like sclerotia on the infected portions.

- Shredding of infected bark. Vine wilts and totally dries off

Management:

- Soil solarization with white polythene mulch for period of 30 days give effective control of collar rot.
- Deep summer ploughing.
- Green manuring with crops like *Sesbania* before planting
- Selection of healthy cuttings for new plantings
- Field sanitation by collection and destruction of infected vines and debris
- Providing good drainage
- Drenching bases of infected vines with Bordeaux mixture 1%
- Drenching carbendazinm 1% on infected and surrounding healthy vines
- Biocontrol with neem cake + *Trchoderma viride (T. viride*, 2kg mixed in 50 kg of FYM)
- Long crop rotation with non hosts

3. Powdery mildew *Oidium piperis*

Disease cycle:

PSI: Dormant mycelium or cleistothecia in infected plant debris in the soil

SSI: Airborne conidia

Mode of spread:

The fungus is mainly air-borne. The spores of the fungus also carried through irrigation water.

Etiology

Hyphae - branched, hyaline, septate

Cleistothecia having myceloid type

Conidiophores - short, club shaped, non septate produce chains of conidia

Conidia - Single celled, hyaline, elliptical and smooth

Symptoms:

- The disease occurs at all stages of its growth and on the tender shoots and leaves
- Initially white to light brown powdery patches appear on the under surface of the leaves.
- Correspondingly the upper leaf surface shows yellowish irregular patches
- Both leaf surfaces covered by white floury mass of fungal growth under favourable conditions.
- When the disease advances, the whitish growth turns to brown blotches and in severe cases, the leaves turn yellow and defoliation occurs.
- The affected leaves get shriveled, deformed and shed from the vines.
- The affected tender shoots and buds are deformed and shrivelled and margins of leaves turn inwards.

Management:

- Wettable sulphur 0.2 to 0.3 per cent spray.

4. Anthracnose or leaf spot: *Colletotrichum capsici.*

Disease cycle:

PSI: Infected plant debris

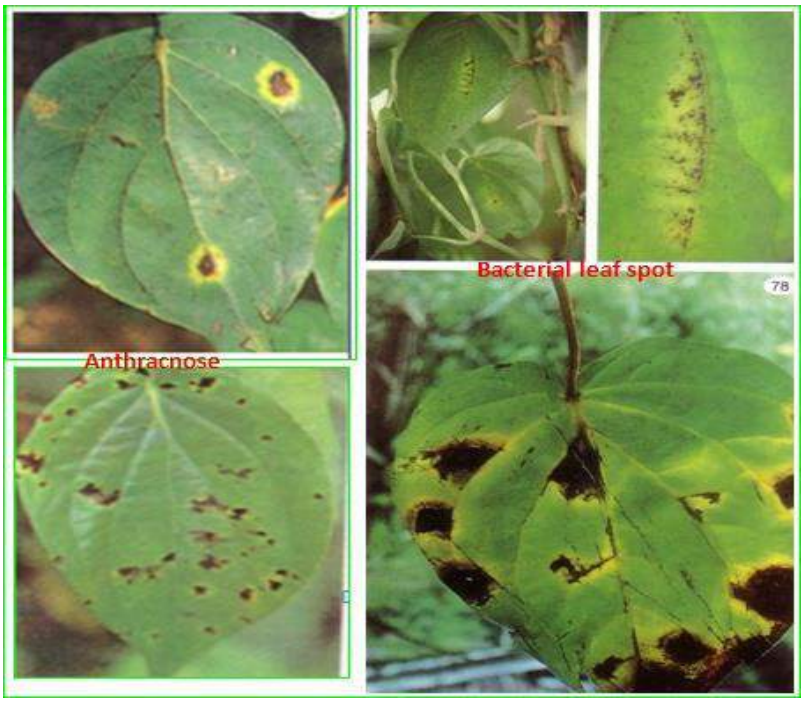
SSI: Wind borne conidia.

Symptoms:

- Only aerial parts of the plant are attacked.
- The leaves show small black circular spots initially which later enlarge and develop to a size of 2 cm in size, become concentric and covered with a yellow halo.
- Spots coalese to form large lesions.
- The affected leaves turn pale yellow and dry up with large black dots in the centre of the spots.
- On the stem, initially small, black, circular specks appear under the green bark and later enlarge rapidly and girdle the stem resulting in withering and drying.
- These blackened lesions generally become depressed due to the death and drying of the infected tissues.

Management:

- Prophylactic sprays with Bordeaux mixture.
- Application of Ziram and Carbendazim at 15 days interval.
- Foliar sprays of Bitertenol 0.05%.



5. Bacterial leaf spot caused by *Xanthomonas campestries* pv.*betlicola* or *Pseudomonas betlis*

Disease cycle:

Bacterium survives in infected debris and also on other species of hosts like black pepper.
It is most serious in southern Kerala.

Favourable conditions:

Cloudy weather with high humidity coupled with inter mittant rains favour the disease development.
More than two years old crop is more vulnerable.

Symptoms: Appears on leaves and stem of betel vine during wet weather

- Minute brown water soaked spots on the under surface of the leaves which later become dark brown circular to angular spots surrounded by yellow zone. Confined to the interveinal space giving angular in shape. Which later increase in size and become necrotic.
- Later these appear on upper surface also. The centre of spots are mottled brown and later turn black.
- On the stem brown spots appear which leads to stem canker.
- Cracking of the stem leading to wilt or death of the plant.
- In severe conditions leaves the leaves lose their lusture, turn yellow and fall off
- Profuse bacterial ooze may be seen on the lower side of the lesion.
- In severe cases, defoliation and stem injury occur, leading to wilting of the plant

Management:

- Cuttings should be dipped in Bordeaux mixture 1% along with streptocycline 200-250 ppm.
- Selection of disease free cuttings.
- Infected vines should be removed and burnt.

6. Mosaic:- ***Piper yellow mottled virus (****PYMoV*)

Observed recently on Kaljedu variety

Symptoms:

- Initially leaves show mosaic mottling and vein clearing symptoms.
- Bright yellow to light green and dark green patches develop on the entire leaf lamina.
- Conspicuous bold blisters are also observed.
- Leaves become puckered and distorted and reduced in size.
- Infected plants appear weak with shortened internodes and show stunted growth.

Pathogen

- *Caulimoviridae, Badnavirus*
- *Piper yellow mottle virus* (*PYMoV*)
- Bacilliform ds DNA.

Disease cycle:

- It is transmitted through sap, cuttings, graftings, cutting knives and implements; and also through insect vectors like mealy bugs in a non-persistent manner.
- Various species of mealy bugs associated are *Planococcus citri* (citrus mealy bug), *Pseudococcus elisae* and *Ferrisia virgata*

Management:

- Selection of cuttings from disease free vines.

DISEASES OF CASHEW

1. Anthracnose : *Colletotrichum gloeosporioides*
2. Pink disease or Die back : *Corticium salmonicolar*

1. Anthracnose: ***Colletotrichum gloeosporioides.***

This disease is called as Soorai in Tamil Nadu

- This disease is most prevalent when there is excessive rainfall coinciding with the appearance of new growth and flowering.
- Pathogen usually infect tender leaves, shoots, inflorescences, young fruits (apples) and young nuts.

Disease cycle:

PSI: Pathogen survives on affected parts of tree.

SSI: Spreads through wind borne conidia.

Symptoms:

- Affects young leaves, twigs, nuts and apples.
- Brown circular or irregular spots with dark margins are formed on leaves.
- Exudation of resin from infected parts.
- As the disease progresses, the lesions enlarge in size, all affected tender leaves wrinkle.
- Infected parts in their early stage show shiny, brown water-soaked lesions which later turn reddish-brown
- Die back of twigs and branches.
- Flowers turn black and get killed.
- Infected nuts and apples shrivel and decay

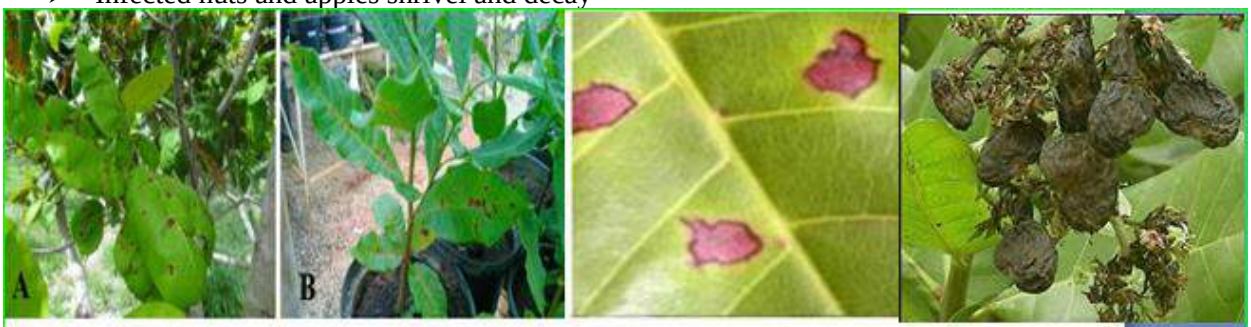


Figure 4 Cashew *Colletotrichum* symptoms on leaves. (A) Symptoms on plants,

Management:

Spraying of 1 per cent Bordeaux mixture at new flush and blossom stage.

Adoption of sanitary measures.

Spray application of COC (Blitox) @ 0.3% (3g/l), Carbendazim or Benomyl or Thiophenate methyl 0.1% (1g/l).

2. Pink disease or Die-back : *Corticium salmonicolor*.

***(Pellicularia salmonicolor, Botryobasidium salmonicolr*)**

This disease usually occurs during the rainy season**.**

Disease cycle:

PSI: Survives as dormant mycelium on affected parts of tree If left unpruned, these branches will also be the site of infestation and multiplication of bark beetles., etc **.**

SSI: Spreads through wind borne conidia.

Symptoms:

- The affected branches show white or pinkish growth of mycelium on the bark.
- The fungus penetrates into the deeper tissues and causes gradual death of the shoots from apex downwards (Die back).

- Towards the cessation of monsoon, a film of silky thread (mycelium) of the fungus is seen on the branches. The mycelium is silvery white in the beginning but changes to a general pink at later stages.
- In advanced stages the bark splits and peels off.
- In a tree either one branch or many branches will be affected.
- On the affected branches, the leaves turn yellow and fall off giving a barren appearance to a portion of the tree.
- Leaf crinkle and sheds, twigs get dried.

Management:
- Pruning of affected branches below the lowest point of infection and burning.
- Protecting the cut surfaces with Bordeaux paste.
- Prophylactic sprays with 1 per cent Bordeaux mixture twice before onset of monsoon and again during September.

DISEASES OF COFFEE

1. **Coffee Rust** : ***Hemileia vastatrix***
2. **Black rot** : ***Corticium koletoga***

1. Coffee Rust : ***Hemileia vastatrix***

Harry Marshall Ward, who had studied Anton de Bary's work on the fungi

Severe epidemics in Srilanka in 1867

Disease cycle:

Survival and spread-

PSI: Survives primarily as mycelium in the living tissues of the host, and since infected leaves drop prematurely or teliospores in infected leaves

SSI: Uredospores dispersed by wind and water. Pycnial and aecial stages are unknown

Secondary spread: By wind, rain splashes and a small proportion by insects.

Favourable conditions

Uredospores germinate only in free moisture

Rainy weather and heavy dew

Severe from July to Dec.

Pathogen

The mycelium is intercellular and sends haustoria into the cells.

The mycelium sends out erumpent stalks through stomata which bear the uredospores.

The uredospores are **reniform or orange segment** like in shape.

The convex side of the spores are echinulated and the lower side is smooth.

The telial stage succeeds the uredial stage in the later stage

Symptoms :
- Rust symptoms are appeared on leaves, petioles, new shoots and young berries.

- Small pale yellow to orange spots on the lower surface of the leaves with the corresponding upper surface shoring characteristic chlorotic lesions which soon turn brown in colour.
- Orange brown round sori with powdery mass of spores are formed.
- The centre of the spot becomes dry and turns brown in colour while the margins continue to expand and produce uredospores.
- The fungus sporulates **through the stomata rather** than breaking through the epidermis, so it does not form the pustules typical of many rusts
- Reduced plant vigour and dieback of twigs and finally plants get killed.
- When young flush is infected, berry size remain small and may not ripe leading to tremendous losses.

Management :

- Planting material should be healthy and quarantine measures should be adopted.
- Pruning and training methods improving the air circulation causes rapid drying of foliage and facilitates less uredospore germination
- Pre monsoon spray with 0.03% a.i plantvax in May-June
- Spray the bushes once with Bordeaux mixture 0.5% or Copper oxychloride 0.25% 2-3 times during monsoon
- Mid monsoon spray (July-August) and post monsoon spray (September October) with either BM or plantvax spreaders and stickers like linseed oil, teepol or casein increased the fungicides efficacy.
- Blossom spray 0.5% B.M in February- March.
- Spraying of Tridemefon 0.05%, Propiconazole 0.1% etc.,
- Biocontrol agents like ***Darluca fulvum,****Verticillim lecanii* (*Lecanellium lecanii*) stain VICENSA
- 'Catimor' varieties that have durable resistance. They include 'Catimor 129', 'Catimor 128'
- Coffee variety 'K7' is tolerant to coffee rust, while 'Ruiru 11', a new hybrid is resistant to both CLR and coffee berry disease (CBD- (*Colletotrichum kahawae*)
- Timor Hybrid (resistant to coffee leaf rust due to its Robusta genetic roots) and Caturra vermelho, Vilasarchi.
-

BLACK ROT/ KOLEROGA/ TREAD BLIGHT : *Corticium koleroga,*
Pellicularia koleroga in 1876, current name *Ceratobasidium noxium*."Koleroga" means "rot disease" in the Kannada.
In south India the disease is severe only in those areas growing with *C. arabica.*
It is influenced by south west monsoon period from June – Sep.

Disease cycle:

PSL: Dormant mycelium or stroma in infected plant parts

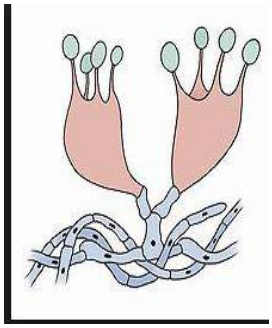
SSI: Wind borne conidia

Mode of spread and survival

- The fungus mostly spreads by contact from leaf to leaf through the vegetative mycelium and also by sclerotium.
- The pathogen spread through infected plant debris.
- Mycelium lies in twigs throughout year.
- Pathogen attacks coffee berries and all plant parts except root
- The pathogen penetrates the leaves through the stomata on the lower side and the hyphae invade intercellularly in the palisade tissue.

Pathogen

The hyphae are hyaline when young and turn light brown with age.
Fructifications arise with numerous basidia and basidiospores.
Basidia are simple, oval rounded or pyriform.
Basidiospores are hyaline, elongated, rounded at one end, slightly concave on one side.
At a later stage the fungus forms sclerotia or hyphal clumps by repeated branching of short cells.



Symptoms :

- Typical signs of the disease includes white mycelial filaments visible on the infected branches and leaves which draw food from the host.
- Infection starts at the base of the branch and extends towards the tip.
- Later blackening and rotting of affected leaves, twigs and developing berries and become black
- Diseased leaves dry up and get detached from branches and hang down by means of slimy fungal strands. Often, necrotic lesions are evident on the affected leaves.
- Affected berries become dry, showing a white web consisting of closely interwoven mycelium.
- Defoliation and berry drop takes place from infected branches in advanced stage.

Management :

- Cultural practices like regulation of shade, avoiding water logging condition, maintaining good aeration and ventilation, proper fertilization, pruning and destruction of infected branches.
- Remove and burn affected parts.
- Spraying chemicals like Carbendazim and Boradaux mixture 1% just before the onset of monsoon and also during break in monsoon (45 days after the first spray) is more effective.

DISEASES OF TEA

Blister blight of tea : *Exobasidium vexans*

This disease was first reported from Assam in 1868, spreading since then to most other tea growing areas of North and South India.

The succulent growth of the plants developing after pruning is highly susceptible to infection. Temperatures above 24°C are fatal to the blisterance and disappearance of this disease.

Disease cycle :

PSI: Prepetuation is from the pre existing infected bushes. Fungus survives in infected plant parts in the form of Basidiospores. Completes life cycle in 11 to 28 days. Many generations of the fungus are completed within a single season.

SSI: Wind borne basidiospores and conidia. It produces conidia and basidiospores in the same blister.

Pathogen

The mycelium is confined to the blistered areas on the leaves.

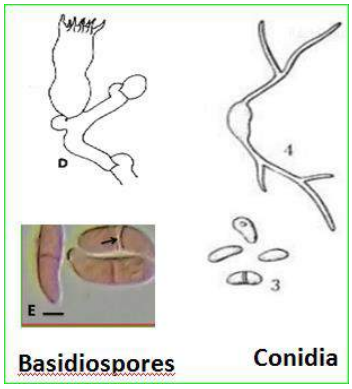
The fungus produces two kinds of spores *viz.*, the conidia and basidiospores.

The conidia are most abundant, borne singly at the tips of long stalks. They are hyaline, elliptical, straight or slightly curved.

Basidia are formed on the surface in larger number but never form a continuous hytneniurn. They are intermingled with 'conidial stalks and sterile hairs.

Each basidium is long, club shaped with usually two short sterigmata at the end, each bearing a basidiospore. The basidia are ovate to oblong, hyaline. Basidiospores are unicellular at first, but a septum develops on maturity

Paraphyses are single, septate and apically rounded.



Favourable conditions

- The disease is favoured by cool day and nights, with wet or humid conditions.
- More severe outbreaks occur under shade or adjacent to jungle or wind breaks or in damp low lying areas where mist persists.
- The succulent growth of the plants developing after pruning is highly susceptible to infection. Temperatures above 24°C are fatal to the blister blight disease, a fact which account for the periodical appearance and disappearance of this disease.

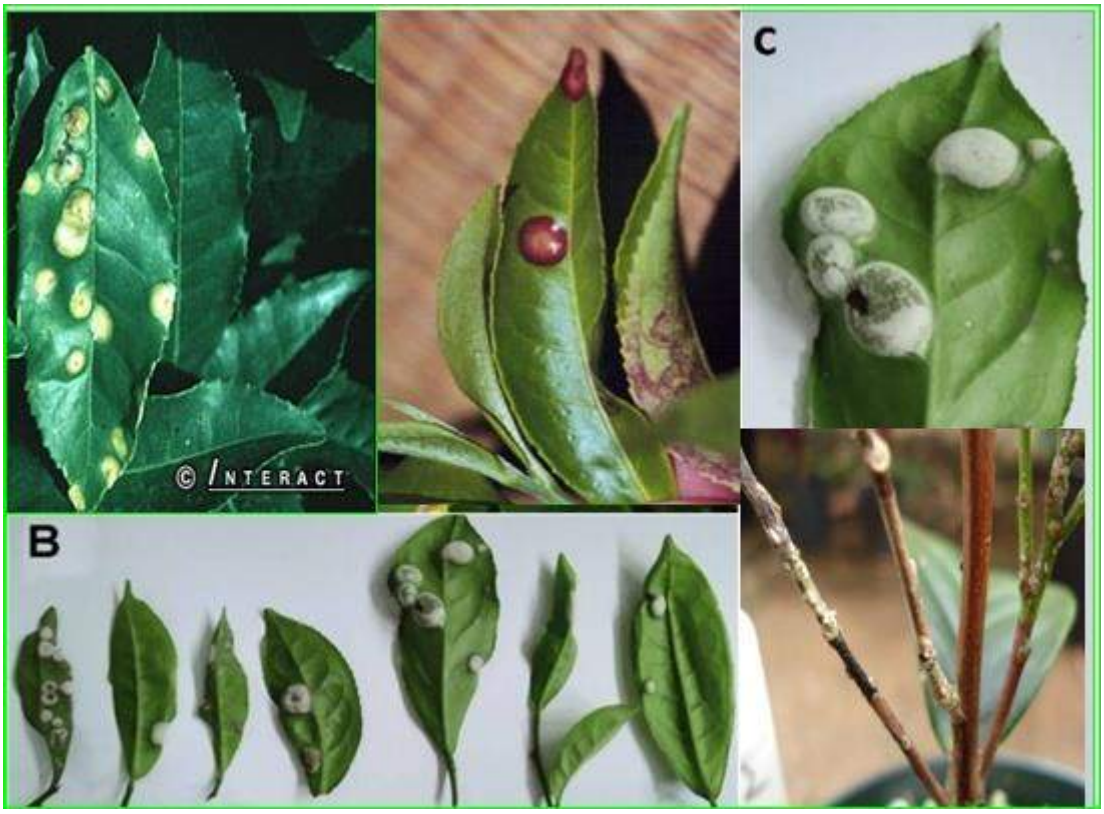
Symptoms :

Seedling infection

- If the disease appears in nursery, when stem is less than 15 cm height, seedlings are stunted and produce many thin stems instead of a single stalk. Repeated attacks cause death of seedlings.

Mature plant

- On the upper surface of the leaf, the spots are light green later turn round pinkish in colour with depressed into a shallow cavity while the under side bulges correspondingly forming a blister like swelling. The lower bulged surface is covered by white growth of the fungus.
- Pink colour later turns to deep red and spots cover the entire foliage leading to blightening of leaves.
- In later stages the blister turns dark brown and shrinks to flattened patch.
- **Old leaves of four weeks and above are immune** and only young leaves are circular shape but becomes elongated along the midrib.
- When many blisters occur near the margin or apex and coalesce, much distortion and curling of the leaf may be caused.
- The infection passes on to petiole and young succulent stem which results in serious damage. On the stem a pale yellow spot appears; this gradually elongates and encircles the whole stem which becomes slightly swollen at this point**.**
- Some times, on the stem, spots without blisters are formed
- The leaves, and buds above the point of attack wilt end to wither.



Management :

- Regulation of shade trees by pollarding and lopping the branches reduces disease severity.
- Removal of affected shoots and leaves by pruning.
- Frequent application of protective copper and eradicant nickel fungicides at short intervals 210 grm of COC +210 gm Nickel chloride/ ha at 5 days interval from June to September and 11 days intervals in Oct-Nov gives economic control.
- Spraying the bushes with 420 g of copper oxychloride and 27 g of Agrimycin 100/ha gives better control over the disease, compared to the treatment with copper oxychloride alone at the same dosage.
- Among the organic fungicides, chlorothalonil gives protectant and therapeutic effect.
- Triademorph, Mancozeb, Baycor, Bayleton and Pyracarbolid offer good disease control.
- Disease control achieved with triademorph at 340 and 560 ml /ha is satisfactory under mild and moderate rainfall conditions.
- Spraying with the systemic fungicide, (Atemi 50 SL) at 400 ml/ha or bitertanol 300 EC at 340 ml/ha at weekly intervals controls the blister blight effectively.

DISEASES OF RUBBER

1. **Powdery mildew : *Oidium heveae***
2. **Root rot diseases**
   a) **White root rot : *Fomes lignosus (Rigidoporus lignosus)***
   b) **Brown root rot : *Fomes noxious (Phellinus noxius)***
   c) **Red root rot : *Ganoderma pseudoferrum***
   d) **Dry root rot/stump rot/ Charcoal rot : *Ustulina zonata***
   e) **Black root disease : *Rosellinia arcuata***
   f) **Armillariella root rot : *Armiilartella mellea***
   g) **Inter root disease : *Botryodiplodia theobromae***

1. Powdery mildew : *Oidium heveae*

- The disease is predominantly noticed on newly formed tender flush during the re-foliation period from January to March.
- It affects the tender leaves produced soon after wintering, causing secondary leaf fall

Disease cycle:

PSI: Survives as dormant mycelium under the bark. At higher elevations pathogen remains active under shade throughout the year.

SSI: Wind borne spores carried for long distances.

Favourable conditions

- Cloudy days with light rains and dew formation during new flush formation favour serious disease outbreaks.

Symptoms:

- The copper brown and apple green leaves are susceptible to infection.
- The disease appears as white powdery patches, as the fungal hyphae grow radially to form extensive circular colonies
- Tender leaves with ashy coating curl, crinkle, and edges roll inwards and fall, leaving the petioles attached to the twigs like a broomstick.
- Colonies appear silvery and shining during active spoorulation.
- Later, the fungus dies, leaving yellow blotches on the leaves.
- Leaves eventually fall and die back symptoms appear.
- Leaf fall and die back of young shoots retards the growth of young plants.
- When semi mature leaves are affected, characteristic translucent brownish yellow spots are clearly seen, and such leaves persist throughout the year with the necrotic spots.

Management:

- Dusting the plants with 325 mesh fine sulphur 10-12.5 kg/ha at 14 days interval during dry season.
- Wettable sulphur 1 kg/400 litres of water sprays at fortnightly intervals.
- Spraying carbendazim @1 g/lit is more effective than sulphur for nurseries and young rubber plantations.
- Alternate use of carbendazim and sulphur is recommended to avoid resistance problem.
- Rubber clones PB 86, GT 1, GLl, PR 107, PB 5/139, RRlM 703, RRII 208 and PB 310 show some tolerance.

2. Root rots:

a) White root rot : *Fomes lignosus*
b) Brown root rot : *Fomes noxious*
c) Red root rot : *Ganoderma pseudoferrum*
d) Dry root rot/stump rot : *Ustulina zonata*

Disease cycle:

PSI: All the pathogens are soil borne and spread to long distances through infected soil and planting material.

SSI: Secondary spread is either by irrigation water or by intercultural operations.

All the pathogens produce, cell wall degrading enzymes (CWDE) causes degradation by results in different colours.

Symptoms:

- In white root rot, tap root of the tree is attacked and fungus spreads on the surface as white rhizomorphs.

- In brown root rot, affected roots are encrusted with mass of soil and small stones held by network of mycelium.
- In red root rot, diseased roots are characterized by formation of red rhizomorpphs on the surface which may coalesce forming red skin.
- In dry root rot or stump rot, root decay is rather dry, bushes die or dry up suddenly with all their leaves attached to the plants.

Management:

- Old and infected stumps should be removed and burnt.
- Drenching of soil with Bordeaux mixture and affected roots with copper sulphate is effective for white root rot management.
- For brown root rot, application of lime @ 2.5 t/ha and for areas of soil with dead trees an additional 25 kg/tree should be given.
- Prophylactic sprays with Emisan solution.
- For red root rot, removal of infected stumps and treating the affected roots with disinfectants.
- Removal and destruction of fruiting bodies present if any.

DISEASES OF MEDICINAL AND AROMATIC CROPS

DISEASES OF PYRETHRUM

1. Damping off	:	*Pythium spp.*
		Phytophthora spp.
2. Wilt	:	*Phytophthora cambivora*
		Rhizoctonia solani
3. Rust	:	*Puccinia horiana*
4. Leaf blotch	:	*Septoria chrysanthemella*
5. Grey mold	:	*Botrytis cinerea*
6. Fusarium wilt	:	*Fusarium oxysporum f.sp.solani*
7. Root rot	:	*Sclerotinia minor*
8. Ray blight	:	*Phoma ligulicola var. inoxydablis,*

1. Damping off : *Pythium* spp. and *Phytophthora* spp.

Symptoms:

- Pre emergence damping off.
- Post emergence damping off. Water soaked lesions on collar region, leading to rottening and damping off.
- Stem size reduction. Toppling over of seedlings & death.
- Post emergence damping off and rotting of roots of stem cuttings.
- Plants will die suddenly.

Disease Cycle:
PSI: Oospores/Chlamydospores in soil.
SSI: Sporangia spread by rain splashes.

Management:

- Pre-plant sterilization of soil medium.

- Better drainage and avoid excessive shade.
- Soil drench with Thiram/Captan @2.5 gm/lit or Copperoxychloride @3g/lit.

2. Root rot and wilt - *Rhizoctonia solani*

Symptoms

Infection appears initially in the form of dark brown, necrotic lesions on the roots and basal parts of the plant, which later turn into root rot and wilt symptoms, leading to premature death and desiccation of infected plants.

Stolons exhibit pinkish brown lesions initially. Which gradually turns into dark brown lesions. Later turns from dark brown to black patches that increase in size resulting in soft decay.

Management

Affected debris destruction

Summer ploughing

Crop rotation

Soil sterilization

Biological agent *Trichoderma* spp.

Carbendizim 0.1% or Mancozeb 0.25% as soil drenching.

Seedlings should be grown in disease-free soil.

Raised nursery beds should be prepared.

Nursery beds should be drenched with 0.15% benomyl or 0.1% zineb.

3. Rust : ***Puccinia horiana*** **(**Chrysanthemum White Rust)

It is a quarantine disease in Canada and many other countries, including the United States.

The fungus produces two types of spores, teliospores and basidiospores. Teliospores are produced on the pustule and remain on the pustule unless they are aggressively brushed off. They can survive up to 8 weeks if they remain in pustules on detached leaves at 50% RH or less. Teliospores germinate to produce basidiospores.

New infections are initiated by basidiospores. They need a film of water on the plant surface to cause infection. They spread from plant to plant by splashing water. Basidiospores can also travel short distances (700 metres) by wind currents during moist weather.

Symptoms

Infected plants can be symptomless, especially during hot and dry conditions. Symptoms usually appear during cooler, wet weather and it may take up to 8 weeks before infected plants show symptoms during periods of hot weather. Symptoms appear as small white to yellow spots up to 4 mm wide on the upper surface of the leaf (Figure 1A). These spots may be slightly dimpled and become brown as the disease progresses. Pustules form on the underside of the leaf just beneath the small spots observed on the upper surface of the leaf (Figure IB). These pustules originally appear as buff to pink-coloured (Figure 2A). As they age, they become white (Figure 4). Pustules are most common on young leaves and flower bracts but can be found on any part of the foliage including flowers.

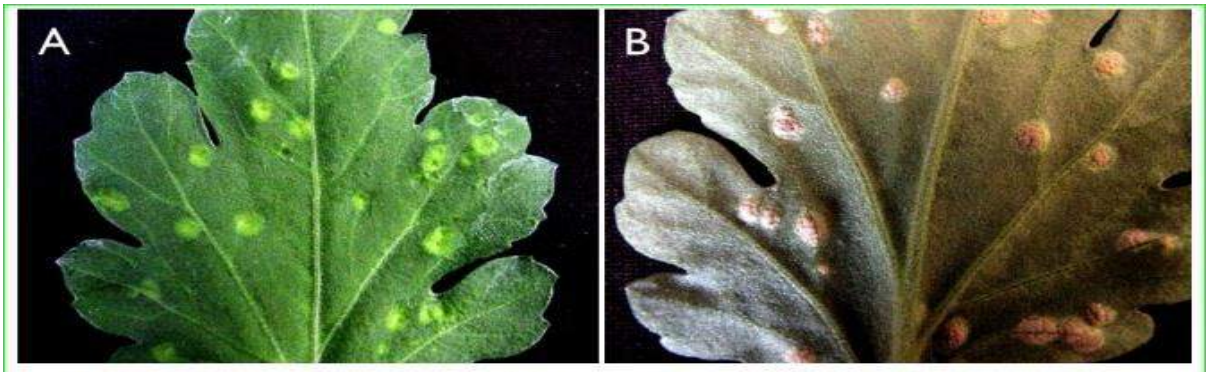


Figure 1. Early symptoms of yellow-white, dimpled spots on the upper leaf surface (A) and young pink-brown pustules on the under leaf surface (B) of chrysanthemum.

Figure 2. Young pink-brown pustules on the under leaf surface (A) and mature white pustules on the under leaf surface of chrysanthemum leaves.

Pathogen- Microcyclic and autoceious rust, Uredial stage is absent

Management:

- Use of healthy suckers for planting.
- Picking off lower leaves at planting and replanting.
- Spraying with Wettable Sulphur 0.2% or Zineb 0.2% or mancozeb 0.25 % or Hexaconazole 0.1 % at 10-15 days gap.
- Crop rotation with non host crop.

4. Leaf blotch : *Septoria chrysanthemella.*

Symptoms:

- Initially small circular/irregular brownish black spots on leaves with an yellow halo.
- Large patches are formed due to merging of such spots covering major portion of leaf.
- Leaves remain small and curl.
- Leaves shed or hang on the stem for some time
- Severe in rainy weather.

Disease Cycle:

PSI: Survives in crop residue or through use of disease suckers.

SSI: Wind borne/Rain spread conidia.
Etiology
Mycelium is septate inter and intrecellular haustoria,
Sexual spores are Ascospores borne in pseudothecium.
Primary source of inoculumis Ascospores (pseudothecium)
Secondary source of inoculums is pycnidiospores
Mode of spread is infected debris in the soil and rain splash.
Management:

- Adoption of sanitary measures.
- Controlled irrigation.
- Spray Carbendazim/Benomyl 0.1% or Mancozeb 0.2% or COC 0.3% at fortnightly intervals.

5. Grey mold : *Botrytis cinerea*

Symptoms:

- Stem girdling, death of upper portions of plant noticed.
- Leaf symptoms start from margins and spread to centre and base a semicircular band.
- Infected stems appear bleached and often are associated with the presence of cottony mycelium and sclerotia. Stem girdling, death of upper plants portions
- On flowers, infection starts as brown water soaked spots on lower petals and flowers covered with greyish brown powdery mass of spores.
- Flower buds do not open.Sterile buds

Plate 4.3.1 - (A) Diseased pyrethrum plant at site 08-1, (B) diseased flowers, (C) rotted out growing points and necrotic foliage colonised with mycelium with soil bound apothecia, and (D) numerous apothecia observed at ground level.

Disease Cycle:
PSI: Has wide host range. Also produces sclerotia.
SSI: Wind/rain spread conidia or contact between healthy and diseased.

Management:

- Adoption of proper spacing.
- Spray of Captan 0.2% or Blitox 0.3% as and when necessary.

Ray Blight - *Phoma ligulicola* var. *inoxydablis*

- Necrosis of ray florets of the flowers
- Distortion of the developing stems of plants followed by necrosis in the growing tips are symptoms of a severe form of ray blight.
- The disease affects all above ground parts of the plant.
- The disease does not affect roots, but the pathogen is able to survive as epiphytic mycelium around root cuttings

Etiology: mycelium is septate, asexual spores are pycnidiospores borne in pycnidia.
MANAGEMENT:

- Healthy seed used for sowing
- Crop rotation with graminacea family
- Affected plants cut and burn
- Chemical: carbendazim, 0.1% as soil drenching

DISEASES OF SENNA

1. Alternaria Leaf spot : *Alternaria alternata*
2. Damping off : *Rhizoctonia bataticola*

1. Alternaria Leaf spot: *Alternaria alternata*

Leaf spot is the most serious among the diseases causing lot of damage to the senna crop.
The infested leaves drop off prematurely and affect the yield considerably.

Etiology

- Septate mycelium, either the conidia or conidiophores are coloured. Conidia have both longitudinal as well as transverse septa (**Dictyosporae**)

Epidemiology

- Nutritionally poor soil,
- Temp 28-35 ° C and RH 85-90% are favourable.
- Susceptible host.

Disease Cycle:

- Primary source of inoculum: Dormant mycelia or Survives in crop residue or through use of diseased seed.
- Secondary source of inoculum: Wind borne/Rain spread conidia.

SYMPTOMS

The disease becomes visible first as minute pale yellow spots on the leaf blade and its margins.
As the spot grow in size they become circular and irregular in shape with their coalesce and turn dark brown to black in colour.
The lesions vary from a minute spot to about 8 mm diameter in size. Concentric rings are formed on the leaves.
Spot size increases and covers the entire leaf resulting in blightening of leaves.
With the advancement of the disease, leaf tips and margins die and the necrotic tissues increase in size.
The disease is known to adversely affect the sennosides content in leaves and it is inversely proportional to the disease intensity.
Loss of sennosides in severely affected leaves was over 60% as compared to that in disease-free leaves.
In advanced stages the pods are also affected with brownish black spots.

Management

- Collect and destroy the infected plant parts
- Use disease free seed.
- Amend the nutritional status of the soil
- Reduce plant population
- Spray mancozeb or Zineb or Rovral 0.25% at 15 day interval.

2. Damping off: *Rhizoctonia bataticola*

Etiology

- Septate mycelia, sub - epidermal haustoria
- Asexual spores are absent and mycelial strands cause infection
- Vegetative structure: Sclerotial bodies

Epidemiology

- Warm weather, soil temp 28-32° C with optimum moisture and neutral pH
- Primary Source of Inoculum :Sclerotial bodies
- Secondery Source of inoculum: Soil borne mycelial strands

Disease cycle

- It is a Deuteromycetes fungus that produces sclerotial bodies which are sesamum seed like vegetative resting structures.
- Under congenial conditions these sclerotial bodies germinate, penetrate inside the host and causes disease.

- The mycelial strands inside the affected host again germinate and cause infection and disease cycle continues.

SYMPTOMS

In the North-Western India, senna is affected by the damping-off disease at seedling stage.
Initially water soaked lesions appear on the collar region of the plant.
After brown discoloration follows the epidermal layer collapse. Seedlings topple down.
The damping-off in seedlings is the most devastating disease.

Management:

- Destruction of diseased debris.
- Deep ploughing during summer.
- Crop rotation with non- host crops
- Soil sterilization
- Biological control with *T. viride*
- Seed treatment with thiram or captafol at 2.5 g/kg of seed gave good protection.
- Soil drenching with Carbendazim (0.1%) or Mancozeb (0.2%).
- Seed treatment with Thiram+Bavistin @1.5 g each /kg seed.

DISEASES OF MINT

1. Stolon rot	:	*Rhizoctonia bataticola* *Macrophomina phaseolina* *Thielaviopsis basicola*
2. Rust	:	*Puccinia menthae*
3. Alternaria leaf blight	:	*Alternaria alternata*
4. Verticillium wilt	:	*Verticillium alboatrum*
5. Powdery mildew	:	*Erysiphe cichoracearum*

1. Stolon rot ***Rhizoctonia bataticola*** **(**Sclerotial stage)
Macrophomina phaseolina **(**Pycnidial stage)
Thielaviopsis basicola (Perfect stage)

The disease was reported on *Mentha cordiaca* by Green and on *M. arvensis* and *M. spicata* by Hussain and Janardhan

Symptoms

Under-ground stolons exhibit pinkish brown lesions in the earlier stages and gradually turn into dark brown to black patches.
These patches increase in size and finally result in the soft decay of a portion or the entire affected stolon.
Primary symptoms are yellowing of leaves and stunted growth of the plants.
The wilting of the plants occurs in advance stages leading to the death of above ground portion.

Management:

- Discard all the diseased stolons while planting
- Treat the stolons with 0.25% Captan or 0.1% Benomyl or zineb or mancozeb .
- Follow 3-year-crop rotation with rice, wheat and mint.
- Deep summer ploughing
- Avoid excess irrigation.
- Uproot and burn the affected plants.
- Soil application of *Trichoderma* sp.
- Drench soil with Kavach 0.2%.

Rust : *Puccinia menthae*

Etiology

- *Puccinia menthae* over winters on mint stubble and on wild and escaped mint.
- *P. menthae* is macrocyclic, autoecious, produces uredosori on leaves, stems, and runners.
- Teliospores are brown, 2 celled, pedicillate, obtuse to slightly pointed.

Symptoms:

- Brown, circular pustules appear on the undersides (Aceiospores) of leaves followed by leaf yellowing.

- A yellow spore stage (uredospores) is in reddish blisters on young shoots in spring, and a black spore stage (teliospores) is on stems and re-growth in fall.
- Telial stage is characterized by swelling on the upper portion of the stem.
- Swollen shoots with elongated, chlorotic internodes are associated with systemic infection.

Management

- Rabbing in April 1 to May 15 is absolutely necessary for rust control.
- Early harvest (before leaves drop due to rust).
- Spray chlorothalonil or propiconazole or tebuconazole or mancozeb or plantovax or diclobutazole.
- Begin applications when plants are 5 to 10 cm high or when conditions become favourable for disease.
- Make second application 10 to 14 days after the first application.
- Crop rotation.
- Removal and destruction of disease affected plants.
- Use disease free planting material.
- Treat planting material with hot water.

3. Alternaria leaf blight : *Alternaria alternata*

Symptoms: -Appear on leaves and stem

- The main symptoms of the disease are appearance of the round to oval or slightly irregular dark brown spots on the upper surface of the leaves with concentric zones surrounding the pale yellow margin.
- Later, these spots coalesce, enlarge and occupy the entire leaf or a portion of the leaf which leads to the dropping of leaves.

Management:

- Spray Mancozeb or Zineb 0.2% or Copper oxy chloride 0.3% at the initial stage.
- Use disease free planting material.
- Treat planting material with hot water.

4. Verticillium wilt : *Verticillium albo atrum*

The pathogen is soil-borne and spread through soil. Roots and stolons used as seed material.
The vegetative mycelium is hyaline, septate, and multinucleate. *Verticillium* is named for this "verticillate" (=whorled) arrangement of the phialides on the conidiophore. Conidia are borne on phialides and each phialide carries a mass of conidia. Conidia are ovoid or ellipsoid and usually single-celled.

Disease cycle:

Disease spreads through infected stolons when used as seed material.

Symptoms:

- Symptoms first appear in the foliage at the top of the plant.
- The main symptoms of the disease are dwarfing, unilateral development of branches, etiolation and wilting
- Upper leaves are sickle-shaped and initially chlorotic or red, soon becoming necrotic.
- Vascular discoloration in stems and roots observed and diseased plants are often stunted.
- Premature defoliation and death of the plant.

Management:

- Applying optimal rates of nitrogen and enhanced dose of potash.
- Use of diseases free stolons
- Follow crop rotation, five years of planting to maize.
- Use bioagents like *Trichoderma sp*
- Cross protection - *V. nigrescens.*
- Deep summer ploughing.
- Well drained field conditions.

5. Powdery mildew : *Erysiphe cichoracearum/ E. biocellata*

This disease was first reported by Ganguli and Pandotra on Japanese mint.
It causes heavy losses in April-May leading to reduction in yield by 20% due to defoliation.

Symptoms:

- Small chlorotic spots appearing on the upper surface and brownish discolouration on the corresponding lower surface prior to the appearance of powdery phase.
- Later, circular white powdery patches appear mostly on the upper portions.
- Later, these white patches turn to purplish black and produce large number of clestothecia.
- Heavy defoliation occurs in severe cases having a banded appearance on stem and branches.
- Banded symptoms appear on stems and branches.

Management:

- Maintain proper row spacing
- Use resistant or tolerant varieties.
- Spraying of wettable sulphur 0.3% or Kerathane 0.05% at 15 day interval.

DISEASES OF OPIUM

1. Downy mildew : *Peronospora arborescens*

Mode of survival

- Primary source of inoculum: Oospores survives on plant debris of infected plants or in the soil for years. Dormant mycelium in crop residue can also initiate infection. Mycelia fragments on capsule may serve as source of infection
- Secondary source of inoculum: Conidia spread through wind, water splash
- The fungus infects *Argemone mexicona, A. platyceras, Meconosis* sp. *Papaver dubium, P. argimone* and *P. rhoeas*.
- The disease is carried from one season to another by means of thick walled oospores. It is not seed borne.

Etiology

- Mycelia- The mycelium is hyaline, coenocytic and intercellular.
- Sporangiophores are erect and very long. They are 7-10 times dichotomously branched and the ultimate branches are fine, curved, sharp, diverging almost at right angles, pointed at the tip.
- At the tip bears a single round –oval conidium .Sporangia are hyaline or pale violet.
- Asexual spores – **Barrel shape conidia** borne on sporangiophore in asexual fruiting body sporangia
- Sexual spore- Oospores borne in oogonium
- Oospores are round yellow, thick walled surrounded by an irregularly thick and reddish brown wall

Epidemiology

Temperature 22-24^{0}C, Relative humidity- 90-95 %, Intermittent rain fall and susceptible host.

- In India it is noticed during a middle of February, When the crop is fully grown.
- It is also noticed during November-February. It appears on seedling and such affected seedlings are killed.
- Yield loss from 7-65%.

Symptoms

- The plants with infection have two kinds of symptoms *viz*., Topical and systemic

In topical infection

- Lower portion is healthy and upper portion diseased.
- In the upper diseased portion of the plant, the leaves remain small, chlorotic, curling downward at the edges and are closely placed.
- Affected tissues curled, thickened and became deformed and necrotic, leading to premature drying of the leaves

In systemic infection

- The leaves near the tips and margins are covered with pale brown spots.
- The growth of severely infected plants was stunted and their leaves curled and twist due to production of high amount of grey - violet fungal growth on the lower side of the affected leaves.
- In the plants with leaf spot infection, the infected area becomes first chlorotic and then necrotic.
- The leaf spots enlarge by coalescing with each other. Sporulation may occur on the underside of spotted area.
- The infection also spread to the stem, branches and even capsules.
- It caused hypertrophy and curvature of the stem and flower stalks leading to premature drying and death of entire infected plants.
- The capsule, if formed, shrinks, gets wrinkled and dries gradually after infection.

- Heavy sporulation occurs on peduncle, buds and capsules which is not persistent.
- Brownish black structures develop in place of normal seeds.
- The infected plants die prematurely.

Management

- All disease plant should be uprooted, collected and destroyed.
- Early sown (30th October) crop was less affected by downy mildew incidence.
- Spraying of protective fungicides such as mancozeb prior to the appearance
- Seed treatment with metalaxyl + four sprays of metalaxyl at 0.2% at 35, 55, 75 and 95 days after sowing.
- Opium poppy cv. 'Rakshit', Jawahar-540 are resistant.
- The poppy line Vo.141, JOP 539 and MOP 540 is found to be highly resistant to the disease.

2. Powdery mildew : *Erysiphe polygoni*

Introduction- Powdery mildew is an endemic disease wherever the opium are grown in the world. The disease has been reported from the American continent, Europe, Africa, Australia and Asia. In India, the disease is most common in North India.

Mode of survival

- Primary source of inoculum : Ascospores, Dormant mycelia. It survives as dormant mycelia and as Cleistothecia on the shoots and buds from season to season.
- Secondary source of inoculum: Conidia.The disease spreads by the air- borne conidia.

Etiology: *Erysiphe polygoni*

- Mycelia is septate, external (ectoparasite) thin mycelia, haustoria is sub epidermal, obligate parasite.
- Asexual spores are barrel shape conidia borne on oidiophore in chains asexual fruiting body oidia.
- Sexual spores are ascospores unicellular inside the Ascus in the ascocarp (Cleistothecium) having myceloid type of appendages.

Epidemiology

The disease occurs in severe from Oct- Nov in North India and Feb- June in South India.

- Disease is favoured by warm winter temperature from 20 to 33.5^{0}C
- Disease development is adversely affected by rain and sunshine.

Symptoms

- The fungus attacks all the green plant parts at all stages of plant growth.
- Powdery mildew is observed in the fields at or after flowering stage *i.e*. nearly 14-16 weeks after sowing.
- Late sown crop (after 15th November) develops heavy powdery mildew disease.
- The base of the stem is attacked first where it appears as small, circular patch consisting of radially arranged mycelium. This enlarges gradually and sporulation occurs soon.
- The fungus produces white to greyish powdery patches on the affected plant parts including capsule but young leaves are most susceptible and develop small whitish patches both on upper as well as lower surface.
- These patches grow in size and coalesce to cover large areas on the leaf lamina.
- Malformation and discoloration of the affected leaves are also common symptom, resulting in distortion.
- Similarly, powdery patches are produced on the stem, tendril, flowers and young fruit branches. The buds on such branches remain rudimentary and wither .

Management:

- Crop sown before 15th October almost escape the disease.

Only one spray of 0.5% wettable sulphur at the time of first appearance of the disease in the field or when the crop has reached the age of 80 days or when the maximum daily temperature has reached 27°C and above or spraying of bayletan or Kerathane (0.05%).

3. Alternaria leaf spot- *Alternaria phragmospora*

Symptoms

- The initial symptoms of the disease were formation of numerous small, dull brown necrotic spots all over the leaves and peduncle.
- In later stages of infection the spots enlarged rapidly, coalesced and turned to brown necrotic patches leading to production of typical blight symptoms.
- Sometimes, dark brown patched were surrounded with chlorotic halo.

- Severely infected peduncle of opium poppy also turned pink to brown.

Management

- Removal and destruction of the plant debris from severely infected fields
- Crop rotation
- Use of disease resistant cultivars.
- Spraying of mancozeb (0.25-0.3%) just after the first appearance of the disease at an interval of 15 days.

4. **Poppy mosaic- *Poppy Mosaic Virus* (*PMV-P)***

Only mosaic disease of opium poppy was observed to date in India.

Mode of spread and survival

- Primary source of inoculum: Affected host plants, weed host (*A. Mexicana*) & self sown crops.
- Secondary source of inoculum: Virus particle transmitted by aphid vector, *Myzus persicae.*, The virus is non persistent type.

Potyviridae, Potyvirus, +ssRNA, flexious rods

Symptoms

Symptoms were more pronounced during February and March.

It causes stunting, vein banding and deformed capsule formation.

The characteristic disease symptoms were mottling, mosaic, irregular necrosis, distortion of capsule and puckering of leaves especially the terminal ones.

Infected plants produced abnormally smaller leaves than those of uninfected healthy plants. In some instances floral organs were converted into malformed structures, consequently, failed to produce normal types of capsules.

The infection caused very low production of alkaloids.

Infected plants eventually lost vigor and showed premature death and drying.

Sometimes the infected plants produced rosette due to the reduction of terminal growth and shortened internodes.

Epidemiology

- Summer season, High temperature, Susceptible host
- Improper management leads to development of this disease.

Management

- Initially spray Dimethoate 0.1% after 15 days and Imidacloprid 0.05 % at another 15 daysand neemazol 0.5 % later.

DISEASES OF DATURA

A. Foliar disease

- Leaf blight and fruit rot- *Alternaria cresaa*
- Leaf blight -*Alternaria alternata*
- Leaf spot – *Alternaria tenuissima*
- Cercospora leaf spot : *Cercospora jamaicenrsis*
- leaf spot- *Cercospora daturicola*
- Phomopsis leaf spot- *Phomopsis venenosa*
- Phyllosticta leaf *spot- Phyllosticta hortorum*
- Leaf spot - *Septoria lycopersici*
- Powdery mildew- *Leveillula taurica*

B. Soil borne diseases

- Collar Root rot/ Southern blight/Wilt - *Sclerotium rolfsii*
- Root rot –*Phytophthora* spp., *Pythium* spp.,
- Cotton root rot, Texas- *Phymatotrichum omnivoram*
- Black root rot -*Thielaviopsis basicola*
- Charcoal rot -*Macrophomina phaseoli*,

Bacterial diseases

- Bacterial wilt -*Pseudomonas solanacearum*
- Phyllody- Phytoplasma

Viral diseases- *D. stramonium* is a susceptible host to more than 60 viruses

- Datura wilt virus -*Tomato spotted wilt virus*
- *Colombian datura virus (CDV)* - aphid non-persistent transmission by *Myzus persicae -Potyvirus*
- *Datura distortion mosaic virus*- non-persistently by *Aphis craccivora* and *Myzus persicae*
- *Datura rugose leaf curl virus- Austroagallia torrida leafhoppers*
- Datura mosaic - *Datura mosaic virus* (DMV)- Virus transmitted by mechanical inoculation; transmitted by grafting;

DISEASES OF BELLADONA (*Atropa belladonna)*

A. Soil borne disease

- Root and stem rot- *Phytophthora parasitica*
- Root rot/wilt -*Fusarium solani*
- Damping off- *Pythium ultimum, P. debaryanum, Rhizoctinia solani* and *Phytophthora parasitica*)

B. Foliar disease

- Leaf spot- *Cercospora atropae*
- Downy mildew -*Peronospora parasitica*
- Leaf necrosis- *Ascochyta atropae*

Viral Diseases

- Mottling of belladonna -*Belladonna mottle virus*- tymovirus -Mechanical
- Belladonna mosaic *Belladonna mosaic virus* - *N*ematodes

DISEASES OF HEMP (*Cannabis sativa*)

1. Foliar diseases

1. Anthracnose-*Colletotrichum coccodes, Colletotrichum dematium*
2. Brown blight- *Alternaria alternata*
3. Curvularia leaf spot- *Curvularia cymbopogonis, Curvularia lunata* [*Cochliobolus lunatus* [teleomorph]
4. Downy mildew - *Pseudoperonospora cannabina, Pseudoperonospora humuli*
5. Rust -*Uromyces inconspicuus*
6. Powdery mildew -*Leveillula taurica, Oidiopsis taurica* [anamorph] *Sphaerotheca macularis*
7. Gray mold- *Botrytis cinerea*
8. White leaf spot -*Phomopsis ganjae*
9. Yellow leaf spot -*Septoria cannabis, Septoria cannabina*
10. Leaf spot: *Cercospora cannabina.*

B. Soil borne disease

1. Southern blight, Sclerotium root and stem rot- *Sclerotium rolfsii, Athelia rolfsii* [teleomorph]

2. Charcoal rot-*Macrophomina phaseolina*
3. Fusarium foot rot and root rot-*Fusarium solani*
4. Fusarium stem canker-*Fusarium sulphureum*
5. Fusarium wilt- *Fusarium oxysporum* f.sp. *cannabis,F. oxysporum* f.sp. *vasinfectum*
6. Hemp canker-*Sclerotinia sclerotiorum*
7. Verticillium wilt -*Verticillium albo-atrum,Verticillium dahliae*
8. Tropical rot -*Lasiodiplodia theobromae,= Botryodiplodia theobromae*
9. Rhizoctonia or sore-shin and root rot -*Rhizoctonia solani*
10. Pink rot -*Trichothecium roseum* (*Cephalothecium roseum)*
11. Phymatotrichum root rot (Cotton root rot) - *Phymatotrichum omnivorum*

C. Bacterial diseases

1. Bacterial blight-*Pseudomonas cannabina*

2. Crown gall- *Agrobacterium tumefaciens*
3. Leaf spot-*Xanthomonas campestris* pv. *cannabis*
4. Witches' broom- *Phytoplasma*

D Viral diseases

- Cucumber mosaic- *Cucumber mosaic virus* (CMV) *Cucumovirus*
- Alfalfa mosaic & Lucerne mosaic- *Alfalfa mosaic virus* (AMV)- *Alfamovirus*
- Arabis mosaic- *Arabis mosaic virus* (ArMV)- *Nepovirus*
- Hemp mosaic- *Sunn-hemp mosaic virus- Tobamovirus*
- Hemp streak-*Hemp streak virus* Transmitted by thrips, most likely a *Tospovirus*

DISEASES OF DIASCORIA

- Leaf spot: *Cercospora dioscoreae*
- Leaf fall & die back: *Colletotrichum sp*
- Leaf Blight: *Glomerella cingulata*
- Tuber Rot: *Fusarium sp., Rhizoctonia sp*
- Damping off: *Pythium spp*

DISEASES OF CROTALARIA

- Anthracnose- *Colletotrichum curvatum*

- Leaf spot- *Pestalotia oxyanth*
- Powdery mildew -*Microsphaera diffusa*
- Rust - *Uromyces* spp.,
- Fusarium wilt : *Fusarium udam* f. sp. *crotalariae*
- Stem rot : *Sclerotium rolfsi*
- Phymatotrichum root rot - *Phymatotrichum omnivorum*
- Root and stem rot- *Sclerotium rolfsii*

DISEASES OF NEEM

1) Powdery mildew: *Oidium azadiractae* (*Erysiphae*)
2) Root rot: *Ganoderma lucidum*
3) Leaf web blight: *Rhizoctonia solani*
4) Leaf spot: *Pseudocercospora subsessilis*
5) Bacterial wilt: *Pseudomonas azadiractae*
6) Angular leaf spot: *Xanthomonas azadiractae*

DISEASES OF COSTUS (*Costus speciosus*)

- Leaf blight- *Curvularia paradissi, Drechslera maydis or D. victoriae*
- Rhizome rot *Fusarium solani*
- Pythium rhizome rot -*Pythium spirosum*

DISEASES OF *Solanum khasianum*

- Leaf-blight- *Pythium butleri*
- Powdery mildew - *Leveillula taurica*
- Fusarium wilt –*Fusarium oxysporum* f. sp. *lacimiti*
- Damping - off - *Pythium butleri*

DISEASES OF WILD INDIGO (*Tephrosia purpurea)*

- Leaf blight- *Alternaria tenuissima*
- Witches'-Broom – Phytoplasma
- Tephrosia yellow vein virus- *Groundnut Stunt Virus*

DISEASES OF CAMPHOR (*Cinnamomum camphora)*

- Branch dieback and decline- *Phytophthora ramorum*
- Canker -*Botryosphaeria dothidea*
- Leaf blight -*Glomerella cingulate*
- Powdery mildew: *Microsphaera alphifoides, Erysiphe cinnamomi*
- Leaf blight: *Alternaria spp*.
- Brown root disease- *Phellinus noxius*
- Verticelium wilt: *Verticillium spp*.
- Bacterial leaf spot: *Pseudomonas spp*.

POSTHARVEST DISEASES OF FRUITS

Postharvest Diseases of Mango:

1. Diplodia stem-end rot: *Diplodia natalensis, Botryodiplodia theobromae*

- Darkening of pericarp near the base of pedicel.
- Affected areas become black cover entire fruits within 2-3 days.
- Pulp of such fruits turn soft and brown.

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2. Black mold rot: *Aspergillus niger*

- Yellowing of fruit base with irregular greyish spots and finally turn dark brown or black.
- In the rotten area mesocarp is soft and depressed finally covering with black spore masses.

3. Soft rot : *Rhizopus arrhizus*

- Causes soft rot of fruits.
- Rotted fruits are covered with profuse growth of fungal sporangiophores and sporangia.

4. Colletotrichum rot: (Anthracnose): *Colletotrichum gloeosporioides*

- Brown to black sunken spots on fruits which later cover the entire fruit surface.
- Production of black fruiting bodies on affected areas i.e. Acervuli
- Production of pink spore masses on rotted portions is often observed under humid conditions.

Management:

- Covering fruits and removing immediately after harvest to ripening houses.
- Care should be taken to prevent snapping off of pedicel.
- Avoiding injuries to fruits.
- Fruits dip in 0.1% Carbendazim/Thiophenate Methyl/Benlate/Prochloraz/Kavach 2gm/litre as pre and post harvest sprays to reduce incidence are effective.
- Dipping fruits in 6% borax solution at 43^0C for 3min.
- Spraying Bordeaux Mixture 1% effective in field.
- Fruit dip is Benomyl 500ppm and TBZ 900 ppm effective.
- HWT at 55^oc for 5 min. preferably along with the above chemicals i.e., Benomyl or TBZ.
- Both pre and post harvest application / treatments are important in reduction of the PH diseases.

Postharvest Diseases of Citrus:

1. Penicillium fruit rot :

Blue mold : *Penicillium italicum*
Green mold : *P. digitatum, P. expansum*

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- Rind of the affected fruits becomes watery and breaks easily when pressed.
- Diseased portions covered with characteristic blue or green mold growth.
- Fruits emit foul smell.
- The disease is more if harvest in wet and humid conditions.

2. Sour rot: *Geotrichum candidum*

- More common in kagzi lime and lemons when fruits ripe or over ripe.
- Fruits emit strong smell of fermented juice.
- Fruits will be very slippery due to rotting.
- White mycelial growth is observed on the rotting fruits.

3. Brown rot : *Phytophthora nicotianae.*

- During storage, healthy looking fruits may develop symptoms of brown rot.
- Fruit rind develops a water soaked areas which are covered with a cottony fungus growth.
- Rotten fruits emit foul smell.
- Mandarins oranges highly affected.
- More in heavy rainfall areas.

4. Diplodia stem-end rot: *Diplodia natalensis, Botryodiplodia theobromae*

- Darkening and rotting of the fruits from the stem end.
- Affected areas become black cover entire fruit surface.
- Pulp of such fruits turn soft and brown.

Management:

- Avoiding harvesting during wet weather.
- Avoiding injuries to fruits.
- Fruit dip in 2% bleaching powder solution for 5 minutes.
- Fruit dip in Boric acid @ 4% is effective for green mold.
- Pre storage dip in 500 ppm of TBZ for 2-3 minutes.

- Fruit dip in Benomyl/Carbendazim 0.1% after harvest.
- Spraying and soil drenching of Bordeaux mixture 1% in the field for brown rot.
- Spraying of Aliette/Metalaxyl/Mancozeb @ 2 gm/litre for managing brown rot.

Postharvest Diseases of Grapes:

1. Grey mould Rot: *Botrytis cinerea.*

- In the early stages tissue beneath the surface of fruit is attacked by the fungus loosening the skin from the flesh. i.e. 'slip skin'.
- Affected area turns brownish,soft, watery mass of decayed tissue.
- In moist weather fungus produces powdery spores and this is known as 'Grey mould stage'.

2. Blue mould rot: *Penicillium spp.*

- Serious in varieties with compact bunches.
- In storage and transit the fungus grows on stems and decays the berries.
- Tissues of the decaying berries become slightly brown and soft watery.
- The pathogen growth is white mouldy in the beginning and bluish green later.

3. Black Mould rot or Stalk end rot: *Aspergillus niger, A. carbonarius*

- More serious in varieties with compact bunches.
- Affected berries show brownish rot at the stalk end.
- Diseased tissues show watery decay and shrink.
- A whitish mycelia growth of the fungus grows in the centre of the diseased area and later black powdery masses appear.

4. Cladosporium rot: *Cladosporium herbarum, C. oxysporum, C. tenuissimum*

- Infection appears as sharply delimited dark spots of decay beneath the skin.
- Infections do not penetrate deeply and the berries are firm and comparatively dry.
- Berries become flat or wrinkled on the affected side.
- Olive green fungal growth is noticed on the infected areas.

5. Soft rot: *Rhizopus nigricans*

- Pathogen enters through cracks or injuries in the skin.
- Berries become soft and show leakage.
- Berries are covered with profuse mycelia growth of the fungus.

Management:

- Increased aeration and exposure to the sun by using appropriate training systems and by removing leaves around fruits.
- Field infection should be avoided by application of Thiophenate methyl 0.1%.
- Prompt cooling and storage of fruits.
- After harvesting fumigation with SO_2 for 20 min prior to storage.
- Fumigation with acetaldehyde 500 ppm vapours for 24 hrs.

Postharvest Diseases of Banana:

1. Anthracnose: ***Colletotrichum musae***

- Circular dark brown sunken areas with diffused edges and turn dark brown to black.
- Pink spore masses appear in humid weather.
- Latent infections from the field are common.

2. Black tip or Finger tip: ***Botryodiplodia theobromae***

- Pulp becomes soft and rots from the tips.

- Rotting of the pulp is faster than the skin.
- Two thirds or even whole finger may get rotted.
- White or light grayish cottony mycelium with pycnidia appears.

3. Crown rot: *Fusarium roseum, Botryodiplodia theobromae, Colletotrichum musae,* etc.

- Darkening of the crown region of the hands.
- Rotting of the infected portions and the hands will not support the fingers.
- Often, whitish/grayish/pinkish mycelia growth is observed.

4. Pink mould rot: *Trichothecium roseum*

- Fruit surface is covered with pink mouldy rot of the fungus.
- Rot is limited to the peel only.

5. Fusarium rot: *Fusarium moniliforme*

- Olive brown spots appear in the middle or the tip of the fruits.
- Fruits turn brown.
- Fruits become pulpy, liquid oozes out and emit bad odour.
- Finally, fruits are covered with whitish to pinkish mycelia growth.

Management:

- Strict sanitation in plantations and packing houses.
- Careful handling and harvesting.
- Cutting hands or bunches and packing carefully in cardboard boxes.
- Cooling fruits after harvest and storing at 10^0 C
- Fruit treatment with Carbendazim 400 ppm or Benomyl 1000 ppm or Aureofunginsol 100 ppm.

Postharvest Diseases of Apple:

1. Blue mold : *Penicillium italicum*

- Rind of the affected fruits become watery.
- Entire fruit rots and emits bad smell.
- Blue fungal growth is seen on the surface of the fruits.

2. Alternaria rot: *Alternaria alternate*

- Round brown to black lesions appear.
- Spots are firm, dry and shallow.
- Surface of spots becomes dark brown to black.
- Rotted tissues become spongy and turns black.

3. Grey mould: *Botrytis cinerea*

- Infected fruits turn slightly brownish.
- The fungus advances into the inner flesh resulting in a soft, watery mass of decayed tissue contained in a slightly intact, brown skin.
- Typical powdery grey mould appears on the fruit surface.

4. Bitter rot: *Glomerella cingulata*

- Small brown circular spots appear on fruit surface.
- Spots become sunken forming a saucer shaped depression.
- In wet weather pink spore masses representing the fruiting bodies of the fungus appear.

5. Brown rot: *Monilinia fructicola*

- Enlarged rots are soft, circular, medium brown in the beginning.

- Decayed area enlarges and small black spots appear at lenticels.
- Entire fruit decayed and under warm conditions turns black and develops a velvety sheen.

Management:

- Pre storage dip in TBZ 500 ppm for 2-3 minutes.
- Fruit dip in 2% bleaching powder for 5 minutes.
- Using sodium bisulphate which releases SO_2 When in contact with moist air with packing material.
- Aromatic compounds such as dichloronitroaniline, sodium ortho phenyl phenate and biphemyl effective in management of apple rots.
- Avoid injuries to the fruits.
- Apples treated in hot water bath immediately after harvest reduces blue mold rot.